Endpoint crossings for the elephant random walk

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Abstract

In the elephant random walk each step copies a uniformly chosen earlier step and reverses it with probability ε. For a walk with an even number N of steps we look for the crossing, a value of ε at which the walk is as likely to end at the origin as at N. We prove that the crossing is unique and that $N\varepsilon$ at the crossing is $2\log N - \log\log N - \log 16 + o(1)$. This is about twice the value for the persistent random walk, which copies its last step instead, because a walk that ends at the origin usually has a reversal among its first few steps. At the crossing the law of the final position has a hump near each of $\pm N$, and for large N the origin is the least likely position other than $\pm N$.

Keywords: elephant random walk; persistent random walk; random recursive tree; Pólya urn; Landau distribution.
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1 Introduction

The elephant random walk of Schütz and Trimper [3] remembers its whole past. Its first step is $+1$ or -1 with probability $\frac{1}{2}$ each. Each later step picks one of the earlier steps uniformly at random and repeats it, except that with probability ε it reverses it. For a walk with N steps we write $x = N\varepsilon$. The expected number of reversals is $(N-1)\varepsilon = x - \varepsilon$, so x is this number up to an error below one. We let A_N be the number of steps equal to $+1$.

We ask when the walk is as likely to end in the middle as at each end. Let N be even. The walk ends at the origin, which we call the middle, when $A_N = N/2$. It ends at N or at $-N$, which we call the ends, when $A_N = N$ or $A_N = 0$. If ε is close to 0 the walk rarely reverses, so the ends win. If $\varepsilon = \frac{1}{2}$ the steps are independent and fair, so the middle wins. The crossing is the value of ε in between at which the middle and each end are equally likely.

In the persistent random walk each step repeats the previous one with probability p and switches with probability $q = 1 - p$. So with $q = \varepsilon$ it is the walk that copies its last step instead of a uniformly chosen one. Its steps are the spins of the Ising chain in zero field with free ends, and its position is the magnetization. For this Markov walk our question is part of Problem 131 of Kagey's Open Problem Collection [1], which is posed for a Galton board. As on that board, we call the values of A_N bins. Let $T = Nq$, which exceeds the expected number $(N-1)q$ of switches by q. The middle catches up with the ends when T is about $\log N$, and [2] locates this crossing within explicit bounds for every $N \geq 2$.

In this paper we find the crossing of the elephant walk. The middle needs about twice as many reversals as the persistent walk needs switches. Namely, the crossing value of x is about $2\log N$,

not $\log N$. The limit theorems of [9, 10] for the elephant walk fix the memory parameter, here $1-\varepsilon$, as N grows, while we let $\varepsilon \to 0$. Kürsten's coupling [4] splits the steps into the clusters of bond percolation on a random recursive tree, and our regime $x \asymp \log N$ lies in the strongly supercritical regime of this percolation [6, 7]. The results there describe cluster sizes, not the probability of the middle (Section 1.3).

Why the elephant needs twice as many reversals. The following argument is heuristic, and Sections 2 and 3 make it precise. The ends are easy. The walk ends in the right end bin exactly when its first step goes right and it never changes direction. This has probability $\frac{1}{2}(1-\varepsilon)^{N-1} \approx \frac{1}{2}e^{-x}$ for the elephant walk, and the same with q in place of ε for the persistent walk. So only the middle differs. In the persistent walk the T switches are spread over the whole walk, and each switch moves the final position by about one run length N/T. So the number of right steps spreads over about $N/\sqrt{T}$ bins, the middle bin has probability of order $\sqrt{T}/N$, and the ends lose when e^{-T} falls to about $1/N$, that is, when $T \approx \log N$.

The elephant walk spreads a reversal in a different way. A reversal at step $t+1$ is copied by later steps, those copies are copied, and so on. The steps that descend from it form a family, whose share of the walk behaves like the share of one ball in Pólya's urn started with one ball against t. So a late reversal has a small family and hardly moves the walk. To end exactly in the middle, the walk usually needs a family that fills about half the walk, so in most such walks a reversal happens among the first few steps. This has probability of order ε, and hitting the exact middle brings another factor of order $1/N$. So the middle bin has probability of order $\varepsilon/N = x/N^2$ (in fact about $4x/N^2$), and the ends lose only when e^{-x} falls to about $1/N^2$, that is, when $x \approx 2\log N$. More precisely, at the crossing of the persistent walk $\frac{1}{2}e^{-T} \approx \sqrt{2T/\pi}/N$, and squaring this relation turns it into the elephant relation $\frac{1}{2}e^{-x} \approx 4x/N^2$ with $x \approx 2T - \log(2\pi)$.

The small families also give the shape of the law. About x/m^2 reversals have a family of m steps, so the bin at distance m from an end has probability about $x/(2m^2)$ over a wide range of m. So the law has a hump near each end, and the middle is where the two tails meet. Indeed, twice $x/(2m^2)$ at $m = N/2$ is the $4x/N^2$ above.

1.1 Setting and notation

The walk takes N steps $X_1, \dots, X_N \in \{+1, -1\}$, where $+1$ means right. Let $A_N = \#\{i \le N : X_i = +1\}$ be the number of right steps, and write $P_N(k) = \mathbb{P}(A_N = k)$ for the probability of bin k. Unless we say otherwise the first step is fair, $\mathbb{P}(X_1 = +1) = \frac{1}{2}$, and the rules treat the two directions alike, so $P_N(k) = P_N(N-k)$. We write $\mathbb{P}_+$ and $\mathbb{E}_+$ for probability and expectation given $X_1 = +1$. For even N we put $h = N/2$ and call $C_N = P_N(h)$ the center and $E_N = P_N(N) = P_N(0)$ the end. Whenever C_N appears, N is even. The crossing is the value of the parameter at which $C_N = E_N$. All logarithms are natural, $L = \log N$, $H_k = \sum_{j \le k} 1/j$ is the k-th harmonic number, and γ is Euler's constant. Limits, $O(\cdot)$ and $o(\cdot)$ refer to $N \to \infty$, over even N when C_N appears. The constants implied by $O(\cdot)$ are absolute unless we name what they depend on.

The elephant walk (Definition 2.1) has the reversal probability $\varepsilon \in (0,1)$ and $x = N\varepsilon$. Its crossing is ε_N, and $x_N = N\varepsilon_N$. The symbols that only one section uses are defined at the start of that section.

For the persistent walk, p, $q = 1-p$ and $T = Nq$ are as above. Its end is $E_N = p^{N-1}/2$, its crossing is p_N, and $z_N = N(1-p_N)$. So $x = N\varepsilon$ corresponds to T, and the crossing x_N corresponds to z_N. We compare the two crossings through the crossing equation of [2, Proposition 3.8], which

says that for every $N \geq 2$

$$z_N + \tfrac{1}{2}\log z_N = L + c_0 + \frac{1}{8z_N} + O(L^{-2}), \qquad c_0 = \tfrac{1}{2}\log\tfrac{\pi}{8}, \tag{1}$$

where the error term is at most $\frac{1}{4L^2} + \frac{L^2+L}{N}$ in absolute value, and that $L/2 \leq z_N \leq L$. This is the only result of [2] that we use, and we use it only to compare the two crossings in Section 2. The letters p and q have their meaning for the persistent walk only in this introduction, and later sections reuse them locally.

1.2 Results

For every even $N \geq 2$ the crossing ε_N exists, is unique and is below $\frac{1}{2}$ (Proposition 2.3). The center has derivative exactly $4/(N+2)$ at $\varepsilon = 0$ (Proposition 2.5), and $C_N = \frac{4\varepsilon}{N+2}(1 + O(\varepsilon \log N))$ for even $N \geq 64$ and $\varepsilon(1+\log N) \leq \frac{1}{100}$, with explicit constants (Theorem 2.6). Let g_N be the root of $g + \log g = 2L - \log 8$. Then $x_N = g_N + O(L^2/N)$, and so $x_N = 2L - \log L - \log 16 + O(\log L/L)$ (Theorem 2.7). Compared with the persistent walk, $x_N = 2z_N - \log(2\pi) + c_*/L + O(\log L/L^2)$ with $c_* = \frac{1}{2}\log(2\pi) - \frac{1}{4}$ (Theorem 2.8). We collect the main statements.

Main Theorem. *For every even $N \geq 2$ the elephant walk has a unique crossing $\varepsilon_N \in (0, \frac{1}{2})$. As $N \to \infty$ over even N, the value $x_N = N\varepsilon_N$ satisfies*

$$x_N = 2\log N - \log\log N - \log 16 + O\Big(\frac{\log\log N}{\log N}\Big), \qquad x_N = 2z_N - \log(2\pi) + \frac{c_*}{\log N} + O\Big(\frac{\log\log N}{\log^2 N}\Big),$$

where z_N is the crossing value of T for the persistent walk and $c_* = \frac{1}{2}\log(2\pi) - \frac{1}{4}$.

We also go one order further. In the same range of ε the center is $\frac{4\varepsilon}{N+2}N^{2\varepsilon}(1 + O(\varepsilon))$, with explicit constants (Proposition 2.9). This gives $x_N = g_N - (2g_N L + g_N^2/2)/(N(1+1/g_N)) + O(L/N)$, so the root g_N is above the crossing for large N (Proposition 2.10). At fixed N the center is a polynomial $c_1\varepsilon + c_2\varepsilon^2 + \cdots$ in ε, and $c_2/c_1 - 2L \to 2\gamma - 2 - \log 2$ (Proposition 2.11).

The remaining results describe the whole law near the crossing. At the crossing the end is below its neighbor for every even $N \geq 8$ (Corollary 3.6 and Proposition 3.8). For $N < 1688$ this rests on a floating-point computation whose rounding error we bound (Lemma 3.7). Let M_N be the number of steps that disagree with the first step. For $N \geq 2$ and $\varepsilon \leq \frac{1}{4}$ the law of M_N is within an explicit δ_B of a compound Poisson law, bin by bin, and $\delta_B = O(x^2 \log^2 N/N)$ when $x^2 \leq N$ (Theorem 3.9). If $x \to \infty$ and $x^3 \log^2 N/N \to 0$, then $(M_N - x\log x)/x$ converges in law to the Landau law, with a bound for each bin (Theorem 3.10). So at the crossing every mode lies at distance $x_N \log x_N + \lambda_0 x_N + O(\sqrt{x_N \log x_N})$ from the nearer end, where $\lambda_0 \approx -0.2228$ is the mode of the Landau density (Corollary 3.12). Corollary 3.17 gives the weaker bound $O(x_N \log^2 x_N)$ for this distance. Farther from the ends, bin $N - m$ has probability $\frac{x}{2m^2}(1 + o(1))$ for $x\log^2(x+1) \ll m \ll N$, when $x \geq 1$ and $x\log^2 N/N \to 0$ (Theorem 3.14). Finally, for large even N the center at the crossing is below every interior bin outside a window of width of order $N^{1/2}\log N$ around it (Theorem 3.15). Inside the window we use second differences. For every $N \geq 3$ and every $0 < \varepsilon < \frac{1}{2}$ the signs of the second difference of the law follow the pattern convex, concave, convex, concave, convex, where some blocks may be empty (Theorem 3.21). So the law has at most two concave stretches. For large even N the law at the crossing is strictly convex between the humps (from $2m^*$ to $N - 2m^*$, with m^* the distance of a mode from the nearer end), and the center is its unique interior minimum (Theorem 3.24 and Corollary 3.25). For odd N we

define the crossing as the value of ε at which the two central bins are as likely as each end. The convexity also holds at this crossing for large odd N, because the two central bins of an odd N are the center of $N+1$ up to an explicit factor (Proposition 3.27). A certified computation gives the convexity for $10 \le N \le 2000$ (Proposition 3.26), and with it the unique interior minimum for every even N in this range (Proposition 3.28). For the N above 2000 that these results do not cover, the convexity is open (Conjecture 3.16).

1.3 Prior work

The asymptotic behavior of the elephant walk for a fixed memory parameter is well understood [22, 21, 23], and Bertoin [24] counts its returns to the origin. For $\varepsilon \le \frac{1}{2}$ the steps of the elephant walk split into clusters, since a step either copies an earlier step exactly, with probability $1-2\varepsilon$, or starts a new cluster with a fair sign. This is Kürsten's coupling with bond percolation on random recursive trees [4], and Baur and Bertoin [5] relate the walk to Pólya-type urns. Our regime $N\varepsilon \asymp \log N$ lies in the strongly supercritical regime of Baur [6], who extends the results of Bertoin [7] for retention probability $1 - t/\log n$. Baur's Corollary 1 gives the sizes of the largest clusters other than the root's (a Poisson limit with intensity $a^{-2}\,da$). In our regime this is the picture of about $2x/m^2$ clusters of size m, since a new cluster starts with probability 2ε. Half of them have the sign opposite to the root, which gives the x/m^2 families of size m behind Theorem 3.14. These results describe cluster sizes, not the center probability, which is a large deviation of the root cluster. Miyazaki and Takei [8] prove limit theorems in the diffusive, critical and superdiffusive regimes for a minimal walk of elephant type whose steps are 0 or +1. Their results also give limit theorems for the size of the root cluster in percolation on uniform random recursive trees. The law that appears in the bulk (Theorems 3.9 and 3.10) is also known. Without its truncation, our compound Poisson law is the Lea–Coulson form of the Luria–Delbrück distribution, and Kessler and Levine [12, Eqs. (14), (18) and (19)] describe its profile $x/(m(m+1))$ for small x and its Landau limit for large x (see also [13]). The same stable law governs the fluctuations of the root cluster when the retention probability is $1 - c/\log n$ [11]. What we add is the comparison of the elephant walk with this law in our regime (with an explicit error in Theorem 3.9) and a certified enclosure of the mode. Guérin, Laulin, Raschel and Simon [9, proof of Theorem 1.1] prove that the law with a fixed first step is unimodal, and their other results fix the memory parameter as N grows. So do the local limit theorem of Trinas and Valle [10, Theorem 3.1] for the superdiffusive walk and their bounds on where log-concavity fails. We did not find an asymptotic formula for the center probability when $\varepsilon \to 0$ as N grows, either in the elephant-walk literature or in the percolation literature.

1.4 Organization

Section 2 reduces the elephant walk to a counting chain, proves that its crossing is unique, finds the crossing, and then gives the center and the crossing to second order. Section 3 describes the whole law near the crossing, namely the dip at each end, the compound Poisson and Landau approximations of the bulk with the place of the modes, the tail $x/(2m^2)$, the window around the center, and the convexity between the humps. Section 4 describes the checks. The longer proofs are in the appendices. Appendix A holds the proofs for Section 2. Appendix B holds the proofs for the tail and the modes in Section 3, and Appendix C holds the proofs for the bulk.

2 The elephant walk and its crossing

We reduce the elephant walk to a counting chain and prove that the crossing is unique. We then compute the center to first order in ε, bound it from both sides, and locate the crossing. The last subsection gives one more term of the center and of the crossing.

Definition 2.1 (elephant walk)**.** Fix $0 < \varepsilon < 1$. Let X_1 be uniform on $\{\pm 1\}$. For $t \geq 1$ let U_t be uniform on $\{1, \dots, t\}$, and put $X_{t+1} = X_{U_t}$ with probability $1 - \varepsilon$ and $X_{t+1} = -X_{U_t}$ with probability ε. All these choices are independent. We call the event $X_{t+1} = -X_{U_t}$ a flip. As in Section 1.1, $A_N = \#\{i \leq N : X_i = +1\}$ and $P_N(k) = \mathbb{P}(A_N = k)$. The crossing ε_N is the value of ε with $C_N = E_N$, and $x_N = N\varepsilon_N$.

A flip is what the introduction called a reversal. In the notation of [3] the memory parameter is $1 - \varepsilon$, and [9] write $a = 1 - 2\varepsilon$. Given the first step, all that matters is how many steps disagree with it. So let M_t be the number of $i \leq t$ with $X_i \neq X_1$, and put

$$r_t(b) = \varepsilon + (1 - 2\varepsilon)\frac{b}{t}, \qquad f_N(m) = \mathbb{P}_+(M_N = m).$$

Lemma 2.2. *The process $(M_t)_{t \geq 1}$ is a Markov chain with $M_1 = 0$ and $\mathbb{P}(M_{t+1} = b + 1 \mid M_t = b) = r_t(b) = 1 - \mathbb{P}(M_{t+1} = b \mid M_t = b)$, and it is independent of X_1. For $0 \leq k \leq N$, $P_N(k) = \frac{1}{2}[f_N(N - k) + f_N(k)]$. In particular $C_N = f_N(h)$, $E_N = \frac{1}{2}(1 - \varepsilon)^{N-1}$, and $P_N(k) = P_N(N - k)$.*

Proof. Given $X_1, \dots, X_t$, the copied step X_{U_t} differs from X_1 with probability M_t/t. So $X_{t+1} \neq X_1$ with conditional probability $(1 - \varepsilon)M_t/t + \varepsilon(1 - M_t/t) = r_t(M_t)$. This depends only on M_t and not on X_1, which proves the first claim. On $\{X_1 = +1\}$ we have $A_N = N - M_N$, and on $\{X_1 = -1\}$ we have $A_N = M_N$. Averaging over X_1 gives the formula for $P_N(k)$, which does not change under $k \mapsto N - k$. Finally $M_N \leq N - 1$, so $f_N(N) = 0$, and $f_N(0) = \prod_{t=1}^{N-1}(1 - r_t(0)) = (1 - \varepsilon)^{N-1}$. □

So the end is that of the persistent walk with switch probability ε. More flips always raise the center relative to the end.

Proposition 2.3. *Let $N \geq 2$ and $1 \leq k \leq N - 1$. Then $f_N(k)/f_N(0)$ and $P_N(k)/E_N$ are strictly increasing in $\varepsilon \in (0, 1)$. In particular C_N/E_N is strictly increasing, the crossing ε_N exists and is unique, and $\varepsilon_N < \frac{1}{2}$. Also $C_N \geq 2E_N$ for $\varepsilon \geq \frac{1}{2}$.*

Proof. We compare each path of M with the path that never flips, one step at a time. The probability of a path of M is a product of $N - 1$ factors, namely $r_s(b_s)$ for an up step at time s and $1 - r_s(b_s)$ otherwise, where b_s is the state at time s. The path that never flips has probability $f_N(0) = (1 - \varepsilon)^{N-1}$, which is one factor $1 - \varepsilon$ for each step. So we divide each factor by $1 - \varepsilon$. With $y = b_s/s \in [0, 1)$ the two kinds of factors become $\frac{\varepsilon + (1-2\varepsilon)y}{1-\varepsilon}$ and $1 - y\frac{1-2\varepsilon}{1-\varepsilon}$, whose derivatives in ε are $\frac{1-y}{(1-\varepsilon)^2} > 0$ and $\frac{y}{(1-\varepsilon)^2} \geq 0$. Both factors are positive, since $r_s(b_s) = \varepsilon(1 - y) + (1 - \varepsilon)y$ lies strictly between 0 and 1. So the ratio for each path is nondecreasing in ε, and it is strictly increasing if the path has an up step. Summing over the paths with $k \geq 1$ up steps proves the claim for $f_N(k)/f_N(0)$. By Lemma 2.2, $P_N(k)/E_N = [f_N(k) + f_N(N - k)]/f_N(0)$, and both k and $N - k$ are at least 1. So the claim for $P_N(k)/E_N$ follows.

It remains to locate the crossing. The ratio C_N/E_N is continuous and strictly increasing in ε. As $\varepsilon \to 0$ it tends to 0, since $C_N \to 0$ (at $\varepsilon = 0$ the chain never leaves 0) and $E_N \to \frac{1}{2}$. For $\varepsilon \geq \frac{1}{2}$ every $r_s(b)$ and every $1 - r_s(b)$ lies in $[1 - \varepsilon, \varepsilon]$, so each path of M that ends at h has probability at least $(1 - \varepsilon)^{N-1} = 2E_N$. Hence $C_N \geq 2E_N$ there. So the ratio equals 1 at exactly one ε, and this ε lies in $(0, \frac{1}{2})$. □

2.1 First order, one early flip

When $\varepsilon = 0$ the chain M never leaves 0. After a first flip, and to first order in ε, it is Pólya's urn.

Lemma 2.4 (Pólya's urn)**.** *Let $2 \le v \le N$. At time v an urn holds one ball of one color and $v-1$ of the other. At each time $k = v, \dots, N-1$ we draw a ball uniformly and return it with one more ball of its color. Let $\beta_v(d)$ be the probability that at time N there are d balls of the first color. Then $\beta_v(d) = \binom{N-d-1}{v-2}/\binom{N-1}{v-1}$ for $1 \le d \le N-v+1$, and $\beta_v(d) = 0$ for other d. Also β_v is nonincreasing in d and $\sum_d d\,\beta_v(d) = N/v$. For $1 \le d \le N-1$,*

$$\sum_{v=2}^{N} \beta_v(d) = \frac{N}{d(d+1)}, \qquad \sum_{v=2}^{N} (v-2)\,\beta_v(d) = \frac{2N(N-d-1)}{d(d+1)(d+2)}.$$

Proof. We compute the probability of one order of the draws and count the orders. At time k the urn holds k balls. So if s of them are of the first color, the next draw is of that color with probability s/k. Take one order of the $N-v$ draws in which i draws are of the first color. The draws of the first color contribute the numerators $1, 2, \dots, i$, the other draws contribute $v-1, v, \dots, N-i-2$, and the denominators are $v, v+1, \dots, N-1$. So this order has probability $\frac{i!\,(v-1)v\cdots(N-i-2)}{v(v+1)\cdots(N-1)} = \frac{(v-1)\,i!\,(N-i-2)!}{(N-1)!}$, whatever the order is. There are $\binom{N-v}{i}$ orders. With $d = i+1$ this gives $\beta_v(d) = a_d(v-1)\binom{N-v}{d-1}$ with $a_d = \frac{(d-1)!(N-d-1)!}{(N-1)!}$, and writing out the factorials shows that this is $\binom{N-d-1}{v-2}/\binom{N-1}{v-1}$. From the last form, $\beta_v(d)/\beta_v(d-1) = 1 - \frac{v-2}{N-d} \le 1$ for $2 \le d \le N-v+1$, so β_v is nonincreasing. The fraction of balls of the first color is a martingale that starts at $1/v$. So the mean number of such balls at time N is N/v.

For the two sums we use the first form of $\beta_v(d)$ and put $i = N-v$, so that $v-1 = \binom{N-1-i}{1}$ and $(v-1)(v-2) = 2\binom{N-1-i}{2}$. The Chu–Vandermonde identity says that $\sum_i \binom{i}{a}\binom{N-1-i}{b} = \binom{N}{a+b+1}$ (sort the subsets of $\{1, \dots, N\}$ of size $a+b+1$ by their element number $a+1$, which we call $i+1$). With $a = d-1$ and $b = 1, 2$ it shows that the two sums are $a_d\binom{N}{d+1}$ and $2a_d\binom{N}{d+2}$, and these are the values in the statement. □

We need some more notation. Let $V_s(b) = \mathbb{P}_+(M_N = h \mid M_s = b)$ be the chance of ending at the center from state b at time s. For $1 \le t \le h$ let

$$\pi_t = \beta_{t+1}(h) = \frac{t}{N-t} \prod_{i=1}^{t-1} \frac{h-i}{N-i}, \qquad c_1 = \sum_{t=1}^{h} \pi_t, \qquad w_t = \frac{\pi_t}{c_1},$$

so that $(w_t)_{1 \le t \le h}$ is a probability vector.

Proposition 2.5. *Let $N \ge 2$ be even.*

(a) $C_N = \sum_{t=1}^{h} \varepsilon(1-\varepsilon)^{t-1} V_{t+1}(1)$.

(b) C_N *is a polynomial in ε with $C_N(0) = 0$. At $\varepsilon = 0$ we have $V_{t+1}(1) = \pi_t$, and $C_N'(0) = c_1 = \frac{4}{N+2}$ exactly.*

(c) $\sum_t w_t t = 3 - \frac{6}{h+2}$, *and $w_t \le 2t2^{-t}$ for $1 \le t \le h$.*

Proof. (a) We condition on the time of the first flip. Under $\mathbb{P}_+$ the first flip happens at step $t+1$ exactly when $M_s = 0$ for $s \le t$ and $M_{t+1} = 1$. Since $r_s(0) = \varepsilon$, this has probability $(1-\varepsilon)^{t-1}\varepsilon$, and by the Markov property the chain then ends at h with probability $V_{t+1}(1)$. A path with no flip ends at $0 \ne h$. So C_N is the sum of $\varepsilon(1-\varepsilon)^{t-1}V_{t+1}(1)$ over $1 \le t \le N-1$. Only $t \le h$ contribute.

Indeed at time $t+1$ there are t steps equal to X_1, this number never decreases, and at the center it is h. So $V_{t+1}(1) = 0$ for $t > h$.

(b) Every transition probability is affine in ε, so C_N and $V_{t+1}(1)$ are polynomials in ε. By (a), $C_N(0) = 0$ and $C_N'(0) = \sum_t V_{t+1}(1)|_{\varepsilon=0}$. At $\varepsilon = 0$ we have $r_s(b) = b/s$. So from time $t+1$ on, M counts the balls of the first color in the urn of Lemma 2.4 with $v = t+1$, and $V_{t+1}(1) = \beta_{t+1}(h) = \pi_t$. Since $\beta_v(h) = 0$ for $v > h+1$, Lemma 2.4 with $d = h$ gives $c_1 = \sum_{v=2}^{N} \beta_v(h) = \frac{N}{h(h+1)} = \frac{4}{N+2}$.

(c) In the same way the second sum of Lemma 2.4, with $d = h$ and $v = t+1$, gives $\sum_t (t-1)\pi_t = \frac{2N(h-1)}{h(h+1)(h+2)}$. Dividing by $c_1 = \frac{N}{h(h+1)}$ gives $\sum_t w_t(t-1) = \frac{2(h-1)}{h+2} = 2 - \frac{6}{h+2}$, and adding $\sum_t w_t = 1$ gives the first claim. For the second claim, each factor $\frac{h-i}{N-i}$ in π_t is at most $\frac{1}{2}$, so $\pi_t \le \frac{t}{N-t} 2^{1-t}$. Hence $w_t = \frac{N+2}{4}\pi_t \le \frac{N+2}{2(N-t)}\, t 2^{-t}$. For $t \le h$ we have $N - t \ge h$ and $\frac{N+2}{2h} = 1 + \frac{2}{N} \le 2$, so $w_t \le 2t2^{-t}$. □

In words, to first order the walk flips once, at some step $t+1$, and then runs as Pólya's urn from t balls of one color and one of the other. As $N \to \infty$ the final fraction of the color that started with one ball tends in law to a Beta$(1, t)$ variable, whose density at $\frac{1}{2}$ is $t2^{1-t}$, and $\sum_t t2^{1-t} = 4$ explains the constant $4/(N+2)$. Since $w_t \le 2t2^{-t}$, only an early flip counts.

2.2 Two-sided bounds on the center

This subsection bounds the center from both sides. The bounds give the crossing in the next subsection.

Theorem 2.6. *Let $N \ge 2$ be even.*

(a) For $0 < \varepsilon < \frac{1}{2}$, $C_N \ge \frac{4\varepsilon}{N+2}(1 - 2\varepsilon)$.

(b) $C_N \le \frac{4\varepsilon}{N+2}\, N^{2\varepsilon} e^{5200\varepsilon}$ whenever $N \ge 64$ and $0 < \varepsilon(1 + \log N) \le \frac{1}{100}$.

In particular $\log C_N = \log \frac{4\varepsilon}{N+2} + O(\varepsilon \log N)$ in the range of (b), with an absolute implied constant.

In words, when $\varepsilon \log N$ is small the first-order term is correct up to a factor $1 + O(\varepsilon \log N)$, which is $1 + O(L^2/N)$ at the crossing.

Idea of the proof. The proof is in Appendix A. After the first flip, M at time s is Pólya's urn with βs extra balls of each color, where $\beta = \varepsilon/(1 - 2\varepsilon)$ (Lemma A.1). They pull the minority fraction toward $\frac{1}{2}$. For (a) we drop them. For (b) we compare M with an urn with $\kappa(s + 10)$ extra balls, where $\kappa = 2\varepsilon/(1 - 4\varepsilon)$. One step of M at time s multiplies the hitting probability of this urn by a factor of at most about $e^{\kappa/s}$, and the product of these factors is about $e^{\kappa H_N} \approx N^{2\varepsilon}$.

2.3 The crossing

By Theorem 2.6, $C_N \approx 4x/N^2$ when $\varepsilon = x/N$ and x is of order $\log N$. Also $E_N = \frac{1}{2}(1 - x/N)^{N-1} \approx \frac{1}{2}e^{-x}$. Setting them equal gives $x + \log x \approx 2 \log N - \log 8$. So let

$$\Lambda_N = 2L - \log 8, \qquad G(x) = x + \log x,$$

and let g_N be the root of $G(g_N) = \Lambda_N$. It exists and is unique, since G increases from $-\infty$ to ∞ on $(0, \infty)$.

Theorem 2.7. *As $N \to \infty$ over even N, the following hold.*

N	x_N	g_N	$x_N - x_{\mathrm{2nd}}$	$x_N/\log N$	$D(N)\log N$
50	3.940487	4.288636	+0.3452	1.007	−0.083
400	7.629217	7.843768	+0.0621	1.273	−0.199
1600	10.240565	10.340051	+0.0179	1.388	+0.215
3200	11.546834	11.610464	+0.0097	1.431	+0.386
12800	14.158845	14.182921	+0.0028	1.497	+0.607
51200	16.778468	16.786946	+0.0008	1.547	+0.707

Table 1: Elephant crossings x_N, computed in double precision from the recursion of Lemma 2.2, and the root g_N of $G(g) = \Lambda_N$. Here x_{2nd} is the formula of Proposition 2.10 without its error term, and $D(N)$ is the difference of Theorem 2.8.

(a) $x_N + \log x_N = \Lambda_N + O(L^2/N)$, *so* $x_N = g_N + O(L^2/N)$.

(b) $x_N = 2\log N - \log\log N - \log 16 + O(\log\log N/\log N)$.

In words, the root g_N locates the crossing to within $O(L^2/N)$, and the expected number of flips at the crossing is about $2\log N$.

Idea of the proof. Let $F(x)$ be the logarithm of C_N/E_N at $\varepsilon = x/N$. It increases in x, and its only zero is x_N. Theorem 2.6 and the formula for E_N show that $F(x)$ is within $O(L^2/N)$ of $G(x) - \Lambda_N$ when x is of order L. Since $G' \geq 1$, the zero of F is within $O(L^2/N)$ of the zero g_N of $G - \Lambda_N$.

Proof. (a) *Step 1: both sides near the crossing.* Put $F(x) = \log C_N(x/N) - \log E_N(x/N)$ for $0 < x < N$, where $C_N(\varepsilon)$ and $E_N(\varepsilon)$ are the center and the end at flip probability ε. By Proposition 2.3, F is strictly increasing and x_N is its only zero. We show that

$$|F(x) - G(x) + \Lambda_N| \leq \rho_N \quad (1 \leq x \leq 2\Lambda_N), \qquad \text{with } \rho_N = O(L^2/N). \tag{$*$}$$

Let $1 \leq x \leq 2\Lambda_N$ and $\varepsilon = x/N$. Then $\varepsilon \leq 4L/N$, so for large N both parts of Theorem 2.6 apply, and together they say that $\log C_N = \log \frac{4\varepsilon}{N+2} + O(\varepsilon L)$. Here $\log \frac{4\varepsilon}{N+2} = \log x + \log 4 - 2L - \log(1 + 2/N)$, and $\log(1 + 2/N) \leq 2/N \leq 2xL/N$ because $x \geq 1$. For the end we use $\log(1 - y) = -y + O(y^2)$ for $0 \leq y \leq \frac{1}{2}$ in $\log E_N = -\log 2 + (N-1)\log(1 - \frac{x}{N})$. So

$$\log C_N = \log x + \log 4 - 2L + O\Big(\frac{xL}{N}\Big), \qquad \log E_N = -\log 2 - x + O\Big(\frac{x^2}{N}\Big).$$

We subtract. Since $\log 4 + \log 2 = \log 8$, the main terms give $x + \log x - 2L + \log 8 = G(x) - \Lambda_N$, and both errors are $O(L^2/N)$ because $x \leq 2\Lambda_N \leq 4L$. This is $(*)$.

Step 2: trap the zero. We have $G' = 1 + 1/x \geq 1$ and $G(g_N) = \Lambda_N$. So $G(x) - \Lambda_N < -\rho_N$ for $x < g_N - \rho_N$, and $G(x) - \Lambda_N > \rho_N$ for $x > g_N + \rho_N$. By $(*)$, $F(x) < 0$ for $1 \leq x < g_N - \rho_N$ and $F(x) > 0$ for $g_N + \rho_N < x \leq 2\Lambda_N$. Both ranges are nonempty for large N, because $\Lambda_N - \log\Lambda_N \leq g_N \leq \Lambda_N$ (note that $G(\Lambda_N - \log\Lambda_N) \leq \Lambda_N \leq G(\Lambda_N)$). Since F is increasing, its zero x_N lies between the two ranges, so $|x_N - g_N| \leq \rho_N$. Then x_N lies in $[1, 2\Lambda_N]$, and $F(x_N) = 0$ in $(*)$ gives $|G(x_N) - \Lambda_N| \leq \rho_N$.

(b) We solve $g_N = \Lambda_N - \log g_N$ by one substitution. Since $\Lambda_N - \log\Lambda_N \leq g_N \leq \Lambda_N$, the ratio g_N/Λ_N is $1 + O(\log\Lambda_N/\Lambda_N)$, so $\log g_N = \log\Lambda_N + O(\log\Lambda_N/\Lambda_N)$. Also $\Lambda_N = 2\log N - \log 8$ gives $\log\Lambda_N = \log 2 + \log\log N + O(1/\log N)$. So $g_N = 2\log N - \log 8 - \log 2 - \log\log N + O(\log\log N/\log N)$, which is (b) for g_N, and (a) gives it for x_N. □

Table 1 gives crossings on the grid $N = 50 \cdot 2^k$, computed from the recursion for f_N in Lemma 2.2. The limit 2 of $x_N/\log N$ is approached slowly, because of the term $-\log\log N$.

Now we compare with the persistent walk. Doubling (1) without its last two terms and adding $\log 2$ gives $2z_N + \log(2z_N) = \Lambda_N + \log(2\pi)$, so $2z_N - \log(2\pi)$ nearly solves $G(x) = \Lambda_N$. What is left is of order $1/L$.

Theorem 2.8. *Let z_N be the crossing value of T for the persistent walk with N steps, as in Section 1.1. As $N \to \infty$ over even N,*

$$D(N) := x_N - \big(2z_N - \log(2\pi)\big) = \frac{c_*}{\log N} + O\Big(\frac{\log\log N}{(\log N)^2}\Big), \qquad c_* = \tfrac{1}{2}\log(2\pi) - \tfrac{1}{4} = 0.668939\ldots$$

In words, the elephant needs twice as many flips as the persistent walk needs switches, less $\log(2\pi) = 1.837\ldots$, and the difference from this tends to 0 like $c_*/\log N$.

Idea of the proof. Both crossings solve an equation of the form $x + \log x = 2L + \text{constant}$ up to small errors. We put the Markov crossing into the equation $G(x) = \Lambda_N$ of the elephant, find that $2z_N - \log(2\pi)$ misses it by c_*/L, and divide by $G' \approx 1$.

Proof. Write $\ell = \log(2\pi)$ and $Y = 2z_N$. We compute $G(Y - \ell)$ in three steps and then compare it with $G(x_N)$.

Step 1: the size of z_N. By Section 1.1, the error term in (1) is $O(L^{-2})$ and $L/2 \le z_N \le L$. So $\log z_N = \log L + O(1)$ and $\frac{1}{8z_N} = O(1)$, and (1) gives $z_N = L - \frac{1}{2}\log L + O(1)$. Hence $Y = 2L - \log L + O(1)$, and

$$\frac{1}{8z_N} = \frac{1}{8L} + \frac{L - z_N}{8Lz_N} = \frac{1}{8L} + O\Big(\frac{\log L}{L^2}\Big), \qquad \frac{1}{Y} = \frac{1}{2L} + O\Big(\frac{\log L}{L^2}\Big).$$

Step 2: the equation for Y. We double (1) and add $\log 2$. The left side becomes $2z_N + \log z_N + \log 2 = Y + \log Y$. On the right, $2c_0 + \log 2 = \log\frac{\pi}{4}$, and $\frac{1}{4z_N} = \frac{1}{4L} + O(\log L/L^2)$ by Step 1. So $Y + \log Y = 2L + \log\frac{\pi}{4} + \frac{1}{4L} + O(\log L/L^2)$.

Step 3: the value $G(Y - \ell)$. We have $\log(Y - \ell) = \log Y + \log(1 - \ell/Y)$, and $\log(1 - \ell/Y) = -\ell/(2L) + O(\log L/L^2)$ by Step 1. With Step 2 and $\log\frac{\pi}{4} - \ell = -\log 8$ this gives

$$G(Y - \ell) = Y + \log Y - \ell + \log\Big(1 - \frac{\ell}{Y}\Big) = \Lambda_N + \frac{1}{4L} - \frac{\ell}{2L} + O\Big(\frac{\log L}{L^2}\Big) = \Lambda_N - \frac{c_*}{L} + O\Big(\frac{\log L}{L^2}\Big).$$

Step 4: compare with x_N. By Theorem 2.7(a), $G(x_N) = \Lambda_N + O(L^2/N)$. So $G(x_N) - G(Y - \ell) = c_*/L + O(\log L/L^2)$. Both x_N and $Y - \ell = 2z_N - \log(2\pi)$ are $2L + O(\log L)$, so $G'(x) = 1 + 1/x = 1 + O(1/L)$ between them. By the mean value theorem we may divide by G', and we get $x_N - (Y - \ell) = c_*/L + O(\log L/L^2)$. □

We computed z_N from the exact law of the persistent walk [2]. The difference $D(N)$ changes sign between $N = 400$ and $N = 1600$, and $D(N)\log N$ is 0.707 at $N = 51200$ against the limit $c_* = 0.669$ (Table 1). So the asymptotic regime is reached slowly.

2.4 The crossing to second order

Theorem 2.6 gives $\log C_N$ up to an error $O(\varepsilon \log N)$. A lower bound that matches part (b) of that theorem brings the error down to $O(\varepsilon)$.

Proposition 2.9 (the center to second order)**.** *Let $N \geq 64$ be even and $0 < \varepsilon(1+\log N) \leq \frac{1}{100}$. Then*

$$-11\varepsilon - \varepsilon^2(84\log N + 1.05\cdot 10^5) \leq \log C_N - \log\frac{4\varepsilon}{N+2} - 2\varepsilon\log N \leq 5200\,\varepsilon.$$

In particular $\bigl|\log C_N - \log\frac{4\varepsilon}{N+2} - 2\varepsilon\log N\bigr| \leq 5200\,\varepsilon$ *there.*

In words, C_N is $\frac{4\varepsilon}{N+2}N^{2\varepsilon}$ times a factor $1+O(\varepsilon)$.

Idea of the proof. The proof is in Appendix A. The upper bound is Theorem 2.6(b). For the lower bound we compare M after the first flip with Pólya's urn conditioned to end at h, which we call the bridge. Then $V_{t+1}(1)/\pi_t$ is the mean over the bridge of a product of factors $1+O(\varepsilon)$, one for each step (Proposition A.5). By Jensen's inequality the mean of the product is at least the exponential of the mean of the sum of the logarithms. The step at time u adds about $2\varepsilon/u$ to this mean, and the sum over $u \leq N$ is about $2\varepsilon\log N$.

With this bound the proof of Theorem 2.7 gives one more term of the crossing.

Proposition 2.10 (the crossing to second order)**.** *As $N \to \infty$ over even N,*

$$x_N = g_N - \frac{2g_N\log N + g_N^2/2}{N(1+1/g_N)} + O\Bigl(\frac{\log N}{N}\Bigr).$$

So $x_N < g_N$ for large N.

In words, the root g_N overshoots the crossing by an amount of order $\log^2 N/N$. The amount is $6\log^2 N/N$ in the limit, but only about 60 percent of that at $N = 51200$ (Table 1).

Proof. We repeat the proof of Theorem 2.7(a) with one more term on each side. The center gains the term $2\varepsilon\log N$ and the end gains the term $-x^2/(2N)$, and both push F up.

Step 1: both sides. Let F be as in the proof of Theorem 2.7, and let $1 \leq x \leq 2\Lambda_N$ and $\varepsilon = x/N$. For large N Proposition 2.9 applies, and its error $5200\,\varepsilon$ is $O(x/N)$, as is the term $\log(1+2/N)$ in $\log\frac{4\varepsilon}{N+2}$. For the end we use $\log(1-y) = -y - \frac{y^2}{2} + O(y^3)$ for $0 \leq y \leq \frac{1}{2}$ with $y = x/N$. So

$$\log C_N = \log x + \log 4 - 2L + \frac{2xL}{N} + O\Bigl(\frac{x}{N}\Bigr), \qquad \log E_N = -\log 2 - x - \frac{x^2}{2N} + O\Bigl(\frac{x}{N}\Bigr).$$

In the second formula we multiplied the expansion by $N-1$, and the terms left over are $O(x/N)$ because x^2/N is bounded for $x \leq 4L$. Subtracting gives

$$F(x) = G(x) - \Lambda_N + \frac{2xL + x^2/2}{N} + O\Bigl(\frac{x}{N}\Bigr) \qquad (1 \leq x \leq 2\Lambda_N).$$

Step 2: solve for x_N. By Theorem 2.7, x_N lies in this range for large N. So $F(x_N) = 0$ gives $G(x_N) = \Lambda_N - \eta_N$ with $\eta_N = (2x_NL + x_N^2/2)/N + O(L/N)$. Since $G(g_N) = \Lambda_N$, the mean value theorem gives $x_N - g_N = -\eta_N/G'(\xi)$ for some ξ between x_N and g_N. Here $G'(\xi) = 1 + 1/\xi = (1+1/g_N)(1+O(1/N))$, since $|\xi - g_N| = O(L^2/N)$ and $g_N \geq L$ for large N. As $\eta_N = O(L^2/N)$, we get $x_N - g_N = -\eta_N/(1+1/g_N) + O(L^2/N^2)$. Replacing x_N by g_N in the main term of η_N changes it by $O(L^3/N^2)$, again because $|x_N - g_N| = O(L^2/N)$. All these errors are $O(L/N)$, which gives the formula. Its main term is of order L^2/N and is larger than the error for large N, so $x_N < g_N$. □

Proposition 2.9 does not identify the term of order ε. For fixed N and small ε we can find it. By Proposition 2.5(b), C_N is a polynomial in ε, and we write $C_N = c_1\varepsilon + c_2\varepsilon^2 + O(\varepsilon^3)$ as $\varepsilon \to 0$ and $k_2(N) = c_2/c_1$.

Proposition 2.11 (the coefficient of ε^2)**.** *As $N \to \infty$ over even N,*

$$k_2(N) - 2\log N \to 2\gamma - 2 - \log 2 = -1.5387\ldots$$

In words, for fixed large N and small ε the factor $1 + O(\varepsilon)$ in Proposition 2.9 is about $1 - 1.54\,\varepsilon$.

Idea of the proof. The proof is in Appendix A. Differentiating the bridge formula at $\varepsilon = 0$ gives $k_2(N) = \sum_t w_t[A_t - (t-1)]$, where A_t is the mean over the bridge of an explicit sum (Lemma A.8). For fixed t and large N the bridge is close to a walk with fair steps, and the sum can be computed in closed form. It is $2\log N + 2(\gamma - 1 - \log 2) + (t-2)H_{t-1} + o(1)$. The weights w_t tend to $t2^{-t-1}$, and the series over t sums to $2\gamma - 2 - \log 2$.

Remark 2.12 (numerical values and open points). From the Taylor coefficients of C_N, computed in double precision (`verify_elephant_crossing.py`), we get $k_2(N) - 2\log N = -1.585878$, -1.546572 and -1.541005 at $N = 1600$, 12800 and 51200. The formula of Proposition 2.10 without its error term is within 0.07 of x_N already at $N = 400$ (Table 1). We do not know whether, for some fixed k, the bound $\log C_N \le \log(c_1\varepsilon) + \varepsilon k_2(N) + O(\varepsilon^2(\log N)^k)$ holds uniformly in N in the range of Proposition 2.9. We also do not know an explicit N beyond which $g_N > x_N$.

3 The elephant walk near its crossing

We now look at the whole law P_N near the crossing (Figure 2). The picture is that of the introduction. Each flip turns a whole family of steps, and most families are small. So the law has a hump near each end, and beyond the humps it falls like $x/(2m^2)$, where m is the distance from the nearer end. Lemmas 3.1 to 3.4, the results of Section 3.2, Theorem 3.9 and the results of Section 3.6 hold under their own hypotheses. Elsewhere in this section and in Appendices B and C we assume, unless we say otherwise, that

$$N \ge 64, \qquad 0 < \varepsilon \le \tfrac{1}{4}, \qquad x = N\varepsilon \ge 1.$$

Recall that $P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)]$. Some statements also use odd N. There x_N is the value of x at which the two central bins are as likely as the ends (it exists and is unique by Proposition 3.27). For a mode $a^* \ge N/2$ of P_N we put $m^* = N - a^*$, and m_h is the largest mode of f_N.

3.1 Families and the first flip

For $k_0 \ge 1$ let $(M_k^{[k_0]})_{k \ge k_0}$ be the Markov chain with $M_{k_0}^{[k_0]} = 0$ and the transition probabilities $r_k(b)$ of M. It is the count of disagreeing steps for a walk whose first k_0 steps are forced to agree. So $M^{[1]}$ has the law of M. We write $Y_d = M_d^{[1]}$, and $Z_t^{(j)} = M_t^{[j-1]}$ for $j \ge 2$ and $t \ge j-1$. In a random recursive tree on $[N] = \{1, \ldots, N\}$, vertex $k+1$ is joined to a uniform vertex of $[k]$, independently for $k = 1, \ldots, N-1$. We write $\mathcal{T}_v$ for the subtree of v, that is, v and its descendants, and we call it the family of v.

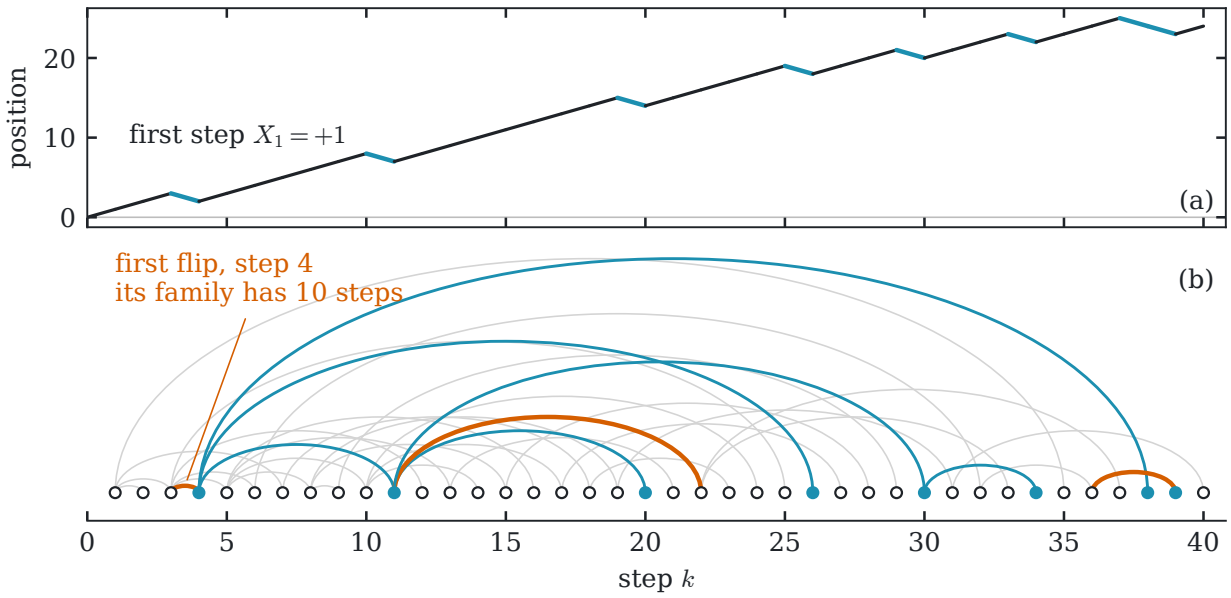


Figure 1: One elephant walk with $N = 40$ steps and $\varepsilon = 0.07$. (a) The position against the step. The steps that disagree with the first step are drawn in blue. (b) The memory tree of the same walk. Step $k+1$ copies the step at the left end of its arc, and the arcs drawn heavier are flips. A step disagrees with the first step exactly when the path from it to step 1 crosses an odd number of flips (Lemma 3.1). The first flip, at step 4, turns its family of ten steps. The flip at step 22, inside that family, turns its own family of three steps back, and the flip at step 39 turns only that step.

Lemma 3.1 (tree and marks)**.** *Let $\mathcal{T}$ be the tree on $[N]$ in which the parent of $k+1$ is U_k. For $2 \le v \le N$ let $\mu_v = 1$ if step v flips the step it copies, and $\mu_v = 0$ otherwise. Then $\mathcal{T}$ is a random recursive tree, the marks μ_v are i.i.d. Bernoulli(ε) and independent of $\mathcal{T}$, and $X_v = X_1(-1)^{\sigma(v)}$, where $\sigma(v)$ is the number of marks on the path from v to the root, the root excluded. So $M_N = \#\{v : \sigma(v) \text{ odd}\}$. If the marks on $2, \dots, k_0$ are set to 0, then the number of odd vertices in $[k]$, $k \ge k_0$, is a copy of the chain $M^{[k_0]}$.*

Proof. The parent U_k of $k+1$ is uniform on $[k]$ and independent of the earlier choices, and the marks are the flip indicators. The path from $k+1$ to the root is $k+1$ followed by the path from U_k, so $\sigma(k+1) = \mu_{k+1} + \sigma(U_k)$ and, by induction on k, $X_{k+1} = X_{U_k}(-1)^{\mu_{k+1}} = X_1(-1)^{\sigma(k+1)}$. Now set $\mu_2 = \cdots = \mu_{k_0} = 0$. If b of the vertices in $[k]$ are odd, $k \ge k_0$, then $k+1$ is odd with conditional probability $(1-\varepsilon)\frac{b}{k} + \varepsilon(1 - \frac{b}{k}) = r_k(b)$, whatever else is known about $[k]$. □

So a single flip at v turns every step of its family (Figure 1). Kürsten's coupling [4] uses bond percolation with retention probability $1 - 2\varepsilon$ instead, which gives the same law of the walk. For $k \ge v$, if $\mathcal{T}_v$ contains s of the vertices in $[k]$, then $k+1$ joins $\mathcal{T}_v$ with probability s/k. This is the urn of Lemma 2.4, so $|\mathcal{T}_v|$ has the law β_v and mean N/v, and on average there are $N/(d(d+1))$ families of size d. If v is the only marked vertex, the odd vertices are those of $\mathcal{T}_v$. Two or more marks have probability $O(\varepsilon^2)$ at fixed N, and no mark gives $M_N = 0$. So at fixed N the part of $f_N(m)$ of first order in ε is exactly $\varepsilon \sum_v \beta_v(m) = x/(m(m+1))$, for $1 \le m \le N-1$ (at $m = h$ this is $c_1\varepsilon$ of Proposition 2.5). This is also proved in Lean (Section 4).

To go beyond first order we cut the tree at the first flip.

Lemma 3.2 (splitting)**.** *Let $\mathcal{T}$ be a random recursive tree on $[N]$, let $2 \le v \le N$ and let $S \subseteq [N]$ with $\min S = v$. Given that $\mathcal{T}_v$ has vertex set S, the restrictions of $\mathcal{T}$ to S and to $[N] \setminus S$, relabeled in increasing order, are independent random recursive trees. They are independent of the parent of v, which is uniform on $[v-1]$.*

Proof. The tree $\mathcal{T}$ is uniform over the $(N-1)!$ increasing trees on $[N]$. Its family $\mathcal{T}_v$ has vertex set S exactly when every $u \in S \setminus \{v\}$ has its parent in S and every $u \notin S$ with $u \ge 2$ has its parent outside S. So, given this event, each element of S other than v picks its parent freely among the smaller elements of S, v picks any element of $[v-1]$, and each element of $[N] \setminus S$ other than 1 picks freely among the smaller elements of $[N] \setminus S$. This is the claim. □

Lemma 3.3 (the first flip)**.** *Let $V = \min\{v \ge 2 : \mu_v = 1\}$, with $V = \infty$ if there is no mark, and let $D = |\mathcal{T}_V|$. Then $\mathbb{P}(V = j,\, D = d) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and for $m \ge 1$*

$$f_N(m) = \sum_{j=2}^{N} \varepsilon(1-\varepsilon)^{j-2} \sum_{d=1}^{N-j+1} \beta_j(d)\, \mathbb{P}\big(d - Y_d + Z^{(j)}_{N-d} = m\big),$$

where Y_d and $Z^{(j)}_{N-d}$ are independent.

Proof. Marks and tree are independent, which gives the law of (V, D). Condition on $V = j$ and on the vertex set S of $\mathcal{T}_j$, with $|S| = d$. The marks on $j+1, \dots, N$ are still i.i.d., and by Lemma 3.2 the trees on S and on $[N] \setminus S$ are independent random recursive trees. The vertices $1, \dots, j-1$ carry no mark, so j is odd, and $u \in S$ is odd exactly when the path from u up to j, with j excluded, has an even number of marks. So by Lemma 3.1 for the tree on S rooted at j, the number of odd vertices in S has the law of $d - Y_d$. Paths from outside S stay outside S, and the tree on $[N] \setminus S$ has the unmarked vertices $1, \dots, j-1$ first. So by Lemma 3.1 its odd count has the law of $Z^{(j)}_{N-d}$. The two counts are independent, and their law depends on S only through d. Sum over j and d, and note that $M_N = 0$ if there is no mark. □

In words, the first flip turns its whole family except the steps of that family that flip back, and the rest of the tree is a walk whose first $j-1$ steps agree with X_1. Only the law of each pair $(Y_d, Z^{(j)}_{N-d})$ enters the lemma. So from now on $(Y_d)_{d \ge 1}$ is a copy of the chain $M^{[1]}$, and $(Z^{(j)}_t)_{t \ge j-1}$ is a copy of $M^{[j-1]}$, independent of Y and of (V, D).

Lemma 3.4 (unimodality given the first step [9, proof of Theorem 1.1])**.** *The function f_N is unimodal.*

This is shown in the proof of Theorem 1.1 of Guérin, Laulin, Raschel and Simon [9], for every N and every memory parameter $1 - \varepsilon$, and it is also proved in Lean (Section 4). So f_N is nondecreasing on $\{0, \dots, m_h\}$ and nonincreasing on $\{m_h, \dots, N\}$. With a fair first step, P_N mixes f_N and its mirror image. When the mode of f_N is far from the center, the mixture can have two humps, one near each end.

3.2 The end and its neighbor

We first say why the end should fall below its neighbor once $x > 2$. Given $X_1 = +1$, the end needs no flip at all. Its neighbor, bin $N-1$, needs one flip that is never copied. A flip at step j is never copied with probability about j/N. Summing $\varepsilon j/N$ over j gives a factor of about $x/2$ in favor of the neighbor.

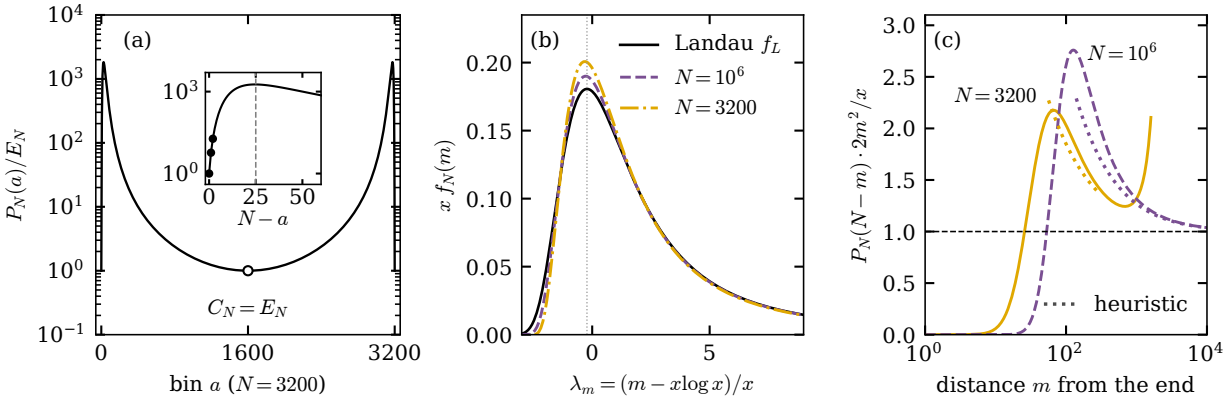


Figure 2: The elephant walk at its crossing. (a) The law P_N for $N = 3200$ at its crossing $x = 11.5468$, divided by the end E_N. The center equals the end, and the law rises from each end to a mode and falls to the center. The inset shows the last 60 bins, with dots at bins N, $N-1$ and $N-2$. The end is below its neighbor, and the modes are 25 steps from each end (dashed). (b) The law of M_N, scaled as $xf_N(m)$, against $\lambda_m = (m - x\log x)/x$, with the Landau density f_L. The dotted line is its mode λ_0. (c) The tail $P_N(N-m)$ divided by $x/(2m^2)$. The dashed line is the limit 1 of Theorem 3.14, and the dotted curves are the heuristic correction $1 + x(2\log m + 2\gamma - 3)/m$ for $2x\log x \le m \le N/x$. The rise of the $N = 3200$ curve near $m = N/2$ is the mirror term $f_N(N-m)$. At $N = 10^6$ we use $x = g_N = 22.4407$.

Theorem 3.5 (endpoint dip)**.** *Let $N \ge 3$ and $0 < \varepsilon \le \frac{1}{2}$, and put $\rho_1 = P_N(N-1)/E_N$. Then*

$$\frac{x}{2(1-\varepsilon)} < \rho_1 \le \frac{x(1+1/N)^2}{2-3\varepsilon} + \frac{\varepsilon}{1-\varepsilon}.$$

Hence the end is strictly below its neighbor if $x \ge 2(1-\varepsilon)$, and strictly above it if the right side is below 1.

In words, when ε is small the neighbor of the end is about $x/2$ times as likely as the end.

Proof. By Lemma 2.2, $\rho_1 = \big(f_N(1) + f_N(N-1)\big)/f_N(0)$ with $f_N(0) = (1-\varepsilon)^{N-1}$. We bound the two terms in turn.

The term $f_N(1)$. On the event $M_N = 1$ the chain moves up exactly once, say from time $j-1$ to time j with $2 \le j \le N$. Before that it stays at 0 for $j-2$ steps, each with probability $1 - r_t(0) = 1 - \varepsilon$. The move up has probability ε. After it the chain stays at 1 at the times $k = j, \dots, N-1$, each with probability $1 - r_k(1) = (1-\varepsilon)(1 - c'/k)$, where $c' = (1-2\varepsilon)/(1-\varepsilon) \in [0,1]$. Multiplying and summing over j gives

$$f_N(1) = \varepsilon(1-\varepsilon)^{N-2} \sum_{j=2}^{N} P_j, \qquad P_j = \prod_{k=j}^{N-1} \Big(1 - \frac{c'}{k}\Big).$$

For the lower bound we use $c' \le 1$. Then $P_j \ge \prod_{k=j}^{N-1}(1 - 1/k) = (j-1)/(N-1)$, and these lower bounds sum to $N/2$ over $2 \le j \le N$. So $f_N(1)/f_N(0) \ge x/(2(1-\varepsilon))$. For the upper bound, $1 - c'/k \le e^{-c'/k}$ and $\sum_{k=j}^{N-1} 1/k \ge \log(N/j)$ give $P_j \le (j/N)^{c'}$. As $(t/N)^{c'}$ is nondecreasing in t, we can compare the sum with an integral, namely

$$\sum_{j=2}^{N} P_j \le \int_2^{N+1} \Big(\frac{t}{N}\Big)^{c'} dt \le \frac{N(1+1/N)^{c'+1}}{c'+1} \le \frac{N(1+1/N)^2(1-\varepsilon)}{2-3\varepsilon},$$

using $c' \le 1$ and $1/(c'+1) = (1-\varepsilon)/(2-3\varepsilon)$. So $f_N(1)/f_N(0) \le x(1+1/N)^2/(2-3\varepsilon)$.

The term $f_N(N-1)$. The event $M_N = N-1$ needs an up move at every time. The first has probability $r_1(0) = \varepsilon$, and the one at time $k \ge 2$ has probability $r_k(k-1) = 1-\varepsilon-(1-2\varepsilon)/k \in [\frac{1}{2}, 1-\varepsilon]$. So $0 < f_N(N-1) \le \varepsilon(1-\varepsilon)^{N-2}$. This makes the lower bound strict and gives the term $\varepsilon/(1-\varepsilon)$ in the upper bound. □

Corollary 3.6 (the dip at the crossing)**.** *Let $N \ge 4$ be even and suppose that $x_N \ge 2(1-\varepsilon_N)$. Then at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal.*

Proof. By Proposition 2.3, $\varepsilon_N < \frac{1}{2}$, so Theorem 3.5 gives $P_N(N-1) > E_N = C_N$. Suppose that P_N is nondecreasing on $\{0, \dots, a_0\}$ and nonincreasing on $\{a_0, \dots, N\}$. It is symmetric about h, so we may replace a_0 by $N - a_0$ and assume $a_0 \le h$. Then $P_N(a) \le C_N$ for every $a \ge h$, including $a = N-1$, which is a contradiction. □

The hypothesis $x_N \ge 2(1-\varepsilon_N)$ says that $\varepsilon_N \ge 2/(N+2)$. By Proposition 2.3 it holds exactly when $C_N \le E_N$ at $\varepsilon = 2/(N+2)$. We now show that this is so for every even $N \ge 8$. For $N \ge 1688$ we use Theorem 2.6(b). For smaller N that theorem does not apply, and we compute C_N/E_N in floating-point arithmetic and bound the rounding error. At $\varepsilon = 2/(N+2)$ the transition probabilities of M are ratios of integers, namely

$$1 - r_t(b) = \frac{N(t-b)+2b}{(N+2)t}, \qquad r_t(b) = \frac{2(t-b)+Nb}{(N+2)t}. \tag{2}$$

By Lemma 2.2, $C_N/E_N = 2f_N(h)/f_N(0)$. Write $f_t(b) = \mathbb{P}_+(M_t = b)$ for the law of M_t at this ε, for $1 \le t \le N$. The law of M_{t+1} follows from the law of M_t by

$$f_{t+1}(b) = f_t(b)\big(1 - r_t(b)\big) + f_t(b-1)\, r_t(b-1), \qquad f_1(0) = 1. \tag{3}$$

Here $f_t(b) > 0$ exactly for $0 \le b \le t-1$. So the first term is absent for $b = t$ and the second for $b = 0$. The values $f_{t+1}(b)$ with $b \le h$ only need the values $f_t(b)$ with $b \le h$. Every number in (3) is positive, and nothing is subtracted. This is what makes the rounding error easy to bound.

Lemma 3.7 (rounding error)**.** *Let $N \ge 2$ be even with $N(N+2) < 2^{53}$, let $\varepsilon = 2/(N+2)$, and put $u = 2^{-53}$. Compute numbers $\hat{f}_t(b)$ for $1 \le t \le N$ and $0 \le b \le \min(t-1, h)$ in IEEE 754 double precision with rounding to nearest, as follows. Put $\hat{f}_1(0) = 1$. In (3) each weight is one division of its numerator in (2) by its denominator $(N+2)t$, each term on the right is one multiplication of a computed value $\hat{f}_t$ by a computed weight, and if both terms are present they are added by one addition. Finally let $\hat{R}$ be twice the computed quotient of $\hat{f}_N(h)$ by $\hat{f}_N(0)$. Suppose that no computed weight, product or sum in the recursion is below 2^{-1000}. Then*

$$(1-u)^{6N-5} \le \frac{C_N/E_N}{\hat{R}} \le (1-u)^{-(6N-5)}.$$

For $N < 1688$ the right side is at most $1 + 3 \cdot 10^{-12}$.

In words, for $N < 1688$ the computed ratio is correct to about 11 digits, because each of the $N-1$ steps of the recursion rounds each term three times and all terms are positive.

Proof. We use the standard model of floating-point arithmetic [17, Section 2.2]. If a and b are floating-point numbers and $\circ$ is addition, multiplication or division, then the computed value of $a \circ b$ is $(a \circ b)(1+\delta)$ with $|\delta| \le u$, as long as the result does not overflow or underflow. Underflow

means that the exact result is nonzero and below the smallest normal number 2^{-1022}, and then the rounded result is at most 2^{-1022}. So a computed result that is at least 2^{-1000} did not underflow, and by hypothesis this covers every operation in the recursion. The integers in (2) are positive and at most $N(N+2) < 2^{53}$, so they are floating-point numbers, and each computed weight is the true weight times some $1+\delta$.

We claim that $\hat{f}_t(b) = f_t(b)(1+\vartheta)$ with $(1-u)^{3(t-1)} \le 1+\vartheta \le (1+u)^{3(t-1)}$. We use induction on t, and the case $t=1$ is clear. In the step from t to $t+1$, each term on the right of (3) is computed as $\hat{f}_t(b')$ times the true weight times $(1+\delta_1)(1+\delta_2)$, and the sum of two terms gets one more factor $1+\delta_3$. All numbers are positive. So $\hat{f}_{t+1}(b)$ lies between $(1-u)^3$ and $(1+u)^3$ times the right side of (3) with $\hat{f}_t$ in place of f_t, and the claim follows. Nothing overflows on the way, because the true weights and the $f_t(b)$ are at most 1 and $(1+u)^{3N} < 2$.

In the last division both numbers lie between 2^{-1000} and 2. So the quotient lies between 2^{-1001} and 2^{1001}, it neither underflows nor overflows, and the division adds one more factor $1+\delta$. Doubling is exact. Hence $(C_N/E_N)/\hat{R}$ lies between $(1-u)^{3(N-1)}(1+u)^{-3(N-1)-1}$ and $(1+u)^{3(N-1)}(1-u)^{-3(N-1)-1}$. Since $1+u \le (1-u)^{-1}$, this gives both bounds with the exponent $6(N-1)+1 = 6N-5$. For $N \le 1686$ put $k = 6N-5 \le 10111$. Then $(1-u)^{-k} \le (1-ku)^{-1} \le 1+2ku \le 1+3\cdot 10^{-12}$. □

Proposition 3.8 (the hypothesis holds from $N=8$ on)**.** *For every even $N \ge 8$ we have $C_N < E_N$ at $\varepsilon = 2/(N+2)$, and hence $x_N > 2(1-\varepsilon_N)$. So for every even $N \ge 8$, at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal. For $N = 2$, 4 and 6 the hypothesis of Corollary 3.6 fails.*

In words, for every even N from 8 on the end is below its neighbor at the crossing, so the law has a hump on each side of the center.

Proof. The range $N \ge 1688$. Take $\varepsilon = 2/N$. Then $\varepsilon(1+\log N) \le \frac{1}{100}$, so Theorem 2.6(b) gives $C_N \le \frac{8}{N(N+2)} N^{4/N} e^{10400/N}$. The right side decreases in N and is below $1.4\cdot 10^{-3}$ at $N = 1688$. Since $\log(1-y) \ge -y-y^2$ for $0 \le y \le \frac{1}{2}$, we also have $E_N = \frac{1}{2}(1-\frac{2}{N})^{N-1} \ge \frac{1}{2}e^{-2-2/N} > 0.06$. So $C_N < E_N$ at $\varepsilon = 2/N$, and Proposition 2.3 gives $\varepsilon_N > 2/N$. Hence $C_N < E_N$ at $\varepsilon = 2/(N+2)$ as well.

The range $8 \le N < 1688$. The script `certify_elephant.py` runs the computation of Lemma 3.7 for each of the 840 even N in this range, with exactly the operations listed there (elementwise operations on arrays of doubles, with no fused operation, and for a sample of N the same bits come out of plain scalar arithmetic). The smallest weight, product or sum that occurs is above 10^{-9}, so the lemma applies. The largest computed ratio is $\hat{R} = 0.947307\ldots$, at $N = 8$, and the next largest is $0.729947\ldots$, at $N = 10$. By Lemma 3.7, $C_N/E_N \le 0.9474$ for every even $8 \le N < 1688$.

The three smallest N. At $\varepsilon = 2/(N+2)$ the ratio C_N/E_N is exactly 2, $\frac{5}{3}$ and $\frac{508}{405}$ for $N = 2$, 4 and 6, so $\varepsilon_N < 2/(N+2)$ there. The last claim of the proposition about $N \ge 8$ is Corollary 3.6. □

As an independent check, the same script runs (3) in exact integer arithmetic for 20 values of N between 8 and 1686. In each case $C_N < E_N$ exactly, and the double-precision ratio differs from the exact one by a relative error below 10^{-13}. At $N = 8$ the exact ratio is $\frac{67903}{71680}$. By Lemma 3.4, P_N is always a mixture of two unimodal laws (namely f_N and its mirror image), as is the limit law in [9, eq. (1)].

3.3 The bulk

The steps that disagree with X_1 lie in the flipped families. Let CP_N be the law of $\sum_{d=1}^{N-1} d\mathcal{N}_d$, where the $\mathcal{N}_d$ are independent Poisson variables with means $x/(d(d+1))$. On average $x/(d(d+1))$ flipped vertices have a family of size d. If no flipped step has a flipped descendant, which is likely when $x^2 \log N/N$ is small, then M_N is the total size of the flipped families, and this is close to the compound Poisson sum.

The law CP_N is known. Without the truncation at $d = N-1$ its generating function is $(1-s)^{x(1-s)/s}$, which is the Lea–Coulson form of the Luria–Delbrück distribution with parameter x [12, Eq. (14)]. Its limit after centering at $x \log x$ and dividing by x is known as well (see [12, Eq. (19)] and [13]), and the same stable law appears for the root cluster of a random recursive tree [11]. So the limit law in the two theorems below is not new. What they add is the comparison with the elephant walk in sup norm, with an explicit error in Theorem 3.9 and an absolute constant in Theorem 3.10. Numerical values of f_L and of its mode are standard (see Kölbig and Schorr [20]). What Lemma 3.11 adds is a certified enclosure.

Theorem 3.9 (compound Poisson bulk)**.** *For $N \geq 2$ and $0 < \varepsilon \leq \frac{1}{4}$,*

$$\sup_{m \geq 0} \bigl| f_N(m) - \mathrm{CP}_N(m) \bigr| \leq \delta_B := \frac{x^2 \log N}{N} + \frac{4x^2}{N} e^{4x^2/N} + \frac{4x^2(H_N^2 + H_N)}{N}.$$

In words, bin by bin the law of the number of disagreeing steps is within δ_B of the compound Poisson law, and $\delta_B = O(x^2 \log^2 N/N)$ when $x^2 \leq N$.

Idea of the proof. The proof is in Appendix C. Let Z_+ be the total size of the flipped families. Then $M_N = Z_+$ unless some flipped step has a flipped descendant, and this has probability at most $x^2 \log N/N$. Given the tree, Z_+ is a sum of independent terms, so its characteristic function is a product, and the product is close to $e^{\varepsilon F}$ with $F = \sum_v (e^{i\theta|\mathcal{T}_v|} - 1)$. The mean of εF is exactly the logarithm of the characteristic function of CP_N. So it remains to show that F is close to its mean. For this we use the Efron–Stein inequality [15]. Changing the parent of one vertex changes the family sizes only along two paths to the root, and these paths have length of order $\log N$.

The families of size up to about x add up to nearly $x \log x$, and the few families of size of order x give fluctuations of order x. So we center M_N at $x \log x$ and divide by x. Put $\lambda_m = (m - x \log x)/x$, and let

$$f_L(\lambda) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\lambda t} e^{B(t)}\, dt, \qquad B(t) = -it \log|t| - \frac{\pi}{2}|t|.$$

This is the Landau density, a stable density with index one. We show in Appendix C that e^B is the characteristic function of a probability law of class L, so that f_L is a unimodal probability density by [14].

Theorem 3.10 (Landau limit)**.** *There is an absolute constant C such that the following hold under the standing assumptions of this section.*

(a) For every integer $m \geq 0$, $\bigl| x\,\mathrm{CP}_N(m) - f_L(\lambda_m) \bigr| \leq C\bigl(\frac{1+\log x}{x} + \frac{x}{N}\bigr)$.

(b) For every integer $m \geq 0$, $\bigl| x f_N(m) - f_L(\lambda_m) \bigr| \leq C\bigl(\frac{1+\log x}{x} + \frac{x}{N}\bigr) + x\delta_B$.

(c) If $x \to \infty$ and $x^3 \log^2 N/N \to 0$, then $(M_N - x \log x)/x$ converges in law to the law with density f_L. Both conditions hold at the crossing.

In words, near the crossing the number of disagreeing steps is about $x \log x$, with fluctuations of order x that follow the Landau law, bin by bin.

Idea of the proof. The proof is in Appendix C. If we allow every jump size $d \geq 1$, the characteristic function of the compound Poisson law is $e^{x\chi(e^{i\theta})}$ with $\chi(s) = (1-s)\log(1-s)/s$, and the jumps $d \geq N$ change it little. For small θ we have $\chi(e^{i\theta}) = -i\theta\log(-i\theta) + O(\theta^2|\log|\theta||)$. With $\theta = t/x$ the exponent becomes $B(t) + it\log x$ plus an error of order $t^2\log x/x$. The term $it\log x$ is the centering at $x\log x$. Both characteristic functions decay like $e^{-c|t|}$, so Fourier inversion turns this into a bound for each bin. Part (b) then follows from Theorem 3.9.

To locate the modes we need to know that f_L is curved at its maximum.

Lemma 3.11 (the mode of the Landau density)**.** *(a) The density f_L has exactly one mode λ_0, and $-0.22278299 < \lambda_0 < -0.22278297$.*

(b) $f_L'(\lambda_0) = 0$, and $-0.0792 \leq f_L''(\lambda) \leq -0.0786$ for $-0.2229 \leq \lambda \leq -0.2227$.

(c) For every real λ, $f_L(\lambda_0) - f_L(\lambda) \geq \kappa_L \min\{1, (\lambda-\lambda_0)^2\}$ with $\kappa_L = 2.6 \cdot 10^{-10}$.

The proof is in Appendix C. It is computer-assisted. We turn the Fourier integral into the real integral $f_L(\lambda) = \frac{1}{\pi}\int_0^\infty e^{-s\log s - \lambda s}\sin(\pi s)\,ds$. Then we enclose f_L' at four points (namely -0.2229, -0.2227, -0.22278299 and -0.22278297) and f_L'' on the interval $[-0.2229, -0.2227]$. We use ball arithmetic at 128 bits (python-flint 0.9.0, which wraps Arb) and the rigorous integration algorithm of [18] on $[2^{-20}, 30]$, with explicit bounds for the rest (`certify_elephant.py`). The run takes under a second. Part (c) follows from (b) and the unimodality of f_L.

Corollary 3.12 (where the modes are)**.** *Let $N \to \infty$ over even N. At the crossing, with $x = x_N$, every mode $a^* \geq N/2$ of P_N has*

$$m^* = x\log x + \lambda_0 x + O\big(\sqrt{x\log x}\big),$$

and so has the largest mode m_h of f_N.

In words, each hump of the law at the crossing sits at distance about $x\log x - 0.2228\,x$ from its end. The proof is in Appendix C. It compares f_N at a mode with f_N at the integer nearest to $x\log x + \lambda_0 x$, and uses Theorem 3.10(b) and Lemma 3.11(c).

Remark 3.13 (how good the bulk approximation is). The bounds of this subsection are far from sharp at the sizes we can compute. At $N = 10^6$ and $x = 22.4407$ the bound δ_B is still about 0.46, but there the computed law has $|f_N(m)/\mathrm{CP}_N(m) - 1| \leq 3.6 \cdot 10^{-4}$ for $1 \leq m \leq 10^4$. At every tested crossing m^* equals the mode of CP_N. The error term in Corollary 3.12 is not effective. Its proof needs the bound of Theorem 3.10(b) to be below $\kappa_L/2$, where κ_L is the constant of Lemma 3.11(c), and numerically the best κ_L is about 0.026. These numbers are computed in double precision and are not used in any proof.

3.4 The tail

This subsection shows that the law falls like $x/(2m^2)$ beyond the humps. We use it for the shape at the center.

Theorem 3.14 (the tail)**.** *Let N be even, and let $x = N\varepsilon \geq 1$ depend on N with $x\log^2 N/N \to 0$. Let $\omega_N \to \infty$ with $\omega_N x\log^2(x+1) = o(N)$, and let $\delta_N' \to 0$. Then*

$$f_N(m) = \frac{x}{m(m+1)}\big(1+o(1)\big) \qquad \text{and} \qquad P_N(N-m) = \frac{x}{2m^2}\big(1+o(1)\big),$$

uniformly for $\omega_N x\log^2(x+1) \leq m \leq \delta_N' N$.

In words, for m well beyond $x\log^2 x$ and well below N, the chance of landing at distance m from an end is about half the expected number of flipped families with m steps, namely $x/(2m^2)$. The half is there because the first step must point to that end.

Idea of the proof. The proof is in Appendix B. For the lower bound we use the first flipped family, which has size m with probability about $x/(m(m+1))$ (Lemma B.4), while the other flips change the count only by a relative amount of order $x\log m/m$. For the upper bound we show $\mathbb{P}(M_N \ge m) \le \frac{x}{m}(1+O(x\log^2 m/m))$, and the unimodality of f_N turns this into a local bound. The mirror term $f_N(N-m)$ is at most $C_N = O(x/N^2)$, which gives the constant $\frac{1}{2}$.

At $N = 10^6$, $x = 22.4407$ and $m = 10^4$ the computed ratio $P_N(N-m)/(x/(2m^2))$ is 1.03778. At moderate m the law CP_N suggests $f_N(m)\,m(m+1)/x \approx 1 + x(2\log m + 2\gamma - 3)/m$, and this factor is still about 2 near $m = 70$ when $N = 3200$ (Figure 2(c)).

3.5 The center

Theorem 3.15 (shape at the crossing)**.** *There are absolute constants C_{w} and N_0 with the following property. Let $N \ge N_0$ be even with $x_N \in [2, N^{1/4}]$. Then at the crossing $P_N(a) > C_N$ for every $1 \le a \le N-1$ with $|a-h| \ge C_{\mathrm{w}}(Nx_N\log N)^{1/2}$.*

By Theorem 2.7, $x_N \in [2, N^{1/4}]$ for large N, so at the crossing the window has width of order $N^{1/2}\log N$. In words, the middle bin is lower than every interior bin except possibly those within about $N^{1/2}\log N$ of it.

Idea of the proof. Up to the mode of f_N the law is above $f_N(1)/2$, which beats $C_N = E_N$ by Theorem 3.5. Farther in, it is close to $x/(2m^2)$, far above the center $4x/N^2$. Near the middle, the main term $x\Pi(m)$ below exceeds its value at h by a factor of about $1 + 12t^2/N^2$ at distance t from h, while we know C_N and the law only up to relative errors of order $x\log N/N$. The two meet at t of order $(Nx\log N)^{1/2}$.

Proof. Write $x = x_N$ and $\varepsilon = \varepsilon_N$, so that $C_N = E_N$. By symmetry we take $a = N-m$ with $1 \le m \le h$, and put $t = h - m$. We compare $P_N(N-m)$ with C_N in three ranges of m. In the first two a crude lower bound is enough. In the third, near the center, we need the comparison with $x\Pi(m)$ and a matching upper bound for C_N. Since $2 \le x \le N^{1/4}$, every error term below is bounded by a function of N alone that tends to 0, and we take N large.

We start with the upper bound for C_N. Put $\Pi(m) = \frac{1}{2}\big[\frac{1}{m(m+1)} + \frac{1}{(N-m)(N-m+1)}\big]$, so that $x\Pi(h) = \frac{4\varepsilon}{N+2}$. As $\varepsilon(1+\log N) \to 0$, Theorem 2.6(b) gives $C_N \le x\Pi(h)(1+\eta)$ with $\eta = N^{2\varepsilon}e^{5200\varepsilon} - 1$. Put $y = \varepsilon(2\log N + 5200)$, so that $\eta = e^y - 1$. Then $y \le \frac{1}{2}$ for large N, and $e^y - 1 \le 2y$ for $0 \le y \le \frac{1}{2}$, so $\eta \le 2y \le 10^4\varepsilon\log N$.

Step 1: $m \le m_h$. Then $f_N(m) \ge f_N(1)$ by Lemma 3.4. The proof of Theorem 3.5 gives $f_N(1) \ge \frac{x}{2(1-\varepsilon)}f_N(0)$, and $f_N(0) = 2E_N = 2C_N$. So $P_N(N-m) \ge \frac{1}{2}f_N(m) \ge \frac{x}{2(1-\varepsilon)}C_N > C_N$, since $x \ge 2$.

Step 2: $m_h < m \le N/8$. Then $f_N(m) \ge f_N(k)$ with $k = \lfloor N/8 \rfloor$, by Lemma 3.4. Proposition B.5(a) at k gives $f_N(k) \ge \frac{x}{k(k+1)}(1 - O(\frac{x\log N}{N}))$. So $P_N(N-m) \ge \frac{32x}{N(N+8)}(1-o(1))$, which exceeds $\frac{4x}{N(N+2)}(1+\eta) \ge C_N$.

Step 3: $N/8 < m \le h$. Then m and $N-m$ are at least $N/8$, and Proposition B.5 at m and at $N-m$ gives $P_N(N-m) \ge x\Pi(m)(1-\bar{e})$ with $\bar{e} = O(x\log N/N)$. It remains to compare $\Pi(m)$ with $\Pi(h)$. We measure the distance from the center by $s = 2t/\bar{N}$ with $\bar{N} = N+1$, so that

$m+\frac{1}{2}=\frac{\bar N}{2}(1-s)$ and $N-m+\frac{1}{2}=\frac{\bar N}{2}(1+s)$. Since $\frac{1}{n(n+1)}\ge(n+\frac{1}{2})^{-2}$ for every $n\ge1$, and $\Pi(h)=\frac{1}{h(h+1)}=(\frac{\bar N}{2})^{-2}(1-\bar N^{-2})^{-1}$, we get

$$\frac{\Pi(m)}{\Pi(h)}\ge\tfrac{1}{2}\big[(1-s)^{-2}+(1+s)^{-2}\big](1-\bar N^{-2})\ge(1+3s^2)(1-\bar N^{-2}).$$

The last step holds because $\frac{1}{2}[(1-s)^{-2}+(1+s)^{-2}]=\sum_{i\ge0}(2i+1)s^{2i}$, and we keep the first two terms of this series. Put $b=\bar e+\bar N^{-2}$, so that $b\le\frac{1}{2}$ and $(1-\bar e)(1-\bar N^{-2})\ge1-b$. Then $P_N(N-m)\ge x\Pi(h)(1+3s^2)(1-b)\ge x\Pi(h)(1+\frac{3}{2}s^2-b)$.

Step 4: the window. The bound of Step 3 exceeds $x\Pi(h)(1+\eta)\ge C_N$ as soon as $\frac{3}{2}s^2>\eta+b$. Now $\eta+b\le A_2x\log N/N$ for an absolute constant A_2, since $x\ge2$, and $s\ge t/N$. So the condition holds when $t\ge C_{\mathrm w}(Nx\log N)^{1/2}$ with $C_{\mathrm w}^2>2A_2/3$. □

We did not compute $C_{\mathrm w}$ or N_0. Inside the window the proof gives nothing, and closing it needs second differences of the law. We study them in Section 3.6, and Corollary 3.25 closes the window for large even N.

Conjecture 3.16 (convexity between the humps)**.** *For every $N\ge10$, $x=x_N$ and every mode $a^*\ge N/2$ of P_N with $m^*=N-a^*$, the second difference $P_N(a-1)-2P_N(a)+P_N(a+1)$ is positive for $2m^*+1\le a\le N-2m^*-1$.*

With the symmetry of the law, the conjecture makes P_N strictly convex on $[2m^*,N-2m^*]$, so h is its unique minimizer there (the two central bins for odd N). The next corollary shows that $m^*=o(N)$ at the crossing, so together with Theorem 3.15 the conjecture would make the center the unique interior minimum for large even N. Section 3.6 proves the conjecture for all large N, even (Theorem 3.24) or odd (Proposition 3.27), and, with a certified computation, for $10\le N\le2000$ (Proposition 3.26). For every even $N\ge8$ and every odd $N\ge1689$ it reduces the conjecture to one inequality (Corollary 3.23 and Proposition 3.27(c)).

Corollary 3.17 (the modes)**.** *For large even N the modes a^* of P_N at the crossing satisfy* $\min(a^*,N-a^*)=O(x_N\log^2x_N)$.

Proof. We show that $f_N(m^*)$ is not much below the maximum of f_N, and then that f_N cannot be that large far to the right of its mode. Write $x=x_N$ and $\varepsilon=\varepsilon_N$. By Theorem 2.7, $x\sim2\log N$, so for large N the hypotheses of Theorem 2.6(b) and Lemma B.7 hold. Let $\ell=\lceil Ax\log(x+2)\rceil$ be as in the proof of Lemma B.7, where it is shown that $f_N(\ell)\ge\frac{x}{2\ell(\ell+1)}$.

Step 1: $f_N(m^*)$ is large. The law is symmetric about h, so $m^*=\min(a^*,N-a^*)$ maximizes $g(m)=f_N(m)+f_N(N-m)$ over $0\le m\le h$. Lemma B.7 gives $m_h\le h$ for large N. So for $m\le h$ we have $N-m\ge h\ge m_h$, and Lemma 3.4 and Theorem 2.6(b) give $f_N(N-m)\le f_N(h)=C_N=O(x/N^2)$. Hence

$$f_N(m^*)\ge g(m^*)-C_N\ge g(m_h)-C_N\ge f_N(m_h)-C_N\ge f_N(\ell)-C_N\ge\frac{x}{4\ell(\ell+1)}$$

for large N, because $\ell=O(\log^2N)$ and so C_N is far below x/ℓ^2.

Step 2: m^* is not far beyond ℓ. Put $m'=\max(m_h,\ell)$ and suppose that $m^*>m'$. To the right of m_h the function f_N is nonincreasing, so Lemma B.1 (with $a=m'$ and $b=m^*$) and Lemma B.2 give

$$(m^*-m'+1)\,f_N(m^*)\le\mathbb P(M_N\ge\ell)\le\frac{xH_\ell}{\ell}.$$

With Step 1 this gives $m^*-m'+1\le4(\ell+1)H_\ell$. Hence $m^*\le\max(m_h,\ell)+4(\ell+1)H_\ell$ in every case. Finally Lemma B.7 gives $m_h=O(x\log^2x)$, and $\ell=O(x\log x)$ and $H_\ell=O(\log x)$. □

In double precision the conjecture also holds for $N = 3200$, 6400, 12800 and 25600. It leaves out the two endpoints $a = 2m^*$ and $a = N - 2m^*$, because there the second difference is not positive for some $N \le 51$. Heuristically, between the humps $P_N(a)$ is close to $\frac{x}{2}\big[\frac{1}{a(a+1)} + \frac{1}{(N-a)(N-a+1)}\big]$, which is strictly convex in a. We leave open the tail between $x \log x$ and $x \log^2 x$, effective constants in Theorems 3.14 and 3.15, and the conjecture for the N between 2000 and the constants N_1 and N_1' of Theorem 3.24 and Proposition 3.27.

3.6 Convexity between the humps

This subsection proves Conjecture 3.16 for all large N, and for every even $N \ge 8$ and every odd $N \ge 1689$ it reduces the conjecture to one inequality. Throughout this subsection $0 < \varepsilon < \frac{1}{2}$ and $c = 1 - 2\varepsilon$, so that $r_t(b) = \varepsilon + cb/t$. For $t \ge 1$ and every integer a we write $P_t(a) = \mathbb{P}(A_t = a)$, which is 0 unless $0 \le a \le t$, and

$$D_t(a) = P_t(a+1) - 2P_t(a) + P_t(a-1).$$

We also put $s_t = D_t(1) = P_t(0) - 2P_t(1) + P_t(2)$ for $t \ge 2$.

Lemma 3.18 (recurrence for the second difference)**.** *For every $t \ge 1$ and every integer a,*

$$P_{t+1}(a) = \big(1 - r_t(a)\big)P_t(a) + r_t(a-1)\,P_t(a-1), \tag{4}$$
$$D_{t+1}(a) = \big(1 - r_t(a+1)\big)D_t(a) + r_t(a-2)\,D_t(a-1). \tag{5}$$

Also $P_t(a) = P_t(t-a)$ and $D_t(a) = D_t(t-a)$. The same two recurrences hold with $f_t(m)$ in place of $P_t(a)$.

Proof. Given $A_t = a$, the copied step is $+1$ with probability a/t, so step $t+1$ is $+1$ with probability $(1-\varepsilon)\frac{a}{t} + \varepsilon(1 - \frac{a}{t}) = r_t(a)$. This gives (4) for $0 \le a \le t+1$, where at $a = 0$ and at $a = t+1$ the formula has one extra term, namely $r_t(-1)P_t(-1)$ or $(1 - r_t(t+1))P_t(t+1)$, and this term is 0 because $P_t(-1) = P_t(t+1) = 0$. For other a every term vanishes. The symmetry of P_t is Lemma 2.2, and it gives the symmetry of D_t. The chain M of Lemma 2.2 has the same transition probabilities, so (4) holds for f_t as well.

For (5) put $\varsigma = c/t$, so that $r_t(b+1) = r_t(b) + \varsigma$. We insert (4) at $a+1$, a and $a-1$ into the definition of $D_{t+1}(a)$ and collect the coefficients of $P_t(a+1)$, $P_t(a)$, $P_t(a-1)$ and $P_t(a-2)$. They are

$$1 - r_t(a+1), \qquad 3r_t(a) - 2, \qquad 1 - 3r_t(a-1), \qquad r_t(a-2).$$

The right side of (5) has the coefficients $1-r_t(a+1)$, $-2+2r_t(a+1)+r_t(a-2)$, $1-r_t(a+1)-2r_t(a-2)$ and $r_t(a-2)$. The second ones agree because $2r_t(a+1) + r_t(a-2) = 3r_t(a)$, and the third ones agree because $r_t(a+1) + 2r_t(a-2) = 3r_t(a-1)$. □

The recurrence (5), and its analogue for differences of any order, is used by Trinas and Valle [10, (4.10)] to bound the law and its differences at a fixed memory parameter. We use it for signs. The coefficients in (5) are positive when $2 \le a \le t-1$. So the second difference moves like a mass that is partly killed, and its sign can change only through what enters at the two edges. The next lemma is the elementary form of the variation diminishing property that we need.

For a finite list z of real numbers let $\mathrm{sp}(z)$ be its sign pattern. We delete the zeros of z, replace each entry by its sign, and merge equal neighbors. So $\mathrm{sp}(z)$ is a list of alternating signs. For two such lists we write $p \preceq q$ if p is a subsequence of q. We put $\Pi_5 = (+,-,+,-,+)$ and $\Pi_3 = (+,-,+)$.

Lemma 3.19 (one step). *Let $\zeta = (\zeta_1, \dots, \zeta_n)$ be real, and let $\eta_i = A_i\zeta_i + B_i\zeta_{i+1}$ for $1 \le i \le n-1$ with $A_i \ge 0$ and $B_i \ge 0$. Let $w = (\zeta_1, \eta_1, \zeta_2, \eta_2, \dots, \eta_{n-1}, \zeta_n)$. Then $\operatorname{sp}(w) = \operatorname{sp}(\zeta)$ and $\operatorname{sp}(\eta) \preceq \operatorname{sp}(\zeta)$.*

Proof. We insert the numbers η_i into ζ one at a time. If $\eta_i = 0$, the sign pattern does not change. If $\eta_i > 0$, then $\zeta_i > 0$ or $\zeta_{i+1} > 0$, so η_i is placed next to an entry of its own sign, and the pattern does not change. The case $\eta_i < 0$ is the same. This proves the first claim. Deleting entries from a list replaces its sign pattern by a subsequence, and η is obtained from w by deleting the entries ζ_i. This proves the second claim. □

Lemma 3.20 (the sequence next to the end). *For every $T \ge 2$ we have $\operatorname{sp}(s_2, s_3, \dots, s_T) \preceq \Pi_3$. Also, s_3 and s_4 do not both vanish.*

Proof. Put $\rho_t = P_t(1)/P_t(0)$. By (4), $P_{t+1}(0) = (1-\varepsilon)P_t(0)$ and $P_{t+1}(1) = (1-\varepsilon-\frac{c}{t})P_t(1)+\varepsilon P_t(0)$. So

$$\rho_{t+1} - \rho_t = \frac{1}{1-\varepsilon}\Big(\varepsilon - \frac{c\rho_t}{t}\Big), \qquad \rho_2 = \frac{2\varepsilon}{1-\varepsilon}.$$

We claim that $\rho_t < \varepsilon t/c$ for $t \ge 2$. For $t = 2$ this holds because $c < 1-\varepsilon$. If it holds for t, then $0 < \rho_{t+1} - \rho_t \le \frac{\varepsilon}{1-\varepsilon} < \frac{\varepsilon}{c}$, and it holds for $t+1$. So ρ_t is strictly increasing for $t \ge 2$, and $\rho_t < 1$ whenever $t \le c/\varepsilon$.

Now (5) with $a = 1$, and $D_t(0) = P_t(1) - 2P_t(0)$, give for $t \ge 2$

$$s_{t+1} = \big(1 - r_t(2)\big)s_t + \sigma_t, \qquad \sigma_t = \Big(\varepsilon - \frac{c}{t}\Big)(\rho_t - 2)\,P_t(0).$$

Here $1 - r_t(2) \ge 1 - \varepsilon - c = \varepsilon > 0$. If $t < c/\varepsilon$, both factors of σ_t are negative and $\sigma_t > 0$. If $t > c/\varepsilon$, the first factor is positive, and the second is first negative and then positive, because ρ_t increases. So $\operatorname{sp}(\sigma_2, \sigma_3, \dots) \preceq \Pi_3$. Let $\Phi_t = \prod_{u=2}^{t-1}(1 - r_u(2)) > 0$. Then

$$\frac{s_t}{\Phi_t} = s_2 + \sum_{u=2}^{t-1} \frac{\sigma_u}{\Phi_{u+1}}, \qquad s_2 = 1 - 3\varepsilon.$$

If $\varepsilon < \frac{1}{3}$, the terms $s_2, \sigma_2, \sigma_3, \dots$ are first nonnegative, then nonpositive, then nonnegative. Their partial sums are nonnegative and nondecreasing in the first stretch, nonincreasing in the second, and nondecreasing in the third. So the sign pattern of the partial sums is a subsequence of Π_3. If $\varepsilon \ge \frac{1}{3}$, then $c/\varepsilon \le 1$, so the terms are first nonpositive and then nonnegative, and the sign pattern of the partial sums is a subsequence of $(-,+)$. This proves the first claim.

For the second claim we compute $P_3(0) = \frac{1}{2}(1-\varepsilon)^2$ and $P_3(1) = P_3(2) = \frac{1}{2} - P_3(0)$, so $s_3 = P_3(0) - P_3(1) = (1-\varepsilon)^2 - \frac{1}{2}$. Suppose that $s_3 = 0$. Then $\varepsilon = 1 - 2^{-1/2}$, $P_3(0) = P_3(1) = \frac{1}{4}$ and $s_4 = \sigma_3 = (\varepsilon - \frac{c}{3})(P_3(1) - 2P_3(0)) = -\frac{5\varepsilon - 1}{12}$. This is not 0, because $\varepsilon \ne \frac{1}{5}$. □

Theorem 3.21 (at most five blocks). *Let $N \ge 3$ and $0 < \varepsilon < \frac{1}{2}$. Then*

$$\operatorname{sp}\big(D_N(1), D_N(2), \dots, D_N(N-1)\big) \preceq (+,-,+,-,+).$$

So the second difference of P_N changes sign at most four times on $\{1, \dots, N-1\}$. The same holds for the second difference of f_N.

In words, for every $N \ge 3$ and every $0 < \varepsilon < \frac{1}{2}$ the law has at most two concave stretches. When it has two, the stretch between them is convex.

Proof. For $3 \le t \le N$ let Z_t be the list

$$Z_t = \big(s_N, s_{N-1}, \dots, s_{t+1},\ s_t, D_t(2), D_t(3), \dots, D_t(t-2), s_t,\ s_{t+1}, \dots, s_{N-1}, s_N\big).$$

It has $2N-t-1$ entries. Its middle part is $(D_t(1), D_t(2), \dots, D_t(t-1))$, because $D_t(t-1) = D_t(1) = s_t$ by symmetry. The list Z_3 is $(s_N, \dots, s_4, s_3, s_3, s_4, \dots, s_N)$, and Z_N is the list in the statement.

We check that Z_{t+1} is obtained from Z_t as in Lemma 3.19, for $3 \le t \le N-1$. The first $N-t$ entries of Z_{t+1} are $s_N, \dots, s_{t+1}$, and they are copies of the first $N-t$ entries of Z_t. So $A_i = 1$ and $B_i = 0$ there. The next $t-2$ entries are $D_{t+1}(b)$ for $2 \le b \le t-1$. By (5),

$$D_{t+1}(b) = r_t(b-2)\, D_t(b-1) + \big(1 - r_t(b+1)\big) D_t(b),$$

and $D_t(b-1)$ and $D_t(b)$ are neighbors in the middle part of Z_t, at the right places. The coefficients satisfy $r_t(b-2) \ge \varepsilon$ and $1 - r_t(b+1) \ge 1 - \varepsilon - c = \varepsilon$, since $2 \le b \le t-1$. The last $N-t$ entries of Z_{t+1} are $s_{t+1}, \dots, s_N$, and each is a copy of the entry of Z_t one place to its right. So $A_i = 0$ and $B_i = 1$ there.

By Lemma 3.19, $\mathrm{sp}(Z_N) \preceq \mathrm{sp}(Z_{N-1}) \preceq \cdots \preceq \mathrm{sp}(Z_3)$. By Lemma 3.20 the sign pattern p of $(s_3, \dots, s_N)$ is a subsequence of Π_3, so it is one of the lists $()$, $(+)$, $(-)$, $(+,-)$, $(-,+)$ and $(+,-,+)$. The sign pattern of Z_3 is p reversed, followed by p, with equal neighbors merged. In the six cases this is $()$, $(+)$, $(-)$, $(-,+,-)$, $(+,-,+)$ and $(+,-,+,-,+)$, and each is a subsequence of Π_5.

For f_N we argue in the same way with $D^f_t(m) = f_t(m+1) - 2f_t(m) + f_t(m-1)$ and the lists

$$Z^f_t = \big(s^f_N, \dots, s^f_{t+1},\ s^f_t, D^f_t(2), \dots, D^f_t(t-1),\ d_t, d_{t+1}, \dots, d_{N-1}\big), \qquad 2 \le t \le N,$$

where $s^f_t = D^f_t(1)$ and $d_u = (1 - r_u(u+1))D^f_u(u) = (\varepsilon - \frac{c}{u}) f_u(u-1)$. The entry $D^f_{t+1}(t)$ of Z^f_{t+1} equals $r_t(t-2)D^f_t(t-1) + d_t$ by (5), and the other entries are as before. The proof of Lemma 3.20 applies to s^f_t without change, with $\rho_t = f_t(1)/f_t(0)$, $\rho_2 = \frac{\varepsilon}{1-\varepsilon}$ and $s^f_2 = 1 - 3\varepsilon$. The numbers d_u are negative for $u < c/\varepsilon$ and positive for $u > c/\varepsilon$. So $\mathrm{sp}(Z^f_2)$ is a subsequence of $(+,-,+)$ followed by $(-,+)$, and $Z^f_N = (D^f_N(1), \dots, D^f_N(N-1))$. As before $\mathrm{sp}(Z^f_N) \preceq \mathrm{sp}(Z^f_2)$, and a subsequence of $(+,-,+)$ followed by $(-,+)$ is a subsequence of Π_5 once equal neighbors are merged. □

Theorem 3.22 (no holes)**.** *Let $N \ge 5$ and $0 < \varepsilon < \frac{1}{2}$. Let $1 \le a_0 < a_1 \le N/2$ be integers with $D_N(a_0) < 0 < D_N(a_1)$. Then $D_N(a) > 0$ for every $a_1 \le a \le N - a_1$.*

In words, once the law has turned convex to the right of its left hump, it stays strictly convex up to the mirror image of that point.

Proof. By symmetry $D_N(N-a_1) > 0 > D_N(N-a_0)$. Suppose that $D_N(a) \le 0$ for some a with $a_1 < a < N - a_1$.

The case $D_N(a) < 0$. Then the entries of Z_N at $a_0 < a_1 < a < N - a_1 < N - a_0$ have the signs $-,+,-,+,-$. So $(-,+,-,+,-) \preceq \mathrm{sp}(Z_N) \preceq \Pi_5$ by Theorem 3.21. This is impossible, because a subsequence of Π_5 with five terms is Π_5 itself.

The case $D_N(a) = 0$. We use the lists Z_t of the proof of Theorem 3.21 and number the entries of Z_t from 1 to $2N-t-1$. So $D_t(b)$ stands at place $N - t + b$ for $1 \le b \le t-1$. For $3 \le t \le N-1$ let W_t be the list w of Lemma 3.19 for $\zeta = Z_t$ and $\eta = Z_{t+1}$. Then $\mathrm{sp}(W_t) = \mathrm{sp}(Z_t) \preceq \Pi_5$. The entry of Z_{t+1} at place k is a combination of the entries of Z_t at the places k and $k+1$, with

nonnegative coefficients. We call these two entries its parents. For $2 \le b \le t-1$ the parents of $D_{t+1}(b)$ are $D_t(b-1)$ and $D_t(b)$, and both coefficients are positive.

For $4 \le t \le N$ let B_t be the set of integers b with $\max(2, a-N+t) \le b \le \min(a, t-2)$. It is not empty, because $2 \le a \le N-2$. We prove by downward induction on t that there are places $p_1 < p_2 < p_4 < p_5$ in Z_t such that the entries there have the signs $-,+,+,-$, that $D_t(b) = 0$ for every $b \in B_t$, and that all the places $N-t+b$ with $b \in B_t$ lie strictly between p_2 and p_4. For $t = N$ we have $B_N = \{a\}$, and we take the places a_0, a_1, $N-a_1$ and $N-a_0$.

Let the claim hold for $t+1$, where $4 \le t \le N-1$, with the places $p'_1 < p'_2 < p'_4 < p'_5$. Let $b \in B_{t+1}$. Then $D_{t+1}(b) = 0$ is a combination of its two parents with positive coefficients. So the parents both vanish, or they have strictly opposite signs. In the second case one parent is negative. In W_t it stands next to $D_{t+1}(b)$, hence between the positive entries of Z_{t+1} at the places p'_2 and p'_4, and these stand between the negative entries at p'_1 and p'_5. So $(-,+,-,+,-) \preceq \operatorname{sp}(W_t) \preceq \Pi_5$, which is impossible. Hence $D_t(b-1) = D_t(b) = 0$ for every $b \in B_{t+1}$. Every element of B_t is of the form b or $b-1$ with $b \in B_{t+1}$, so D_t vanishes on B_t.

The entry of Z_{t+1} at p'_1 is negative, so one of its parents is negative. Let $p_1 \in \{p'_1, p'_1+1\}$ be its place in Z_t, and choose p_2, p_4 and p_5 in the same way. Then $p_1 \le p'_1+1 \le p'_2 \le p_2$, and $p_1 \ne p_2$ because the signs differ. In the same way $p_4 < p_5$. Let q^- and q^+ be the smallest and the largest place $N-t-1+b$ with $b \in B_{t+1}$. The places $N-t+b$ with $b \in B_t$ lie between q^- and $q^+ + 1$. Since $p_2 \le p'_2 + 1 \le q^-$ and $p_4 \ge p'_4 \ge q^+ + 1$, and the entries at p_2 and p_4 are not zero, these places lie strictly between p_2 and p_4. This proves the claim for t.

The induction step with $t = 4$ shows that both parents of $D_5(b)$ vanish for every $b \in B_5 \subseteq \{2,3\}$. The parents of $D_5(2)$ are s_4 and $D_4(2)$, and those of $D_5(3)$ are $D_4(2)$ and s_4. So $s_4 = D_4(2) = 0$. By (5) with $t = 3$ and $a = 2$, and since $D_3(1) = D_3(2) = s_3$, we have $D_4(2) = r_3(0)s_3 + \big(1 - r_3(3)\big)s_3 = 2\varepsilon s_3$. So $s_3 = 0$ as well, which contradicts Lemma 3.20. □

Corollary 3.23 (one inequality is enough)**.** *Let $N \ge 8$ be even and $x = x_N$, or let $N \ge 5$ and $0 < \varepsilon < \frac{1}{2}$ be such that $P_N(0) < P_N(1)$. Let $a^* \ge N/2$ be a mode of P_N and $m^* = N - a^*$. If $D_N(a_1) > 0$ for some integer a_1 with $m^* < a_1 \le 2m^* + 1$ and $a_1 \le N/2$, then $D_N(a) > 0$ for every $2m^* + 1 \le a \le N - 2m^* - 1$. In particular the statement of Conjecture 3.16 holds for this N, this ε and this m^* as soon as $D_N(2m^* + 1) > 0$.*

Proof. At the crossing with $N \ge 8$ even, $P_N(0) = E_N < P_N(N-1) = P_N(1)$ by Proposition 3.8. Let a_0 be the smallest maximizer of P_N. Then $1 \le a_0 \le m^*$, because $P_N(0) < P_N(1)$ and $P_N(m^*) = P_N(a^*)$ is the maximum. Also $P_N(a_0 - 1) < P_N(a_0) \ge P_N(a_0 + 1)$, so $D_N(a_0) < 0$. Now Theorem 3.22 applies to $a_0 < a_1$. □

The condition $a_1 \le N/2$ costs nothing at $a_1 = 2m^* + 1$, because $2m^* + 1 \le N/2$ whenever the range $2m^* + 1 \le a \le N - 2m^* - 1$ is not empty.

Theorem 3.24 (convexity between the humps for large N)**.** *There is N_1 such that Conjecture 3.16 holds for every even $N \ge N_1$. That is, for even $N \ge N_1$, $x = x_N$ and every mode $a^* \ge N/2$ of P_N with $m^* = N - a^*$,*

$$P_N(a-1) - 2P_N(a) + P_N(a+1) > 0 \qquad \text{for } 2m^* + 1 \le a \le N - 2m^* - 1.$$

Idea of the proof. By Corollary 3.23 it is enough to find one point of strict convexity between m^* and $2m^* + 1$. If there were none, P_N would be concave on $[m^*, 2m^* + 2]$, and then its value halfway would be at least half of its maximum. The Landau limit shows that the value halfway is much smaller.

Proof. Let N be even and large, and let $x = x_N$ and $\varepsilon = \varepsilon_N$. By Theorem 2.7, $x \to \infty$ and $x = O(\log N)$. So the standing assumptions of Section 3 hold, and $(1+\log x)/x$, x/N and $x\delta_B = O(x^3 \log^2 N/N)$ tend to 0. So by Theorem 3.10(b) there are numbers $\eta_N \to 0$ with

$$\bigl|x f_N(m) - f_L(\lambda_m)\bigr| \le \eta_N \qquad \text{for every integer } m \ge 0. \tag{6}$$

The density f_L is continuous, because it is the Fourier transform of an integrable function. It is unimodal with mode λ_0 (Lemma 3.11), so it is nonincreasing on $[\lambda_0, \infty)$, and it is integrable. So $f_L(\lambda) \to 0$ as $\lambda \to \infty$. We write $o(1)$ for a quantity that is bounded in absolute value by a function of N alone that tends to 0.

Step 1: the mirror term. Let $0 \le m \le N/2$. Then $\lambda_{N-m} \ge \frac{N}{2x} - \log x \to \infty$. So (6) gives $x f_N(N-m) \le f_L(\lambda_{N-m}) + \eta_N = o(1)$. Since $P_N(m) = P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)]$, we get

$$2xP_N(m) = f_L(\lambda_m) + o(1) \qquad \text{uniformly for } 0 \le m \le N/2.$$

Step 2: the mode. Let m_0 be the integer part of $x\log x + \lambda_0 x$. Then $0 \le m_0 \le N/2$ and $|\lambda_{m_0} - \lambda_0| \le 1/x$, so Step 1 and the continuity of f_L give $2xP_N(m_0) = f_L(\lambda_0) + o(1)$. Now $P_N(m^*) \ge P_N(m_0)$ and $m^* \le N/2$, so Step 1 at m^* gives $f_L(\lambda_{m^*}) \ge f_L(\lambda_0) - o(1)$. By Lemma 3.11(c) this forces $|\lambda_{m^*} - \lambda_0| \le 1$ for large N. Hence

$$x\log x + (\lambda_0 - 1)x \le m^* \le x\log x + (\lambda_0+1)x, \qquad 2xP_N(m^*) \ge f_L(\lambda_0) - o(1).$$

In particular $m^* \to \infty$ and $2m^* + 2 \le N/2$ for large N.

Step 3: halfway. Let $v = m^* + \lfloor m^*/2 \rfloor$. Then $v \ge \frac{3}{2}m^* - 1$, so by Step 2

$$\lambda_v \ge \tfrac{1}{2}\log x + \tfrac{3}{2}(\lambda_0 - 1) - \tfrac{1}{x} \to \infty.$$

Also $v \le N/2$. So Step 1 gives $2xP_N(v) = f_L(\lambda_v) + o(1) = o(1)$. Since $f_L(\lambda_0) > 0$, Step 2 shows that $P_N(v) < \frac{1}{2}P_N(m^*)$ for large N.

Step 4: a point of convexity. Suppose that $D_N(a) \le 0$ for every $m^* < a \le 2m^* + 1$. Then P_N is concave on the integers of $[m^*, 2m^*+2]$. With $u = m^*$ and $w = 2m^* + 2$ this gives

$$P_N(v) \ge \frac{w-v}{w-u}P_N(u) + \frac{v-u}{w-u}P_N(w) \ge \frac{w-v}{w-u}P_N(m^*) \ge \tfrac{1}{2}P_N(m^*),$$

because $w - v \ge \frac{1}{2}(m^*+2) = \frac{1}{2}(w-u)$. This contradicts Step 3. So $D_N(a_1) > 0$ for some $m^* < a_1 \le 2m^* + 1$, and $a_1 \le N/2$ by Step 2. Corollary 3.23 gives the claim. □

Corollary 3.25 (the center is the unique interior minimum)**.** *For every large even N, at the crossing, $P_N(a) > C_N$ for every $1 \le a \le N-1$ with $a \ne h$.*

Proof. Let N be large and even, and let m^* be as in Theorem 3.24. By that theorem and the symmetry of P_N, the first differences $P_N(a+1) - P_N(a)$ are strictly increasing for $2m^* \le a \le N - 2m^* - 1$, and $P_N(h+1) - P_N(h) = \frac{1}{2}D_N(h) > 0$. So P_N is strictly increasing on $[h, N-2m^*]$ and strictly decreasing on $[2m^*, h]$. By Step 2 of the last proof, $m^* = O(\log N \log\log N)$. So for large N every a with $1 \le a \le N-1$ and $|a-h| < C_{\mathrm{w}}(Nx_N \log N)^{1/2}$ lies in $[2m^*, N-2m^*]$, and the other a are covered by Theorem 3.15. □

Proposition 3.26 (small N, computer-assisted)**.** *Conjecture 3.16 holds for every $10 \le N \le N_{\mathrm{c}} = 2000$, even or odd.*

Proof. The script `certify_convex.py` uses ball arithmetic with 160 bits (the library Arb). For each N it encloses ε_N in an interval of relative width at most $2 \cdot 10^{-8}$. It does so by evaluating the center and the end at the two ends of the interval, and Proposition 2.3 then places the crossing between them. It then runs (4) with ε equal to this interval. Each step only multiplies and adds enclosures of numbers in $[0,1]$, so the radii stay small. From the resulting enclosure of P_N it lists every index that can be a mode, and for each of them it checks that the enclosure of $D_N(a)$ is positive for every $2m^*+1 \le a \le N-2m^*-1$. It also checks that $D_N(a_0) < 0$ for some $a_0 \le 2m^*$ and that $D_N(2m^*+1) > 0$. By Theorem 3.22 these two inequalities already imply the claim, so the test of every a is a second check. □

Proposition 3.27 (odd N)**.** *Let $N \ge 3$ be odd, and write $\widetilde{C}_N = P_N(\frac{N-1}{2}) = P_N(\frac{N+1}{2})$ for the two central bins. The ratio $\widetilde{C}_N/E_N$ is strictly increasing in ε, and it equals 1 at exactly one $\varepsilon_N \in (0, \frac{1}{2})$. Put $x_N = N\varepsilon_N$.*

(a) For every $0 < \varepsilon < \frac{1}{2}$,

$$\widetilde{C}_N = \Big(1 - \frac{c}{N}\Big)^{-1} C_{N+1},$$

where C_{N+1} is the center of the walk with $N+1$ steps and the same ε.

(b) As $N \to \infty$ over odd N, $x_N + \log x_N = \Lambda_N + O(L^2/N)$. So both parts of Theorem 2.7 hold for odd N as well.

(c) For every odd $N \ge 1689$ we have $x_N > 2$, and at the crossing $P_N(0) < P_N(1)$. So Corollary 3.23 applies to these N with $x = x_N$.

(d) There is N_1' such that Conjecture 3.16 holds for every odd $N \ge N_1'$.

In words, the center of a walk with an even number of steps is reached from the two central bins one step earlier, and from each of them with the same probability. So the two central bins of an odd N are the center of $N+1$ up to the factor $1 - c/N$, and what we know about the crossing for even N carries over.

Proof. The crossing. By Proposition 2.3, $P_N(k)/E_N$ is strictly increasing in ε for $k = \frac{N+1}{2}$. It tends to 0 as $\varepsilon \to 0$. At $\varepsilon = \frac{1}{2}$ the steps are independent and fair, so $\widetilde{C}_N = \binom{N}{k} 2^{-N} > 2^{-N} = E_N$. So the ratio equals 1 at exactly one ε_N, and $\varepsilon_N < \frac{1}{2}$.

(a) Put $k = \frac{N+1}{2}$. Since $\varepsilon + \frac{c}{2} = \frac{1}{2}$, we have $r_N(k) = \frac{1}{2} + \frac{c}{2N}$ and $r_N(k-1) = \frac{1}{2} - \frac{c}{2N}$. So (4) with $t = N$ and $a = k$ gives

$$C_{N+1} = P_{N+1}(k) = \Big(\frac{1}{2} - \frac{c}{2N}\Big)\big(P_N(k) + P_N(k-1)\big) = \Big(1 - \frac{c}{N}\Big)\widetilde{C}_N.$$

(b) We follow the proof of Theorem 2.7(a), with $\widetilde{F}(x) = \log \widetilde{C}_N - \log E_N$ at $\varepsilon = x/N$ in place of F. Let $1 \le x \le 2\Lambda_N$. For large N, Theorem 2.6 applies to the even number $N+1$ at this ε and gives $\log C_{N+1} = \log \frac{4\varepsilon}{N+3} + O(\varepsilon L)$. By (a), $\log \widetilde{C}_N = \log C_{N+1} + O(1/N)$. Here $\log \frac{4\varepsilon}{N+3} = \log x + \log 4 - 2L + O(1/N)$, and $1/N \le xL/N$. So

$$\log \widetilde{C}_N = \log x + \log 4 - 2L + O\Big(\frac{xL}{N}\Big),$$

which is the estimate for $\log C_N$ in Step 1 of that proof. The formula $E_N = \frac{1}{2}(1-\varepsilon)^{N-1}$ holds for every N (Lemma 2.2). So $(*)$ holds for $\widetilde{F}$, and Step 2 of that proof gives $|x_N - g_N| = O(L^2/N)$ and the claim. Part (b) of Theorem 2.7 follows from part (a) as before.

(c) We argue as in the proof of Proposition 3.8. Let $N \geq 1689$ be odd and take $\varepsilon = 2/N$. Then $\varepsilon(1+\log(N+1)) \leq \frac{1}{100}$, so Theorem 2.6(b) for $N+1$ and part (a) give

$$\widetilde{C}_N \leq \frac{N}{N-1}\, C_{N+1} \leq \frac{N}{N-1} \cdot \frac{8}{N(N+3)} (N+1)^{4/N} e^{10400/N}.$$

The right side decreases in N and is below $1.4 \cdot 10^{-3}$ at $N = 1689$. Also $E_N = \frac{1}{2}(1-\frac{2}{N})^{N-1} > 0.06$, as in that proof. So $\widetilde{C}_N < E_N$ at $\varepsilon = 2/N$, and hence $\varepsilon_N > 2/N$ and $x_N > 2 \geq 2(1-\varepsilon_N)$. Theorem 3.5 then gives $P_N(N-1) > E_N$ at the crossing, which is $P_N(0) < P_N(1)$ by symmetry. This is the hypothesis of the second case of Corollary 3.23.

(d) The proof of Theorem 3.24 uses that N is even in two places only. The first is $x \to \infty$ and $x = O(\log N)$, which holds for odd N by (b). The second is Corollary 3.23 at the end of Step 4, which applies by (c). The rest of that proof uses Theorem 3.10, Lemma 3.11 and $P_N(m) = \frac{1}{2}[f_N(m) + f_N(N-m)]$, and these hold for every N. □

Proposition 3.28 (the center for $10 \leq N \leq 2000$)**.** *(a) Let $N \geq 8$ be even and $x = x_N$. Let $a^* \geq N/2$ be a mode of P_N and $m^* = N - a^*$. Suppose that $2m^*+1 \leq N/2$ and that $D_N(a) > 0$ for every $2m^* + 1 \leq a \leq N - 2m^* - 1$. Then $P_N(a) > C_N$ for every $1 \leq a \leq N-1$ with $a \neq h$.*

(b) For every even $10 \leq N \leq 2000$, at the crossing, $P_N(a) > C_N$ for every $1 \leq a \leq N-1$ with $a \neq h$.

In words, if the law is strictly convex between the humps and the stretch between the humps contains the center, then the center is the unique lowest interior bin. So Corollary 3.25 holds for every even N from 10 to 2000, and not only for large N.

Idea of the proof. Between the humps the law is strictly convex and symmetric, so the center is its only minimum there. Outside, a bin as low as the center would be a second valley. A valley needs a point of convexity with points of concavity on both sides, and with the mirror image this gives more changes of sign of the second difference than Theorem 3.21 allows.

Proof. (a) Write $b = 2m^*$, so that $b+1 \leq h$, and put $\delta(a) = P_N(a+1) - P_N(a)$.

Between the humps. Since $D_N(a) = \delta(a) - \delta(a-1) > 0$ for $b+1 \leq a \leq N-b-1$, the differences $\delta(a)$ are strictly increasing for $b \leq a \leq N-b-1$. By symmetry $\delta(h-1) = -\delta(h)$, and $\delta(h-1) < \delta(h)$ because $b \leq h-1$. So $\delta(h-1) < 0 < \delta(h)$. Hence P_N is strictly decreasing on $\{b, \dots, h\}$ and strictly increasing on $\{h, \dots, N-b\}$, and $P_N(a) > C_N$ for every $b \leq a \leq N-b$ with $a \neq h$. In particular $P_N(b) > C_N$.

Near the ends. By symmetry it is enough to treat $1 \leq a \leq b-1$. Suppose that $P_N(a) \leq C_N$ for some such a, and let k be the smallest index in $\{1, \dots, b\}$ at which P_N takes its least value on this set. By Proposition 3.8, $P_N(1) > P_N(0) = E_N = C_N$, and $P_N(b) > C_N$. So $2 \leq k \leq b-1$, and $P_N(k-1) > P_N(k) \leq P_N(k+1)$, which gives $D_N(k) > 0$. Next let a' be the smallest index at which P_N takes its largest value on $\{0, \dots, k\}$. This value is at least $P_N(1)$, which exceeds both $P_N(0)$ and $P_N(k)$. So $1 \leq a' \leq k-1$ and $P_N(a'-1) < P_N(a') \geq P_N(a'+1)$, which gives $D_N(a') < 0$. In the same way let a'' be the largest index at which P_N takes its largest value on $\{k, \dots, h\}$. This value is at least $P_N(b)$, which exceeds both $P_N(k)$ and $P_N(h) = C_N$. So $k+1 \leq a'' \leq h-1$ and $P_N(a''-1) \leq P_N(a'') > P_N(a''+1)$, which gives $D_N(a'') < 0$. Finally $D_N(h) > 0$ by hypothesis.

So D_N has the signs $-,+,-,+$ at $a' < k < a'' < h$. By the symmetry $D_N(a) = D_N(N-a)$ it has the signs $-,+,-$ at $N-a'' < N-k < N-a'$. Hence the sign pattern of $(D_N(1), \dots, D_N(N-1))$ has a subsequence of seven alternating signs. This contradicts Theorem 3.21, which allows at most five.

(b) Let $10 \le N \le 2000$ be even. By Proposition 3.26, $D_N(a) > 0$ for $2m^*+1 \le a \le N-2m^*-1$ and every mode. The script `certify_center.py` encloses ε_N and the law P_N in the same way as `certify_convex.py`, lists every index $a^* \ge N/2$ that can be a mode, and checks that $2(N-a^*)+1 \le N/2$ for each of them. This holds for every even N in the range. So (a) applies. □

Part (a) also gives Corollary 3.25 from Theorem 3.24 without Theorem 3.15, since $m^* = o(N)$ at the crossing. As a second check, for each even N in the range the script also tests directly that the enclosure of $P_N(a) - P_N(h)$ is positive for every $a \ne h$, and this succeeds for all 996 values of N. The smallest value of $N/2 - (2m^*+1)$ is 1, at $N = 12$.

Remark 3.29 (what is still open). The constant N_1 of Theorem 3.24 is not explicit, because the constant C in Theorem 3.10 is not. (Lemma 3.11(c) gives the explicit value $\kappa_L = 2.6 \cdot 10^{-10}$, but it is far too small to help.) Even with explicit constants, Step 3 needs $f_L(\frac{3}{2}\lambda_0 + \frac{1}{2}\log x) < \frac{1}{2} f_L(\lambda_0)$ up to the error η_N, and at the sizes we can compute the ratio $P_N(v)/P_N(m^*)$ is still about 0.8. Numerically f_L falls to half of its maximum at $\lambda = 2.432$. So even without the error η_N, Step 3 needs $\frac{3}{2}\lambda_0 + \frac{1}{2}\log x > 2.432$, that is, $x > 252.8$. Since $x_N + \log x_N$ is close to $2\log N - \log 8$ (Theorem 2.7), this means that N is above about $3 \cdot 10^{56}$. So the N between N_c and the constants N_1 and N_1' of Theorem 3.24 and Proposition 3.27 are open, and no computation can reach these constants as the proof stands. By Corollary 3.23 and Proposition 3.27(c) the whole range would be closed by an effective proof of the single inequality $D_N(a_1) > 0$ at one point $m^* < a_1 \le 2m^* + 1$. Numerically the convex stretch starts near $1.5\,m^*$. This agrees with the Landau limit, because the inflection point of f_L to the right of its mode is at $\lambda = 1.104$ numerically, and $x\log x + 1.104\,x$ is below twice the mode $x\log x + \lambda_0 x$ as soon as $x \ge 4.8$. A proof along these lines would need the second differences of P_N near $2m^*$ with an error below their size, which is of order $1/x^3$. The main error in Theorem 3.10 is $C(1+\log x)/x$ for each value of $x f_N$, with a constant C that is not explicit, and this is far too large for the N in question.

4 Verification

All computations are in Python, and the main ones were done by two independent methods (for example, forward and backward recursions for the elephant center agree to about 10^{-14} at $N = 12800$). The proofs use computation in four places only. First, some constants are values of explicit sums and integrals, such as the 60-term sum in the proof of Theorem 2.6(b), which we evaluated in 30-digit arithmetic (`verify_elephant_crossing.py`). Its value 102.58640 is below the bound 102.59 used in the proof by more than $3 \cdot 10^{-3}$, even with the tail added, and this margin is far larger than any rounding error in a sum of 60 positive terms at this precision. Second, Lemma 3.11 rests on enclosures of f_L' at four points and of f_L'' on an interval, computed in ball arithmetic with rigorous error bounds. Third, Proposition 3.8, which makes Corollary 3.6 hold for every even $N \ge 8$, rests for $N < 1688$ on a double-precision computation of C_N/E_N at $\varepsilon = 2/(N+2)$ whose rounding error is bounded in Lemma 3.7. The second and third computations are in `certify_elephant.py`, which also repeats the third one in exact integer arithmetic for 20 values of N. Fourth, Proposition 3.26 rests on a computation in ball arithmetic (`certify_convex.py`, about an hour for the whole range, with its output in `data/certify_convex_output.txt`), and so does Proposition 3.28(b) (`certify_center.py`, about half an hour, with its output in

`data/certify_center_output.txt`). The tables, the figures and the numbers quoted outside the proofs rest on computation alone, and so does the evidence for Conjecture 3.16 beyond what Section 3.6 proves. The script `verify_elephant_odd.py` checks the identity of Proposition 3.27(a) in exact rational arithmetic for six values of N, and it computes the numbers of Remark 3.29 for the Landau density in floating point.

Some of the exact and finite statements are also proved in Lean 4, among them Proposition 2.3, the value $C'_N(0) = 4/(N+2)$ of Proposition 2.5(b), Lemma 3.4 and Theorem 3.5. The Lean files are in the folder `kagey-problems/problem131/lean/PaperD`. The file `MAP.md` in that folder lists each formalized statement with its Lean name. The asymptotic results were not formalized.

The scripts and the Lean files are in the repository named under "Data availability" at the end of the paper, which also says how to rerun the checks.

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A note on the use of AI

All mathematical claims, proofs and computations in this paper are my responsibility.

A Proofs for Section 2

We first prove Theorem 2.6. Then Section A.1 proves Proposition 2.9, and Section A.2 proves Proposition 2.11. Throughout, $N \geq 2$ is even and we condition on $X_1 = +1$. By Proposition 2.5(a) we need to bound $V_{t+1}(1)$, and we compare V with explicit hitting probabilities of Pólya urns. Let $\mathcal{L}_s F(b) = (1 - r_s(b))F(b) + r_s(b)F(b+1)$, so that $V_s = \mathcal{L}_s V_{s+1}$ and $V_N(b) = \mathbf{1}\{b = h\}$. Both M_t and $t - M_t \geq 1$ are nondecreasing. So from $b \geq 1$ the chain stays at states ≥ 1, and for $b \geq 1$ we have $V_s(b) = 0$ unless b is feasible, that is, $b \in \mathcal{S}_s = \{b \in \mathbb{Z} : 1 \leq b \leq h,\ 1 \leq s - b \leq h\}$. Note that $\mathcal{S}_N = \{h\}$.

For $c \geq 0$ the Pólya(c) chain moves at time s from b to $b+1$ with probability $r_s^c(b) = \frac{b+c}{s+2c}$. It is Pólya's urn with weights $b+c$ and $s-b+c$ for the two colors, that is, with c extra balls of each color. For $b + c > 0$ and $s - b + c > 0$ let $\Phi_s^c(b) = \binom{N-s}{h-b}\mathrm{B}(h+c, h+c)/\mathrm{B}(b+c, s-b+c)$, where B is the beta function and the binomial coefficient is 0 unless $0 \leq h - b \leq N - s$. Theorem 2.6(a) will follow from Lemma A.1(d), and (b) from Proposition A.4.

Lemma A.1. *Let $c \geq 0$, $1 \leq s \leq N$, $b + c > 0$ and $s - b + c > 0$.*

(a) $\Phi_s^c(b)$ is the probability that the Pólya(c) chain at b at time s is at h at time N. So $\Phi_N^c(b) = \mathbf{1}\{b = h\}$, and $\Phi_s^c(b) = r_s^c(b)\Phi_{s+1}^c(b+1) + (1 - r_s^c(b))\Phi_{s+1}^c(b)$ for $s < N$.

(b) If $s < N$ and $\Phi_s^c(b) > 0$, then $r_s^c(b)\Phi_{s+1}^c(b+1)/\Phi_s^c(b) = \bar{r}_s(b) := \frac{h-b}{N-s}$.

(c) If $0 < \varepsilon < \frac{1}{2}$ and $\beta = \frac{\varepsilon}{1-2\varepsilon}$, then $r_s(b) = r_s^{\beta s}(b)$ for $0 \leq b \leq s$.

(d) If $0 < \varepsilon < \frac{1}{2}$ and $0 \leq c \leq \beta s$, then $V_s(b) \geq \Phi_s^c(b)$ for $1 \leq b \leq s - 1$. In particular $V_{t+1}(1) \geq \Phi_{t+1}^0(1) = \pi_t$ for $1 \leq t \leq h$.

Proof. (a) Let $x = b + c$, $y = s - b + c$ and $(x)_k = x(x+1)\cdots(x+k-1)$. As in the proof of Lemma 2.4, each order of the $N - s$ draws with $h - b$ draws of the first color has probability

$(x)_{h-b}(y)_{N-s-h+b}/(x+y)_{N-s} = \mathrm{B}(h+c,h+c)/\mathrm{B}(x,y)$, and there are $\binom{N-s}{h-b}$ such orders. The recursion is the Markov property.

(b) Use $\binom{N-s-1}{h-b-1}/\binom{N-s}{h-b} = \frac{h-b}{N-s}$ and $\mathrm{B}(x,y)/\mathrm{B}(x+1,y) = \frac{x+y}{x} = 1/r_s^c(b)$.

(c) Since $1+2\beta = 1/(1-2\varepsilon)$, we have $\frac{b+\beta s}{s+2\beta s} = (1-2\varepsilon)(\frac{b}{s}+\beta) = r_s(b)$.

(d) We check the hypotheses of Lemma A.2(b) below for $F_u = \Phi_u^c$ on $\{1,\dots,u-1\}$, where $s \le u \le N$. The reason is that M has $\beta u \ge c$ extra balls at time u, so it is pulled toward $\frac{u}{2}$ at least as strongly as the Pólya(c) chain, and from a state closer to $\frac{u}{2}$ it is easier to end at h.

By (a), $F_N(b) = \mathbf{1}\{b=h\}$, and F_u vanishes off $\mathcal{S}_u$ because the binomial coefficient does. Let $s \le u < N$ and $b \in \mathcal{S}_u$. Subtracting the recursion in (a) from the definition of $\mathcal{L}_u$ gives

$$\mathcal{L}_u F_{u+1}(b) - F_u(b) = \big(r_u(b) - r_u^c(b)\big)\big(F_{u+1}(b+1) - F_{u+1}(b)\big).$$

We show that the two factors on the right never have opposite signs. For the first factor, $r_u^c(b) = \frac{1}{2} - (\frac{u}{2}-b)/(u+2c)$, and by (c) the same formula with βu in place of c gives $r_u(b)$. So the first factor is $\frac{u}{2}-b$ times $\frac{1}{u+2c} - \frac{1}{u+2\beta u}$. The second of these is nonnegative because $c \le \beta s \le \beta u$. Hence the first factor has the sign of $\frac{u}{2}-b$ or vanishes. For the second factor there are three cases. If $b=h$ then $b \ge \frac{u}{2}$ and $F_{u+1}(b+1) = 0$, so the factor is ≤ 0. If $u-b=h$ then $b \le \frac{u}{2}$ and $F_{u+1}(b) = 0$, so the factor is ≥ 0. Otherwise both values are positive, and the formula for Φ gives $F_{u+1}(b+1)/F_{u+1}(b) = \frac{(h-b)(u-b+c)}{(h-u+b)(b+c)}$. With $b = \frac{u}{2} - d$ the numerator minus the denominator is $2d(h+c)$, so the ratio is ≥ 1 if $b \le \frac{u}{2}$ and ≤ 1 if $b \ge \frac{u}{2}$. In all three cases the second factor has the sign of $\frac{u}{2}-b$ or vanishes. Hence $\mathcal{L}_u F_{u+1} \ge F_u$ on $\mathcal{S}_u$, and Lemma A.2(b) with $s_0 = s$ gives $V_s \ge F_s$. Finally $\Phi_{t+1}^0(1) = \beta_{t+1}(h) = \pi_t$ by (a) and Lemma 2.4. □

Lemma A.2. *Let $1 \le s_0 \le N$, and for $s_0 \le s \le N$ let $F_s \ge 0$ be a function on $\{1,\dots,s-1\}$.*

(a) If $F_N(h) \ge 1$ and $\mathcal{L}_s F_{s+1} \le K_s F_s$ on $\mathcal{S}_s$ for $s_0 \le s < N$, with constants $K_s \ge 0$, then $V_s \le \big(\prod_{u=s}^{N-1} K_u\big) F_s$ on $\mathcal{S}_s$.

(b) If $F_N(b) \le \mathbf{1}\{b=h\}$, each F_s vanishes off $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} \ge F_s$ on $\mathcal{S}_s$ for $s_0 \le s < N$, then $V_s \ge F_s$.

Proof. We use backward induction on s, from $V_N(b) = \mathbf{1}\{b=h\}$ and $V_s = \mathcal{L}_s V_{s+1}$. Note that $\mathcal{L}_s$ is an average of two values, so it keeps inequalities between functions. In (a), suppose that $V_{s+1} \le \prod_{u>s} K_u\, F_{s+1}$ on $\mathcal{S}_{s+1}$. This also holds off $\mathcal{S}_{s+1}$, because there $V_{s+1} = 0$ and $F_{s+1} \ge 0$. So on $\mathcal{S}_s$ we get $V_s = \mathcal{L}_s V_{s+1} \le \prod_{u>s} K_u\, \mathcal{L}_s F_{s+1} \le \prod_{u \ge s} K_u\, F_s$. In (b), suppose that $V_{s+1} \ge F_{s+1}$ on $\{1,\dots,s\}$. Then $V_s = \mathcal{L}_s V_{s+1} \ge \mathcal{L}_s F_{s+1} \ge F_s$ on $\mathcal{S}_s$, and $F_s = 0 \le V_s$ off $\mathcal{S}_s$. □

Idea of the proof of the upper bound. We compare M with an urn whose number of extra balls, $\kappa(s+10)$ of each color at time s, grows a little faster than the βs of M. Its hitting probability $\Phi_s^{\kappa_s}$ is explicit. One step of M applied to it gives it back times a factor Δ_s, and Proposition A.4 shows that $\log \Delta_s \le \kappa/s$ up to a smaller term. Two effects enter Δ_s. Raising the extra balls from κ_s to κ_{s+1} changes Φ, and we bound this through the derivative of Φ_s^τ in τ (Lemma A.3). Also M pulls toward the middle less than the comparison urn, which helps. The product of the factors is about $e^{\kappa H_N}$. Let ψ be the digamma function.

Lemma A.3. *Let $2 \le s \le N$, $b \in \mathcal{S}_s$ and $\tau \ge 0$. Put $X = b+\tau$, $Y = s-b+\tau$, $\Sigma = X+Y$ and $z = (s-2b)^2/\Sigma^2$. Then*

$$\partial_\tau \log \Phi_s^\tau(b) < -\log(1-z) + \frac{1+z}{(1-z)\Sigma} + \frac{2(1+z)}{3\Sigma^2(1-z)^2}.$$

Proof. We differentiate the beta functions and bound the digamma function by the first terms of its expansion. Only the two beta functions in $\Phi_s^\tau(b)$ depend on τ, and $\log \mathrm{B}(x,y) = \log\Gamma(x)+\log\Gamma(y)-\log\Gamma(x+y)$. So the derivative is $2\psi(\Sigma)-\psi(X)-\psi(Y)$ minus the difference $2\psi(N+2\tau)-2\psi(h+\tau)$. The duplication formula [19, 5.5.5] gives $2\psi(2y) = \psi(y)+\psi(y+\frac12)+2\log 2$, and ψ is increasing. With $y = h+\tau$ this shows that the difference we subtract is more than $2\log 2$. For $x>0$, [19, 5.9.15] writes $\log x - \frac{1}{2x} - \psi(x)$ as $2\int_0^\infty \frac{t\,dt}{(t^2+x^2)(e^{2\pi t}-1)}$, which lies in $(0, \frac{1}{12x^2})$ since $\int_0^\infty \frac{t\,dt}{e^{2\pi t}-1} = \frac{1}{24}$. So $\psi(\Sigma) < \log\Sigma - \frac{1}{2\Sigma}$, and $\psi(x) > \log x - \frac{1}{2x} - \frac{1}{12x^2}$ for $x = X$ and $x = Y$. Hence the derivative is less than $\log\frac{\Sigma^2}{4XY} + (\frac{1}{2X}+\frac{1}{2Y}-\frac{1}{\Sigma}) + \frac{1}{12}(\frac{1}{X^2}+\frac{1}{Y^2})$. Now $X-Y = 2b-s$, so $4XY = \Sigma^2-(X-Y)^2 = \Sigma^2(1-z)$ and $X^2+Y^2 = \Sigma^2(1+z)/2$. So the three terms are $-\log(1-z)$, $\frac{X^2+Y^2}{2XY\Sigma} = \frac{1+z}{(1-z)\Sigma}$ and $\frac{X^2+Y^2}{12X^2Y^2} = \frac{2(1+z)}{3\Sigma^2(1-z)^2}$. □

Proposition A.4. *Let $0<\varepsilon<\frac14$, $\kappa = \frac{2\varepsilon}{1-4\varepsilon}$, $\kappa_s = \kappa(s+10)$ and $\Sigma_s = s+2\kappa_s$. For $1\le s\le N-1$ and $b\in\mathcal{S}_s$ let $\Delta_s(b) = \mathcal{L}_s\Phi_{s+1}^{\kappa_{s+1}}(b)/\Phi_s^{\kappa_s}(b)$.*

(a) Let $c=\kappa_s$, $c'=\kappa_{s+1}$, $d=\frac{s}{2}-b$, $u=\frac{d}{s+2c'}$, $v=\frac{d}{N-s}$ and $\rho = \frac{s+2c'}{s(1+2\beta)}$. Then $|u|<\frac12$ and

$$\Delta_s(b) = \mathcal{J}\,\frac{\Phi_s^{c'}(b)}{\Phi_s^{c}(b)}, \qquad \mathcal{J} = 1-(\rho-1)\frac{u(u+v)}{\frac14-u^2}.$$

(b) $\log\Delta_s(b) \le \frac{\kappa}{\Sigma_s} + \frac{2\kappa}{3\Sigma_s^2}$.

(c) For $1\le t\le h$, $V_{t+1}(1) \le \pi_t \exp\big(\kappa\big[H_{N-1}-H_t+\frac{2}{3t}\big] + \kappa_{t+1}\big[\log\frac{(t+1)^2}{4t}+\frac23\big]\big)$.

(d) $C_N \le U_N := \varepsilon c_1 e^{\kappa H_{N-1}}\sum_{t=1}^h w_t(1-\varepsilon)^{t-1}e^{\kappa\mathcal{G}(t)}$, where $\mathcal{G}(t) = (t+11)\big(\log\frac{(t+1)^2}{4t}+\frac23\big) - H_t + \frac{2}{3t}$.

The slope κ solves $2(\kappa-\beta)/(1+2\beta) = \kappa$, so that in (b) the gain in $\mathcal{J}$ cancels the main loss from raising the extra balls from c to c'.

Proof. (a) Write $r' = r_s^{c'}(b)$, $\bar r = \bar r_s(b)$ and $r = r_s(b)$. By Lemma A.1(b) with c', and the recursion in Lemma A.1(a), we have $r'\Phi_{s+1}^{c'}(b+1) = \bar r\,\Phi_s^{c'}(b)$ and $(1-r')\Phi_{s+1}^{c'}(b) = (1-\bar r)\Phi_s^{c'}(b)$. So the numerator of $\Delta_s(b)$ divided by $\Phi_s^{c'}(b)$ is $\frac{(1-r)(1-\bar r)}{1-r'} + \frac{r\bar r}{r'}$. Here $r' = \frac12-u$, $\bar r = \frac12+v$ and $r = \frac12-\rho u$, since $\frac12 - r = (1-2\varepsilon)\frac{d}{s} = \rho u$. Over the denominator $r'(1-r') = \frac14-u^2$ the numerator is $f(u,v)+f(-u,-v)$ with $f(u,v) = (\frac12+\rho u)(\frac12-v)(\frac12-u)$, and this equals $\frac14-\rho u^2-(\rho-1)uv$. Also $|d|\le\frac{s}{2}-1$, so $|u|<\frac12$.

(b) *Step 1: the factor $\mathcal{J}$.* Write $c=\kappa_s$, $c'=c+\kappa$ and $\Sigma=\Sigma_s$. Since $c'-\beta s = (\kappa-\beta)s+11\kappa$, the choice of κ gives $\rho-1 = \kappa+\nu_s$ with $\nu_s = \frac{22\kappa}{s(1+2\beta)}$. As u and v have the sign of d, $u(u+v)\ge u^2$. So $\mathcal{J}\le 1-(\kappa+\nu_s)W'$ with $z' = 4u^2$ and $W' = \frac{z'}{1-z'}$. Also $\mathcal{J}>0$ by (a), since $0<r<1$ and $\bar r$, $1-\bar r$ are not both 0. So $\log\mathcal{J}\le -(\kappa+\nu_s)W'$.

Step 2: the ratio $\Phi_s^{c'}/\Phi_s^c$. The logarithm $\log(\Phi_s^{c'}(b)/\Phi_s^c(b))$ is the integral over $\tau\in[c,c']$ of the derivative in Lemma A.3. Its bound increases in z and decreases in Σ, while z decreases and Σ increases in τ. So we may take the bound at $\tau=c$, where $X+Y=\Sigma$, and multiply it by the length κ of the interval. Put $W = \frac{z}{1-z}$. Then $\frac{1}{1-z} = 1+W$, $\frac{1+z}{1-z} = 1+2W$ and $-\log(1-z)\le W$, so

$$\log\Delta_s(b) \le \kappa\Big[W + \frac{1+2W}{\Sigma} + \frac{2(1+2W)(1+W)}{3\Sigma^2}\Big] - (\kappa+\nu_s)W'. \tag{7}$$

Step 3: what is left to show. The terms of (7) without W and W' are $\frac{\kappa}{\Sigma}+\frac{2\kappa}{3\Sigma^2}$, which is the bound we want. Since $(1+2W)(1+W)=1+3W+2W^2$, the other terms are

$$\kappa(W-W')+\frac{2\kappa W}{\Sigma}+\frac{2\kappa W}{\Sigma^2}+\frac{4\kappa W^2}{3\Sigma^2}-\nu_s W',$$

and we must show that they add up to at most 0. If $W=0$ then $d=0$, so $W'=0$ and there is nothing to do. So let $W>0$. The first four terms are positive, and we show that each is a small multiple of $\frac{\kappa}{s}W$. The last term is negative (it comes from the shift 10 in κ_s), and it is a larger multiple of $\frac{\kappa}{s}W$. We need three facts, with $X=b+c$ and $Y=s-b+c$ as in Step 2.

(i) $1+W\le\frac{\Sigma}{2(1+c)}$. Indeed $X,Y\ge 1+c$, since $b\ge 1$ and $s-b\ge 1$, and the larger of X and Y is at least $\frac{\Sigma}{2}$. So $XY\ge(1+c)\Sigma/2$, and $1+W=\frac{1}{1-z}=\frac{\Sigma^2}{4XY}$.

(ii) $W-W'\le\frac{2\kappa}{1+c}W$, and so $W'\ge\frac{9}{11}W$. Indeed z and z' have the same numerator $(s-2b)^2$, and their denominators are Σ^2 and $(s+2c')^2=(\Sigma+2\kappa)^2$. So $z'=\lambda z$ with $\lambda=(\frac{\Sigma}{\Sigma+2\kappa})^2\ge 1-\frac{4\kappa}{\Sigma}$. Then

$$W-W'=\frac{(1-\lambda)z}{(1-z)(1-\lambda z)}\le(1-\lambda)W(1+W)\le\frac{4\kappa}{\Sigma}\,W\,\frac{\Sigma}{2(1+c)}=\frac{2\kappa}{1+c}W,$$

where we used $1-\lambda z\ge 1-z$ and then (i). Since $c=\kappa(s+10)\ge 11\kappa$, we have $\frac{2\kappa}{1+c}\le\frac{2}{11}$.

(iii) $\Sigma\ge s$, $1+c>\kappa s$ and $\Sigma^2\ge 3s$. The first two are clear. For the third, $W>0$ means $s\ne 2b$, and with $b\ge 1$ and $s-b\ge 1$ this forces $s\ge 3$.

Step 4: the terms with W. We bound the four positive terms one by one.

$$\begin{aligned}
\kappa(W-W')&\le\frac{2\kappa^2}{1+c}W\le\frac{2\kappa}{s}W && \text{by (ii) and } 1+c>\kappa s,\\
\frac{2\kappa W}{\Sigma}&\le\frac{2\kappa}{s}W && \text{by } \Sigma\ge s,\\
\frac{2\kappa W}{\Sigma^2}&\le\frac{2\kappa}{3s}W && \text{by } \Sigma^2\ge 3s,\\
\frac{4\kappa W^2}{3\Sigma^2}&\le\frac{4\kappa W(1+W)}{3\Sigma^2}\le\frac{2\kappa W}{3\Sigma(1+c)}\le\frac{2\kappa}{3s}W && \text{by (i) and } \Sigma\ge s.
\end{aligned}$$

The four bounds add up to $\frac{16\kappa}{3s}W$. For the negative term, (ii) gives $-\nu_s W'\le-\frac{9}{11}\nu_s W=-\frac{18\kappa}{s(1+2\beta)}W$. So

$$\log\Delta_s(b)\le\frac{\kappa}{\Sigma}+\frac{2\kappa}{3\Sigma^2}+W\Big[\frac{16\kappa}{3s}-\frac{18\kappa}{s(1+2\beta)}\Big],$$

and the bracket is negative because $1+2\beta<2$ for $\varepsilon<\frac14$, so $18/(1+2\beta)>9>16/3$.

(c) We first compare $V_{t+1}(1)$ with $\Phi_{t+1}^{\kappa_{t+1}}(1)$, and then this with $\pi_t=\Phi_{t+1}^0(1)$. For the first step apply Lemma A.2(a) with $s_0=t+1$, $F_s=\Phi_s^{\kappa_s}$ and $K_s=\exp(\frac{\kappa}{s}+\frac{2\kappa}{3s^2})$. This is allowed, since $F_N(h)=1$, $F_s>0$ on $\mathcal{S}_s$, and $\mathcal{L}_sF_{s+1}=\Delta_sF_s\le K_sF_s$ on $\mathcal{S}_s$ by (b) and $\Sigma_s\ge s$. The product of the K_s over $t<s<N$ is the exponential of $\kappa(H_{N-1}-H_t)+\frac{2\kappa}{3}\sum_{s>t}s^{-2}$, and $\sum_{s>t}s^{-2}\le\frac1t$. Since $1\in\mathcal{S}_{t+1}$, we get $V_{t+1}(1)\le e^{\kappa[H_{N-1}-H_t+2/(3t)]}\,\Phi_{t+1}^{\kappa_{t+1}}(1)$.

For the second step, $\log(\Phi_{t+1}^{\kappa_{t+1}}(1)/\pi_t)$ is the integral over $\tau\in[0,\kappa_{t+1}]$ of the derivative in Lemma A.3 with $s=t+1$ and $b=1$. As in Step 2 of (b), the bound of that lemma is largest at $\tau=0$. There $X=1$, $Y=t$ and $\frac{1}{1-z}=\frac{(t+1)^2}{4t}$, and the bound is $\log\frac{(t+1)^2}{4t}+\frac{t^2+1}{2t(t+1)}+\frac{t^2+1}{12t^2}\le\log\frac{(t+1)^2}{4t}+\frac23$, because the two fractions are at most $\frac12$ and $\frac16$ for $t\ge1$. Multiplying by the length κ_{t+1} of the interval gives (c).

(d) Insert (c) into Proposition 2.5(a), and use $\kappa_{t+1}=\kappa(t+11)$ and $\pi_t=c_1w_t$. □

Proof of Theorem 2.6. (a) By Proposition 2.5(a) and Lemma A.1(d), $C_N \ge \varepsilon c_1 \sum_t w_t (1-\varepsilon)^{t-1}$. Now $(1-\varepsilon)^{t-1} \ge 1-(t-1)\varepsilon$ and $\sum_t w_t(t-1) = 2 - \frac{6}{h+2}$ by Proposition 2.5(c). So $C_N \ge \varepsilon c_1 (1-2\varepsilon)$, and $c_1 = \frac{4}{N+2}$.

(b) *Step 1: the constants.* Let $N \ge 64$ and $\varepsilon(1+\log N) \le \frac{1}{100}$. Then $\varepsilon < 0.002$ and $\kappa \le 2\varepsilon(1+8\varepsilon) \le 2.1\varepsilon$. Since $\log \frac{(t+1)^2}{4t} \ge 0$ and $\frac{2}{3}(t+11) > 1 + \log t \ge H_t$, we have $\mathcal{G}(t) > 0$. Since $\frac{(t+1)^2}{4t} \le \frac{t+1}{2}$, $\frac{2}{3} < \log 2$ and $H_t \ge \frac{2}{3t}$, also $\mathcal{G}(t) \le (t+11)\log(t+1) \le 12t\log(t+1)$. So $\kappa\mathcal{G}(t) \le 25.2\,\varepsilon t \log(t+1)$, and this is at most $\frac{t}{2}\log 2$ for $t \le h$, since then $25.2\,\varepsilon \log(t+1) \le 25.2\,\varepsilon \log N \le 0.252 < \frac{1}{2}\log 2$.

Step 2: the sum over t. Since $e^y - 1 \le y e^y$ for $y \ge 0$, Step 1 gives $e^{\kappa\mathcal{G}(t)} - 1 \le \kappa \mathcal{G}(t) 2^{t/2}$. By Proposition A.4(d), $(1-\varepsilon)^{t-1} \le 1$, $\sum_t w_t = 1$ and $w_t \le 2t2^{-t}$ (Proposition 2.5(c)),

$$C_N \le \varepsilon c_1 e^{\kappa H_{N-1}} \Big(1 + 50.4\,\varepsilon \sum_{t \ge 1} t^2 \log(t+1) 2^{-t/2}\Big),$$

and we bound the series as follows. The ratio of its consecutive terms is $(1+\frac{1}{t})^2 \frac{\log(t+2)}{\log(t+1)} 2^{-1/2}$, which decreases in t and is below $\frac{3}{4}$ at $t = 60$. So the terms with $t > 60$ add up to less than four times the term at $t = 61$, and this is below 10^{-4}. The first 60 terms add up to 102.5864 to four decimals (computed in 30-digit arithmetic by `verify_elephant_crossing.py`), so the series is below 102.59.

Step 3: the factor $e^{\kappa H_{N-1}}$. Finally $H_{N-1} \le 1 + \log N$, so $\kappa H_{N-1} \le 2\varepsilon \log N + 2\varepsilon + 16\varepsilon^2(1 + \log N) \le 2\varepsilon \log N + 3\varepsilon$. With $1 + y \le e^y$ this proves (b), since $3 + 50.4 \cdot 102.59 < 5200$. □

A.1 The bridge and the lower bound

Fix $1 \le t \le h$, and start the Pólya(c) chain at state 1 at time $t+1$. By Lemma A.1(b), given that it is at h at time N it moves up at time u from b with probability $\bar r_u(b) = \frac{h-b}{N-u}$, whatever c is. So its $h-1$ up steps are placed uniformly at random among the $N-t-1$ steps after time $t+1$. We call this law of the path $(M_u)_{t+1 \le u \le N}$ the bridge, and we write $\mathbb{P}_t^{\mathrm{br}}$ and $\mathbb{E}_t^{\mathrm{br}}$ for it. For $1 \le b \le u-1$ let

$$\varpi_u^+(b) = 1 + \varepsilon \frac{u-2b}{b}, \qquad \varpi_u^-(b) = 1 - \varepsilon \frac{u-2b}{u-b},$$

$$\theta_u(b) = \frac{N(u-2b)^2}{2(N-u)\,b(u-b)}, \qquad Q_u(b) = (u-2b)^2 \Big[\frac{\bar r_u(b)}{b^2} + \frac{1-\bar r_u(b)}{(u-b)^2}\Big],$$

and let $\varpi_u = \varpi_u^+(M_u)$ if $M_{u+1} = M_u + 1$ and $\varpi_u = \varpi_u^-(M_u)$ otherwise. Put

$$A_t = \sum_{u=t+1}^{N-1} \mathbb{E}_t^{\mathrm{br}}\, \theta_u(M_u), \qquad B_t = \sum_{u=t+1}^{N-1} \mathbb{E}_t^{\mathrm{br}}\, Q_u(M_u).$$

Both are nonnegative, and neither depends on ε.

Proposition A.5 (bridge formula)**.** *Let $0 \le \varepsilon \le 1$ and $1 \le t \le h$. Then*

$$V_{t+1}(1) = \pi_t\, \mathbb{E}_t^{\mathrm{br}} \Big[\prod_{u=t+1}^{N-1} \varpi_u \Big].$$

If $0 < \varepsilon \le \frac{1}{2}$, then $\log V_{t+1}(1) \ge \log \pi_t + \varepsilon A_t - \varepsilon^2 B_t$.

Proof. We divide V by the hitting probability of Pólya's urn. The quotient satisfies a recursion whose weights are the step probabilities of the bridge, and each step leaves one factor ϖ_u. For $u \ge t+1$ and $b \in \mathcal{S}_u$ we have $\Phi^0_u(b) > 0$, and we put $R_u(b) = V_u(b)/\Phi^0_u(b)$. By Lemma A.1(a) and (b) with $c = 0$, $\frac{b}{u}\Phi^0_{u+1}(b+1) = \bar r_u(b)\Phi^0_u(b)$ and $(1-\frac{b}{u})\Phi^0_{u+1}(b) = (1-\bar r_u(b))\Phi^0_u(b)$. Dividing $V_u = \mathcal{L}_u V_{u+1}$ by $\Phi^0_u(b)$ gives

$$R_u(b) = (1-\bar r_u(b))\,\frac{1-r_u(b)}{1-b/u}\,R_{u+1}(b) + \bar r_u(b)\,\frac{r_u(b)}{b/u}\,R_{u+1}(b+1),$$

where a term with an infeasible state is absent, and its bridge probability is 0 (if $b = h$ then $\bar r_u(b) = 0$, and if $u - b = h$ then $\bar r_u(b) = 1$). Also $\frac{r_u(b)}{b/u} = \varpi^+_u(b)$ and $\frac{1-r_u(b)}{1-b/u} = \varpi^-_u(b)$. So $R_u(b) = \mathbb{E}^{\mathrm{br}}_t[\varpi_u R_{u+1}(M_{u+1}) \mid M_u = b]$, and $R_N(h) = 1$. Iterating from $u = t+1$ and using $\Phi^0_{t+1}(1) = \pi_t$ gives the formula.

Now let $0 < \varepsilon \le \frac12$. We take the logarithm of the formula and bound it from below in three steps. First, every factor is at least $\frac12$. Indeed if $b \le u/2$ then $\varpi^+_u(b) \ge 1$ and $\varpi^-_u(b) - 1 \in [-\varepsilon, 0]$, and if $b \ge u/2$ the same holds with $+$ and $-$ exchanged. Second, by Jensen's inequality for the logarithm,

$$\log \mathbb{E}^{\mathrm{br}}_t\Big[\prod_u \varpi_u\Big] \ge \sum_u \mathbb{E}^{\mathrm{br}}_t \log \varpi_u \ge \sum_u \mathbb{E}^{\mathrm{br}}_t\big[(\varpi_u - 1) - (\varpi_u - 1)^2\big].$$

The second inequality is $\log(1+y) \ge y - y^2$ for $y \ge -\frac12$, which holds since the difference vanishes at 0 and its derivative $y(1+2y)/(1+y)$ has the sign of y. Third, we compute the two means given $M_u = b$. Write $\bar r = \bar r_u(b)$. The bridge moves up with probability $\bar r$, and then $\varpi_u - 1 = \varepsilon\frac{u-2b}{b}$. Otherwise $\varpi_u - 1 = -\varepsilon\frac{u-2b}{u-b}$. Since $u\bar r - b = \frac{N(u-2b)}{2(N-u)}$, this gives

$$\mathbb{E}^{\mathrm{br}}_t[\varpi_u - 1 \mid M_u = b] = \varepsilon(u-2b)\frac{u\bar r - b}{b(u-b)} = \varepsilon\,\theta_u(b), \qquad \mathbb{E}^{\mathrm{br}}_t[(\varpi_u - 1)^2 \mid M_u = b] = \varepsilon^2 Q_u(b).$$

So the sum above is $\varepsilon A_t - \varepsilon^2 B_t$, which is the bound. □

Lemma A.6 (the weights w_t)**.** *For each fixed t, $w_t \to t2^{-t-1}$ as $N \to \infty$. For every even $N \ge 2$, $\sum_t w_t(t+1)^3 \le 124$.*

Proof. Each fraction in $\pi_t = \frac{t}{N-t}\prod_{i=1}^{t-1}\frac{h-i}{N-i}$ tends to $\frac12$, so $N\pi_t \to t2^{1-t}$, and $Nc_1 \to 4$. This is the first claim. For the second claim we write w_t and $(t+1)^3$ through binomial coefficients in t. By the proof of Lemma 2.4 (with $v = t+1$ and $d = h$), π_t is a constant times $\binom{t}{1}\binom{N-1-t}{h-1}$, and the Chu–Vandermonde identity gives $\sum_t \binom{t}{j}\binom{N-1-t}{h-1} = \binom{N}{h+j}$. With $j = 1$ this gives the constant, namely $w_t = \binom{t}{1}\binom{N-1-t}{h-1}/\binom{N}{h+1}$. Now $t(t+1)^3 = 8\binom{t}{1} + 38\binom{t}{2} + 54\binom{t}{3} + 24\binom{t}{4}$, because both sides are polynomials of degree 4 that agree at $t = 0,1,2,3,4$. So $\sum_t w_t(t+1)^3$ is the sum of 8, 38, 54 and 24 times $\binom{N}{h+j}/\binom{N}{h+1}$ for $j = 1,2,3,4$. Since $\binom{N}{h+j} \le \binom{N}{h+1}$ for $j \ge 1$, it is at most $8 + 38 + 54 + 24 = 124$. □

Under $\mathbb{P}^{\mathrm{br}}_t$ let $K = M_u - 1$ and $K' = m - K$ be the numbers of up steps and of other steps among the $m = u - t - 1$ steps between times $t+1$ and u. So K is hypergeometric, with m draws from $N - t - 1$ steps of which $h - 1$ are up steps. Let $\Omega_u = \{u/8 \le M_u \le 7u/8\}$.

Lemma A.7. *Let $N \ge 16$, $1 \le t \le h$ and $t+1 \le u \le N-1$.*

(a) $$\mathbb{E}^{\mathrm{br}}_t(u - 2M_u)^2 = \frac{(t-1)^2(N-u)^2}{(N-t-1)^2} + \frac{4(u-t-1)(h-1)(h-t)(N-u)}{(N-t-1)^2(N-t-2)} \le (t-1)^2 + u.$$

(b) If $t \le \frac{N+2}{4}$, then $A_t \ge 2(H_{N-1} - H_t - 1 - \frac{1}{t}) \ge 2(\log N - \log t - 3)$.

(c) If $t \le \frac{N}{8}$ and $u \ge 8(t+1)$, then $\mathbb{P}_t^{\mathrm{br}}(\Omega_u^c) \le 2e^{-0.0943u}$.

(d) If $t \le \frac{N}{8}$, then $B_t \le 84(t+1+\log N) + 683(t+1)^3 + 19100$.

Proof. (a) Here $u - 2M_u = u - 2 - 2K$ and $u - 2 = m + t - 1$. Since $N - 2 = 2(h-1)$, the mean $\mathbb{E}K = \frac{m(h-1)}{N-t-1}$ gives

$$\mathbb{E}(u - 2M_u) = \frac{(m+t-1)(N-t-1) - m(N-2)}{N-t-1} = \frac{(t-1)(N-t-1-m)}{N-t-1} = \frac{(t-1)(N-u)}{N-t-1}.$$

The variance of a hypergeometric variable with m draws from $N-t-1$ steps, of which $h-1$ are up steps and $h-t$ are not, is $\operatorname{Var} K = \frac{m(h-1)(h-t)(N-t-1-m)}{(N-t-1)^2(N-t-2)}$, and $N-t-1-m = N-u$. So the second term is $4\operatorname{Var} K$, and the identity is $\mathbb{E}(u-2M_u)^2 = (\mathbb{E}(u-2M_u))^2 + 4\operatorname{Var} K$. For the bound, the first term is at most $(t-1)^2$ because $N - u \le N - t - 1$. In the second term $4(h-1)(h-t) = (N-2)(N-2t) = (N-t-1)^2 - (t-1)^2 \le (N-t-1)^2$, so it is at most $\frac{m(N-u)}{N-t-2}$. This is 0 for $u = t+1$, and for $u \ge t+2$ it is at most u, because $m \le u$ and $N - u \le N - t - 2$.

(b) We bound θ_u from below by a multiple of $(u-2b)^2$ and use only the variance term of (a). Since $b(u-b) \le u^2/4$, $\theta_u(b) \ge \frac{2N(u-2b)^2}{(N-u)u^2}$. By (a), $\mathbb{E}_t^{\mathrm{br}}(u-2M_u)^2 \ge 4\operatorname{Var} K$, and in $4\operatorname{Var} K$ the factor $N-u$ cancels. With $4(h-1)(h-t) = (N-2)(N-2t)$ this gives $A_t \ge P_t \sum_{u=t+1}^{N-1} \frac{u-t-1}{u^2}$ with $P_t = \frac{2N(N-2)(N-2t)}{(N-t-1)^2(N-t-2)}$. We show that $P_t \ge 2$. Here $P_1 = \frac{2N}{N-3} > 2$, and for real $t \in [1, \frac{N+2}{4}]$, $\frac{d}{dt}\log P_t = -\frac{2}{N-2t} + \frac{2}{N-t-1} + \frac{1}{N-t-2} \ge \frac{3}{N-t-1} - \frac{2}{N-2t} \ge 0$, since the last inequality is equivalent to $4t \le N+2$. Next, the sum is $H_{N-1} - H_t - (t+1)\sum_{u=t+1}^{N-1} u^{-2}$, which is at least $H_{N-1} - H_t - \frac{t+1}{t}$. The sum is also nonnegative. So multiplying by $P_t \ge 2$ gives the first inequality, whether this lower bound is positive or not. For the second inequality use $H_{N-1} - H_t \ge \log\frac{N}{t+1} \ge \log N - \log t - \frac{1}{t}$.

(c) Here $m \ge \frac{7}{8}u$, so $\frac{u}{8} \le \frac{m}{7}$. Since $t \ge 1$, $\mathbb{E}K \ge \frac{m}{2}$, and since $t \le \frac{N}{8}$, $\mathbb{E}K' = \frac{m(h-t)}{N-t-1} \ge \frac{3m}{8}$. Hoeffding's inequality also holds for sampling without replacement [16, Section 6], so $\mathbb{P}(K - \mathbb{E}K \le -\sigma m) \le e^{-2\sigma^2 m}$ for $\sigma > 0$, and the same holds for K'. Now $\{M_u < \frac{u}{8}\} \subset \{K - \mathbb{E}K < -\frac{5m}{14}\}$, and since $u - M_u = t + K'$, also $\{u - M_u < \frac{u}{8}\} \subset \{K' - \mathbb{E}K' < -\frac{13m}{56}\}$. So $\mathbb{P}(\Omega_u^c) \le e^{-2(5/14)^2 m} + e^{-2(13/56)^2 m} \le 2e^{-0.10778m} \le 2e^{-0.0943u}$.

(d) We split each term $\mathbb{E}_t^{\mathrm{br}} Q_u(M_u)$ of B_t into the part on Ω_u and the part off Ω_u, and the three terms of the bound come from the parts on Ω_u, the parts off Ω_u with $u < 8(t+1)$, and the parts off Ω_u with $u \ge 8(t+1)$. In all of them we use that $\bar{r}(u-b)^2 + (1-\bar{r})b^2 \le u^2$, so that $Q_u(b) \le \frac{(u-2b)^2 u^2}{b^2(u-b)^2}$. On Ω_u we have $b(u-b) \ge \frac{7u^2}{64}$, so $Q_u \le \frac{4096}{49}\frac{(u-2b)^2}{u^2} \le 83.6\frac{(u-2b)^2}{u^2}$. By (a) these parts add up to at most $83.6\sum_{u>t}(\frac{(t-1)^2}{u^2} + \frac{1}{u}) \le 83.6(t + H_{N-1}) \le 84(t+1+\log N)$. Off Ω_u we only have $b(u-b) \ge u - 1 \ge \frac{u}{2}$, which gives $Q_u \le 4(u-2b)^2 \le 4u^2$. For $u < 8(t+1)$ we bound the probability of Ω_u^c by 1, and these parts add up to at most $\sum_{u<8(t+1)} 4u^2 \le \frac{4}{3}(8(t+1))^3 \le 683(t+1)^3$. For $u \ge 8(t+1)$, (c) bounds the part by $8u^2 e^{-0.0943u}$, and with $q = e^{-0.0943}$ we have $8\sum_{u\ge 1} u^2 q^u = 8q(1+q)/(1-q)^3 < 19100$. □

Proof of Proposition 2.9. The upper bound is Theorem 2.6(b). For the lower bound let $t_0 = \lfloor N/8 \rfloor \ge 8$. By Proposition 2.5(a), $C_N = \varepsilon c_1 \sum_t w_t (1-\varepsilon)^{t-1} V_{t+1}(1)/\pi_t$. This is εc_1 times a mean over the probability vector (w_t). By Jensen's inequality the logarithm of the mean is at least the mean of the logarithms. For $t \le t_0$ we bound $\log(V_{t+1}(1)/\pi_t)$ from below by Proposition A.5. For $t > t_0$ we use only that it is ≥ 0, which is Lemma A.1(d). This gives

$$\log C_N \ge \log(\varepsilon c_1) + \sum_t w_t (t-1)\log(1-\varepsilon) + \sum_{t \le t_0} w_t\big(\varepsilon A_t - \varepsilon^2 B_t\big).$$

The first sum. By $\log(1-\varepsilon) \geq -\varepsilon-\varepsilon^2$ and Proposition 2.5(c), it is at least $-(\varepsilon+\varepsilon^2)\sum_t w_t(t-1) \geq -2\varepsilon - 2\varepsilon^2$.

The terms with A_t. Since $t_0 \leq \frac{N+2}{4}$, Lemma A.7(b) gives $A_t \geq 2(\log N - \log t - 3)$ for $t \leq t_0$. Let $w_{\leq} = \sum_{t\leq t_0} w_t$. By Jensen's inequality $\sum_t w_t \log t \leq \log \sum_t w_t t \leq \log 3$, so

$$\sum_{t\leq t_0} w_t\, \varepsilon A_t \geq 2\varepsilon(\log N - 3)w_{\leq} - 2\varepsilon \log 3 \geq 2\varepsilon \log N - 2\varepsilon(1-w_{\leq})\log N - 6\varepsilon - 2\varepsilon\log 3.$$

By Proposition 2.5(c), $1 - w_{\leq} \leq 2\sum_{t>t_0} t2^{-t} = 2(t_0+2)2^{-t_0}$, and $N < 8(t_0+1)$. As $(t_0+2)2^{-t_0}\log(8t_0+8)$ decreases for $t_0 \geq 8$, we get $2\varepsilon(1-w_{\leq})\log N \leq 4\cdot 10\cdot 2^{-8}\log 72\,\varepsilon < 0.67\,\varepsilon$.

The terms with B_t. Here $t \leq t_0 \leq \frac{N}{8}$, so Lemma A.7(d) applies. With $\sum_t w_t(t+1) \leq 4$ (Proposition 2.5(c)) and Lemma A.6 it gives $\sum_{t\leq t_0} w_t B_t \leq 84(3+1+\log N) + 683\cdot 124 + 19100 = 84\log N + 104128$.

Collecting terms, and using $c_1 = \frac{4}{N+2}$,

$$\log C_N - \log\frac{4\varepsilon}{N+2} - 2\varepsilon\log N \geq -(2+0.67+6+2\log 3)\,\varepsilon - \varepsilon^2(84\log N + 104130),$$

which gives the lower bound since $2+0.67+6+2\log 3 < 11$. For the last claim, the hypothesis on ε gives $\varepsilon^2(84\log N + 1.05\cdot 10^5) \leq \varepsilon\,\frac{84\log N + 1.05\cdot 10^5}{100(1+\log N)}$. The fraction decreases in N, and its value at $N = 64$ is below 205. Since $11 + 205 < 5200$, the two-sided bound follows. □

A.2 The coefficient of ε^2

We prove Proposition 2.11 in three parts. Lemma A.8(a) writes $k_2(N)$ through the sums A_t. Lemma A.8(b) finds the limit of $A_t - 2\log N$ for fixed t, and (c) bounds it uniformly in t. Then we sum over t.

Lemma A.8. *(a) For every even $N \geq 2$, $k_2(N) = \sum_{t=1}^{h} w_t\big[A_t - (t-1)\big]$.*

(b) For each fixed $t \geq 1$, as $N \to \infty$ over even N, $A_t - 2\log N \to A_t^{\infty} := 2(\gamma - 1 - \log 2) + (t-2)H_{t-1}$.

(c) For even $N \geq 64$ and $1 \leq t \leq h$, $-2\log(4t) - 6 \leq A_t - 2\log N \leq 2 + 2(t+11)\log(t+1)$.

Idea of the proof of (b). Fix t and let N be large. At a fixed time u the bridge looks like a walk with fair steps that starts at state 1 at time $t+1$. For such a walk $(u-2M_u)^2$ is about u and $M_u(u-M_u)$ is about $u^2/4$, so θ_u is about $2/u$, and the sum over $u < N$ is about $2\log N$. To find the constant we split θ_u into a part that is quadratic in $u - 2M_u$, whose mean over the bridge we know exactly (Lemma A.7(a)), and a remainder whose means are summable over u, uniformly in N. So $A_t - 2\log N$ has the same limit as the corresponding quantity for the fair walk. For the fair walk the mean of θ_u is an explicit integral over $[0,1]$, and we can sum these integrals over u.

Proof. We prove (c) before (b), because (c) is short.

(a) By Proposition 2.5(a) and (b), $c_2 = \sum_t[-(t-1)\pi_t + \partial_\varepsilon V_{t+1}(1)|_{\varepsilon=0}]$. In Proposition A.5 the bridge does not depend on ε, and each ϖ_u is affine in ε and equals 1 at $\varepsilon = 0$. So by the conditional mean computed in its proof, $\partial_\varepsilon V_{t+1}(1)|_{\varepsilon=0} = \pi_t \sum_u \mathbb{E}_t^{\mathrm{br}}[(\varpi_u - 1)/\varepsilon] = \pi_t A_t$. Divide by c_1.

(c) Both sides of Proposition A.4(c) are differentiable in ε on $[0, \frac{1}{4})$ and equal π_t at $\varepsilon = 0$, and $\kappa = 2\varepsilon/(1-4\varepsilon)$ has derivative 2 at 0. Comparing the derivatives at 0 and using the proof of (a)

gives $A_t \le 2H_{N-1} + 2\mathcal{G}(t)$. By Step 1 in the proof of Theorem 2.6(b), $\mathcal{G}(t) \le (t+11)\log(t+1)$, and $H_{N-1} \le 1 + \log N$. This is the upper bound. For $t \le \frac{N+2}{4}$ the lower bound follows from Lemma A.7(b). For $t > \frac{N+2}{4}$ we use $A_t \ge 0$ and $N < 4t$.

(b) Fix t and let $N \ge 16(t+1)$. Recall that $m = u - t - 1$ and $K = M_u - 1$. Steps 1 to 5 show that the limit exists and is a limit for the fair walk. Steps 6 to 9 compute it.

Step 1: split θ_u. For $1 \le b \le u-1$ we have $u^2 = 4b(u-b) + (u-2b)^2$, so $\frac{1}{2b(u-b)} = \frac{2}{u^2} + \frac{(u-2b)^2}{2b(u-b)u^2}$. Multiplying by $\frac{N(u-2b)^2}{N-u}$ gives

$$\theta_u(b) = \frac{N}{N-u}\Big[\frac{2(u-2b)^2}{u^2} + \vartheta_u(b)\Big], \qquad \vartheta_u(b) = \frac{(u-2b)^4}{2b(u-b)u^2} \ge 0.$$

The first term in the bracket is what we get when we replace $b(u-b)$ by its largest value $u^2/4$, and ϑ_u is the correction. So $A_t = A'_t + A''_t$, where

$$A'_t = \sum_{u=t+1}^{N-1} \frac{2N\,\mathbb{E}_t^{\mathrm{br}}(u-2M_u)^2}{(N-u)u^2}, \qquad A''_t = \sum_{u=t+1}^{N-1} \frac{N}{N-u}\,\mathbb{E}_t^{\mathrm{br}}\vartheta_u(M_u).$$

Step 2: the quadratic part A'_t. Write $\varsigma_t = \sum_{u>t} u^{-2}$. We show that $A'_t = 2\log N + 2\gamma - 2H_t + 2((t-1)^2 - t - 1)\varsigma_t + o(1)$. We insert the formula of Lemma A.7(a). In its second term the factor $N-u$ cancels, and we get

$$A'_t = \sum_{u=t+1}^{N-1} \frac{2N(t-1)^2(N-u)}{(N-t-1)^2u^2} + P_t \sum_{u=t+1}^{N-1} \frac{u-t-1}{u^2},$$

with P_t as in the proof of Lemma A.7(b). In the first sum the factor $\frac{2N(N-u)}{(N-t-1)^2}$ is at most 8 (because $N-t-1 \ge \frac{N}{2}$) and tends to 2 for fixed u. So this sum tends to $2(t-1)^2\varsigma_t$ by dominated convergence. The second sum is $H_{N-1} - H_t - (t+1)\sum_{u=t+1}^{N-1} u^{-2} = \log N + \gamma - H_t - (t+1)\varsigma_t + o(1)$, and $P_t = 2 + O(1/N)$. Adding the two gives the claim.

Step 3: the terms of A''_t converge. Fix u. The variable K counts the up steps among m steps drawn without replacement from $N-t-1$ steps, of which $h-1$ (about half) are up steps. So as $N \to \infty$ its law tends to the binomial law with m trials and success probability $\frac{1}{2}$. Let ν_u have this binomial law and put $\bar{M}_u = 1 + \nu_u$. This is the walk with fair steps from state 1 at time $t+1$. Since $\frac{N}{N-u} \to 1$, the u-th term of A''_t tends to $\mathbb{E}\vartheta_u(\bar{M}_u)$.

Step 4: the terms of A''_t are dominated. To exchange the limit and the sum we need a bound for the u-th term of A''_t that is summable over u and does not depend on N. We use four facts.

First, $\mathbb{E}_t^{\mathrm{br}}(u-2M_u)^4 \le 64\min(u, N-u)^2 + 8t^4$. To see this, note that $u - 2M_u - \mathbb{E}(u - 2M_u) = -2(K - \mathbb{E}K)$, where K is hypergeometric with $m \le u$ draws. The last $N-u$ steps of the bridge also form a uniform sample, so $h - M_u$ is hypergeometric with $N-u$ draws, and $u - 2M_u - \mathbb{E}(u-2M_u) = 2(h - M_u - \mathbb{E}(h - M_u))$. By Hoeffding's comparison with sampling with replacement [16, Theorem 4], applied to the convex function $x \mapsto (x - x_0)^4$, the fourth central moment of a hypergeometric variable with n draws is at most that of the binomial variable with n trials and the same mean. The latter is $np(1-p)(1 + 3(n-2)p(1-p)) \le n^2/2$, where p is the success probability. We may take $n = m$ or $n = N - u$, so the fourth central moment of $u - 2M_u$ is at most $8\min(u, N-u)^2$. Now use $|\mathbb{E}(u - 2M_u)| \le t$ (by the proof of Lemma A.7(a)) and $(x_1 + x_2)^4 \le 8(x_1^4 + x_2^4)$.

Second, $\frac{N}{N-u} \le 2u$ and $\frac{N}{N-u}\min(u, N-u)^2 \le 2u^2$. The first holds because $u(N-u) \ge N - 1 \ge \frac{N}{2}$. The second is clear for $u \le \frac{N}{2}$, and for $u > \frac{N}{2}$ it follows from $N(N-u) \le 2u^2$. Third, on Ω_u

we have $2M_u(u-M_u)u^2 \ge \frac{7}{32}u^4$, so $\vartheta_u(M_u) \le \frac{32}{7u^4}(u-2M_u)^4$. By the first two facts the part of the u-th term on Ω_u is at most $\frac{32}{7u^4}(128u^2+16t^4u)$. Fourth, off Ω_u we only have $M_u(u-M_u) \ge \frac{u}{2}$, which gives $\vartheta_u \le u$. So the part of the u-th term off Ω_u is at most $2u^2\,\mathbb{P}_t^{\mathrm{br}}(\Omega_u^c)$. This is at most $4u^2e^{-0.0943u}$ for $u \ge 8(t+1)$ by Lemma A.7(c), and at most $2u^2$ for the finitely many smaller u. The sum of the two bounds is summable over u and does not depend on N.

Step 5: the limit as a sum for the fair walk. By Steps 2 to 4 and dominated convergence,

$$A_t - 2\log N \to \tilde{A}_t := 2\gamma - 2H_t + 2\big((t-1)^2 - t - 1\big)\varsigma_t + \sum_{u>t} \mathbb{E}\vartheta_u(\bar{M}_u).$$

We now undo the split of Step 1 for the fair walk. Let $\bar{\theta}_u = \mathbb{E}\frac{(u-2\bar{M}_u)^2}{2\bar{M}_u(u-\bar{M}_u)} = \mathbb{E}\big[\frac{2(u-2\bar{M}_u)^2}{u^2} + \vartheta_u(\bar{M}_u)\big]$, where the second form is the identity of Step 1. This is the mean of θ_u for the fair walk, without the factor $\frac{N}{N-u}$. The variable $u - 2\bar{M}_u = (m+t-1) - 2\nu_u$ has mean $t-1$ and variance m, so $\mathbb{E}(u-2\bar{M}_u)^2 = (t-1)^2 + (u-t-1)$. So the computation of Step 2 gives $\sum_{u=t+1}^{u_1} \mathbb{E}\frac{2(u-2\bar{M}_u)^2}{u^2} = 2\log u_1 + 2\gamma - 2H_t + 2((t-1)^2 - t - 1)\varsigma_t + o(1)$ as $u_1 \to \infty$. We subtract $2\log u_1$ and add the convergent sum of the $\mathbb{E}\vartheta_u(\bar{M}_u)$. Hence

$$\tilde{A}_t = \lim_{u_1\to\infty}\Big(\sum_{u=t+1}^{u_1} \bar{\theta}_u - 2\log u_1\Big). \tag{8}$$

Step 6: $\bar{\theta}_u$ as an integral. It remains to compute the limit in (8). Write $\nu = \nu_u$, and for $0 \le q \le 1$ let $y = \frac{1+q}{2}$, so that $\mathbb{E}q^\nu = y^m$. Since $(u-2b)^2 = u^2 - 4b(u-b)$ and $\frac{1}{b(u-b)} = \frac{1}{u}(\frac{1}{b} + \frac{1}{u-b})$, we have $\frac{(u-2b)^2}{2b(u-b)} = \frac{u}{2}(\frac{1}{b} + \frac{1}{u-b}) - 2$. We take $b = \bar{M}_u = 1 + \nu$. Then $u - b = t + (m-\nu)$, and $m - \nu$ has the law of ν. So $\bar{\theta}_u = \frac{u}{2}\mathbb{E}[\frac{1}{1+\nu} + \frac{1}{t+\nu}] - 2$. For an integer $a \ge 1$ we have $\frac{1}{a+\nu} = \int_0^1 q^{a-1+\nu}\,dq$, so $\mathbb{E}\frac{1}{a+\nu} = \int_0^1 q^{a-1}y^m\,dq$. Hence

$$\bar{\theta}_u = u\int_0^1 \frac{1+q^{t-1}}{2}\,y^m\,dq - 2.$$

Step 7: split off the part with q^{t-1}. Let $\sigma(q) = \sum_{j=0}^{t-2} q^j$, with $\sigma = 0$ for $t = 1$. Since $1 - y = \frac{1-q}{2}$, we have $(1-y)\sigma(q) = \frac{1-q^{t-1}}{2}$, that is, $\frac{1+q^{t-1}}{2} = 1 - (1-y)\sigma(q)$. Also $\int_0^1 y^m\,dq = \frac{2(1-2^{-m-1})}{m+1}$ and $u = m + 1 + t$. So

$$\bar{\theta}_u = \Big[\frac{2t}{m+1} - \Big(1 + \frac{t}{m+1}\Big)2^{-m}\Big] - \mathcal{T}_m, \qquad \mathcal{T}_m = u\int_0^1 (1-y)\sigma(q)y^m\,dq.$$

To compute $\mathcal{T}_m$ we split $u = (m+1) + t$. For the part with $m+1$ we integrate by parts, using $\frac{d}{dq}y^{m+1} = \frac{m+1}{2}y^m$. For $j \ge 0$ this gives

$$(m+1)\int_0^1 q^j\,\frac{1-q}{2}\,y^m\,dq = -\delta_{j0}2^{-m-1} + \int_0^1 y^{m+1}\big((j+1)q^j - jq^{j-1}\big)\,dq,$$

where the first term on the right is the boundary term at $q = 0$. We sum over $0 \le j \le t-2$. The integrands telescope to $(t-1)q^{t-2}y^{m+1}$. For the part with t we use $(1-y)\sigma(q) = \frac{1-q^{t-1}}{2}$ again. With $I(j,k) = \int_0^1 q^j y^k\,dq$ this gives

$$\mathcal{T}_m = -[t \ge 2]\,2^{-m-1} + (t-1)I(t-2, m+1) + \frac{t}{2}\int_0^1 (1-q^{t-1})y^m\,dq,$$

where $[t \ge 2]$ is 1 if $t \ge 2$ and 0 if $t = 1$, and the term with I is absent for $t = 1$.

Step 8: sum over u. We sum the formulas of Step 7 over $0 \le m \le m_0$, so that $u_1 = m_0 + t + 1$. The terms $\frac{2t}{m+1}$ add up to $2tH_{m_0+1}$, and the terms 2^{-m-1} add up to $1 - 2^{-m_0-1}$. Under the integrals we use $\sum_{m \le m_0} y^m = \frac{2(1-y^{m_0+1})}{1-q}$ and $\frac{1-q^{t-1}}{1-q} = \sigma(q)$. So

$$\sum_{u=t+1}^{u_1} \bar\theta_u = 2tH_{m_0+1} - \sum_{m=0}^{m_0}\Big(1 + \frac{t}{m+1}\Big)2^{-m} + [t \ge 2](1 - 2^{-m_0-1}) \\ - (t-1)\sum_{k=0}^{m_0} I(t-2, k+1) - t\int_0^1 \sigma(q)\big(1 - y^{m_0+1}\big)\,dq.$$

Step 9: let $m_0 \to \infty$. We go through the terms of Step 8. The first sum tends to $\sum_{m \ge 0}(1 + \frac{t}{m+1})2^{-m} = 2 + 2t\log 2$, since $\sum_{m \ge 0} \frac{2^{-m}}{m+1} = 2\log 2$. The last integral tends to $\int_0^1 \sigma\, dq = H_{t-1}$. The sum of the $I(t-2, k+1)$ grows like $2\log m_0$, and we need it up to $o(1)$. For $t \ge 2$ we sum the geometric series as before and write $q^{t-2} = 1 - (1 - q^{t-2})$. So the sum is twice the difference of $\int_0^1 y\frac{1-y^{m_0+1}}{1-q}dq$ and $\int_0^1 (1-q^{t-2})y\frac{1-y^{m_0+1}}{1-q}dq$. In the first integral we substitute $q = 2y - 1$. It becomes $\int_{1/2}^1 (y + \cdots + y^{m_0+1})\,dy = \sum_{j=2}^{m_0+2} \frac{1-2^{-j}}{j} = H_{m_0+2} - \frac{1}{2} - \log 2 + o(1)$. In the second integral $\frac{1-q^{t-2}}{1-q} = \sum_{j=0}^{t-3} q^j$ is bounded, so by dominated convergence it tends to $\int_0^1 \sum_{j=0}^{t-3} q^j \frac{1+q}{2} dq = \frac{1}{2}(H_{t-2} + H_{t-1} - 1)$. So $\sum_{k \le m_0} I(t-2, k+1) = 2H_{m_0+2} - 2\log 2 - H_{t-2} - H_{t-1} + o(1)$. We collect the terms. Since $2tH_{m_0+1} - 2(t-1)H_{m_0+2} = 2H_{m_0+1} + o(1)$, we get

$$\sum_{u=t+1}^{u_1} \bar\theta_u = 2H_{m_0+1} - 2 - 2\log 2 + [t \ge 2] + (t-1)H_{t-2} - H_{t-1} + o(1).$$

Here $[t \ge 2] + (t-1)H_{t-2} - H_{t-1} = (t-2)H_{t-1}$ for all $t \ge 1$, where the term $(t-1)H_{t-2}$ is read as 0 when $t = 1$ (for $t \ge 2$ use $H_{t-2} = H_{t-1} - \frac{1}{t-1}$), and $H_{m_0+1} = \log u_1 + \gamma + o(1)$. So by (8), $\tilde A_t = 2(\gamma - 1 - \log 2) + (t-2)H_{t-1} = A_t^\infty$. □

Proof of Proposition 2.11. We pass to the limit under the sum over t and then sum the series. Since $\sum_t w_t = 1$, Lemma A.8(a) gives $k_2(N) - 2\log N = \sum_{t \le h} w_t[A_t - 2\log N - (t-1)]$. For fixed t the t-th term tends to $t2^{-t-1}[A_t^\infty - (t-1)]$ by Lemma A.8(b) and Lemma A.6. By Lemma A.8(c) and $w_t \le 2t2^{-t}$, for $N \ge 64$ the t-th term is at most $C_0(1 + t\log(t+1))\,t2^{-t}$ in absolute value, with a constant C_0 that does not depend on N or t. This bound is summable over t. So by dominated convergence,

$$k_2(N) - 2\log N \to \sum_{t \ge 1} t2^{-t-1}\big[A_t^\infty - (t-1)\big].$$

To sum the series, note that $\sum_t t2^{-t-1} = 1$ and $\sum_t t(t-1)2^{-t-1} = 2$. Let $\mathcal{H}(q) = \sum_{k \ge 0} H_k q^k = \frac{\ell}{1-q}$ with $\ell = -\log(1-q)$, and let $D = \frac{d}{dq}$. With $k = t - 1$,

$$\sum_{t \ge 1} t(t-2)2^{-t-1}H_{t-1} = \frac{1}{4}\sum_{k \ge 0}(k^2 - 1)H_k 2^{-k} = \frac{1}{4}\big[(qD)^2\mathcal{H} - \mathcal{H}\big]_{q=1/2}.$$

Here $(qD)^2\mathcal{H} = q\big[\frac{1+\ell}{(1-q)^2} + \frac{q}{(1-q)^3} + \frac{2q(1+\ell)}{(1-q)^3}\big]$ equals $8 + 6\log 2$ at $q = \frac{1}{2}$, and $\mathcal{H}(\frac{1}{2}) = 2\log 2$. So the sum is $2 + \log 2$, and the limit is $2(\gamma - 1 - \log 2) + (2 + \log 2) - 2 = 2\gamma - 2 - \log 2$. □

B Proofs for the tail and the modes

We keep the conventions of Section 3. We write C for an absolute constant whose value may change from line to line, and $\bar{c}$, A_0, A_1 for fixed absolute constants. Since M is independent of X_1, $f_N(m) = \mathbb{P}(M_N = m)$. Every path of a chain $M^{[k_0]}$ is nondecreasing in time, and we use the paths of Y and $Z^{(j)}$ fixed after Lemma 3.3. The family $\mathcal{T}_v$ has the law β_v of Lemma 2.4. Let $\bar{\beta}_v(s) = \mathbb{P}(|\mathcal{T}_v| \geq s)$, with $\bar{\beta}_v(s) = 1$ for $s \leq 1$. By Pascal's rule $\binom{N-s}{v-1} - \binom{N-s-1}{v-1} = \binom{N-s-1}{v-2}$ the tail of β_v telescopes. So, for $1 \leq s \leq N-1$,

$$\bar{\beta}_v(s) = \binom{N-s}{v-1} \Big/ \binom{N-1}{v-1} = \prod_{i=1}^{v-1} \Big(1 - \frac{s-1}{N-i}\Big) \leq e^{-(v-1)(s-1)/(N-1)}, \qquad \beta_v(s) = \frac{v-1}{N-s}\,\bar{\beta}_v(s), \tag{9}$$

where the product vanishes as soon as one factor does. Where they are defined, the ratios are

$$\frac{\beta_v(d)}{\beta_v(d-1)} = 1 - \frac{v-2}{N-d}, \qquad \frac{\bar{\beta}_v(s-1)}{\bar{\beta}_v(s)} = 1 + \frac{v-1}{N-s-v+2}. \tag{10}$$

Lemma B.1 (monotone sandwich)**.** *Let $0 \leq a \leq b \leq N$. If $m_h \leq a$, then $(b-a+1)f_N(b) \leq \mathbb{P}(a \leq M_N \leq b)$. If $b \leq m_h$, then $(b-a+1)f_N(a) \leq \mathbb{P}(a \leq M_N \leq b)$.*

Proof. By Lemma 3.4, $f_N(\ell) \geq f_N(b)$ for $a \leq \ell \leq b$ in the first case, and $f_N(\ell) \geq f_N(a)$ in the second. Sum over ℓ. □

Lemma B.2 (tail)**.** *Let $0 < \varepsilon < 1$, $k_0 \geq 1$ and $t \geq k_0$. For every integer $z \geq 1$,*

$$\mathbb{P}\big(M_t^{[k_0]} \geq z\big) \leq \frac{\varepsilon t H_z}{z}, \qquad \mathbb{E}M_t^{[k_0]} \leq \varepsilon t \log \frac{t}{k_0}.$$

In particular $\mathbb{P}(M_N \geq z) \leq xH_z/z$, and $\mathbb{P}(Z_s^{(j)} \geq z) \leq xH_z/z$ for $j-1 \leq s \leq N$.

Proof. We use the tree of Lemma 3.1 on $[t]$, with marks only on $v > k_0$. Every odd vertex lies in the family of a marked vertex, so $M_t^{[k_0]} \leq \sum_{v \text{ marked}} |\mathcal{T}_v|$. We call a family large if its size is at least z, and small otherwise. If $M_t^{[k_0]} \geq z$, then some marked family is large, or the small marked families have total size at least z. We use the union bound for the first event and Markov's inequality for the second. Marks and tree are independent, so

$$\mathbb{P}\big(M_t^{[k_0]} \geq z\big) \leq \varepsilon \sum_v \mathbb{P}(|\mathcal{T}_v| \geq z) + \frac{\varepsilon}{z} \sum_v \mathbb{E}\big[|\mathcal{T}_v|; |\mathcal{T}_v| < z\big].$$

By the first sum in Lemma 2.4, with t in place of N, on average $\frac{t}{d(d+1)}$ vertices have a family of size d. So the first sum over v is $\sum_{d \geq z} \frac{t}{d(d+1)} \leq \frac{t}{z}$, and the second is $\sum_{d=1}^{z-1} \frac{t}{d+1} = t(H_z - 1)$. Together they give $\frac{\varepsilon t H_z}{z}$. For the mean, only the vertices $v > k_0$ can be marked and $\mathbb{E}|\mathcal{T}_v| = t/v$, so $\mathbb{E}M_t^{[k_0]} \leq \varepsilon \sum_{v=k_0+1}^{t} \frac{t}{v} \leq \varepsilon t \log \frac{t}{k_0}$. □

Lemma B.3 (late start)**.** *Let $j \geq 2$, $t \geq j-1$ and $z \geq 0$. Then $\mathbb{P}(M_t^{[j-1]} \geq z) \leq \exp\{-\frac{j-1}{2}(\frac{z}{t} - \varepsilon \log \frac{2t}{j-1})\}$.*

Proof. This is a Chernoff bound with an exponent that changes with time. A chain that starts at 0 at the late time $j-1$ is like an urn that already holds $j-1$ balls of one color, so the fraction $M_k^{[j-1]}/k$ moves slowly. For this reason we take an exponent θ_k of size about $(j-1)/(2k)$.

Write $G_k = M_k^{[j-1]}$ and $\nu = (j-1)/2$. For $k \ge j-1$ let $\theta_k = \log \frac{k}{k-\nu}$, so that $e^{\theta_k} - 1 = \frac{\nu}{k-\nu}$. We claim that for $k \ge j-1$

$$\mathbb{E} e^{\theta_{k+1} G_{k+1}} \le e^{\varepsilon\nu/(k+1-\nu)}\, \mathbb{E} e^{\theta_k G_k}.$$

To see this, condition on $G_k = c$. The chain moves up with probability $r_k(c) \le \varepsilon + c/k$, and $1 + y \le e^y$. So

$$\mathbb{E}\big[e^{\theta_{k+1} G_{k+1}} \;\big|\; G_k = c\big] = e^{\theta_{k+1} c}\big(1 + r_k(c)(e^{\theta_{k+1}} - 1)\big) \le \exp\Big\{c\Big(\theta_{k+1} + \frac{\nu}{k(k+1-\nu)}\Big) + \frac{\varepsilon\nu}{k+1-\nu}\Big\}.$$

It remains to check that $\theta_{k+1} + \frac{\nu}{k(k+1-\nu)} \le \theta_k$. Now $\theta_k - \theta_{k+1} = \log(1+w)$ with $w = \frac{\nu}{(k-\nu)(k+1)}$, and $\log(1+w) \ge \frac{w}{1+w} = \frac{\nu}{k(k+1-\nu)}$. This proves the claim.

We start from $G_{j-1} = 0$ and use the claim for $k = j-1, \dots, t-1$. Since $\sum_{k=j}^{t} \frac{1}{k-\nu} \le \log \frac{t-\nu}{\nu} \le \log \frac{2t}{j-1}$, we get $\mathbb{E} e^{\theta_t G_t} \le \exp\{\varepsilon\nu \log \frac{2t}{j-1}\}$. Finally $\theta_t = -\log(1 - \frac{\nu}{t}) \ge \frac{\nu}{t}$, so Markov's inequality gives $\mathbb{P}(G_t \ge z) \le e^{-\nu z/t}\, \mathbb{E} e^{\theta_t G_t}$, which is the claim. □

Lemma B.4 (the first flipped family)**.** *For $1 \le d \le N-1$,*

$$\frac{x}{d(d+1)}\Big(1 - \frac{2x(N-d-1)}{N(d+2)}\Big) \le \mathbb{P}(D = d) \le \frac{x}{d(d+1)}, \qquad \mathbb{P}(D \ge d) \le x\Big(\frac{1}{d} - \frac{1}{N}\Big).$$

Proof. By Lemma 3.3, $\mathbb{P}(D = d) = \sum_j \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and $1 - (j-2)\varepsilon \le (1-\varepsilon)^{j-2} \le 1$. Apply the two sums of Lemma 2.4, and sum the upper bound over $d' \ge d$. □

Proposition B.5 (local lower bound)**.** *There is an absolute constant C such that the following hold.*

(a) For $2 \le m \le N/4$, $f_N(m) \ge \frac{x}{m(m+1)}(1 - e_1(m))$ with $e_1(m) = C\big(\frac{x \log(m+1)}{m} + \varepsilon \log N + m e^{-m/9}\big)$.

(b) For $N/4 < m \le N - 16$ and $r = N - m$, $f_N(m) \ge \frac{x}{m(m+1)}(1 - e_2(m))$ with $e_2(m) = C\big(\frac{x \log N}{r} + \frac{x \log N}{m} + N e^{-r/40}\big)$.

Idea of the proof. We use the formula of Lemma 3.3. Given the time j of the first flip, the count M_N is $d - Y_d + Z^{(j)}_{N-d}$, where d is the size of the first flipped family. As d grows by one this quantity moves by at most one, so it hits m at some d close to m unless Y or Z is large. (If no step of the family flips back and the rest of the tree has no flip, then $Y = Z = 0$ and the quantity is d itself, which hits m at $d = m$.) The law β_j of d is nearly flat near m, and this gives almost the full weight $\beta_j(m)$. Then we sum over j. The errors e_1 and e_2 collect the chance that Y or Z is large, the small slope of β_j, and the first flips that come too late to be treated this way.

Proof. (a) We bound the inner sum of Lemma 3.3 from below for each early first flip, namely for $2 \le j \le J := \lfloor N/8 \rfloor$. The terms with $j > J$ are nonnegative and we drop them.

Step 1: one value of j. Fix $2 \le j \le J$ and put $\Gamma_d = d - Y_d + Z^{(j)}_{N-d}$ for $1 \le d \le N - j + 1$. The inner sum in Lemma 3.3 is $Q_j(m) = \mathbb{E} S_j$ with $S_j = \sum_d \beta_j(d) \mathbf{1}\{\Gamma_d = m\}$. Let $\mathcal{E}$ be the event $\{Z^{(j)}_{N-1} \le m-1,\ Y_{2m} \le m\}$. We claim that

$$S_j \ge \beta_j(m)\Big(\mathbf{1}_{\mathcal{E}} - \frac{2(j-2)}{N}\, Y_{2m}\Big).$$

Off $\mathcal{E}$ the right side is at most 0 and $S_j \ge 0$. So let $\mathcal{E}$ hold. As d grows by one, Y_d grows by 0 or 1 and $Z^{(j)}_{N-d}$ falls by 0 or 1. So every step of Γ lies in $\{-1,0,1\}$, and Γ cannot jump over m. It starts at $\Gamma_1 = 1 + Z^{(j)}_{N-1} \le m$ (note that $Y_1 = 0$). At $d^\circ = m + Y_{2m} \le 2m$ we have $\Gamma_{d^\circ} \ge d^\circ - Y_{d^\circ} \ge d^\circ - Y_{2m} = m$, since $Z^{(j)} \ge 0$ and Y is nondecreasing. (The index d° is allowed, because $2m + j \le N$.) So $\Gamma_d = m$ for some $d \le d^\circ$, and as β_j is nonincreasing, $S_j \ge \beta_j(d^\circ) = \beta_j(m + Y_{2m})$. It remains to compare $\beta_j(m+y)$ with $\beta_j(m)$ for $0 \le y \le m$. By (10),

$$\frac{\beta_j(m+y)}{\beta_j(m)} = \prod_{i=1}^{y}\Big(1 - \frac{j-2}{N-m-i}\Big) \ge 1 - \frac{2y(j-2)}{N}.$$

Here $N - m - i \ge N/2$, so each fraction is at most $\frac{2(j-2)}{N} \le \frac14$, and a product of factors $1-q_i$ with $0 \le q_i \le 1$ is at least $1 - \sum_i q_i$. This proves the claim. Now $\mathbf{1}_{\mathcal{E}} \ge 1 - \mathbf{1}\{Y_{2m} > m\} - \mathbf{1}\{Z^{(j)}_{N-1} \ge m\}$. Taking means in the claim gives

$$Q_j(m) \ge \beta_j(m)\big[1 - \mathbb{P}(Y_{2m} > m) - \mathbb{P}(Z^{(j)}_{N-1} \ge m)\big] - \frac{2(j-2)}{N}\,\mathbb{E}Y_{2m}\,\beta_j(m).$$

Step 2: the chance that Y or Z is large. Lemma B.2 gives

$$\mathbb{P}(Y_{2m} > m) \le 2\varepsilon H_{m+1}, \qquad \mathbb{P}(Z^{(j)}_{N-1} \ge m) \le \frac{xH_m}{m}, \qquad \mathbb{E}Y_{2m} \le 2\varepsilon m \log N.$$

None of these bounds depends on j. Put $p = 2\varepsilon H_{m+1} + xH_m/m$, so that $p \le C(\frac{x\log(m+1)}{m} + \varepsilon \log N)$. If $p > 1$, then $e_1(m) > 1$ (once C is large) and there is nothing to prove. So let $p \le 1$.

Step 3: sum over j. By Lemma 3.3, $f_N(m) \ge \sum_{j=2}^{J} \mathbb{P}(V=j)\,Q_j(m)$. With Steps 1 and 2 this gives

$$f_N(m) \ge (1-p)\sum_{j=2}^{J}\mathbb{P}(V=j)\,\beta_j(m) - \frac{4\varepsilon m\log N}{N}\sum_{j=2}^{J}\mathbb{P}(V=j)\,(j-2)\,\beta_j(m).$$

In the second sum we use $\mathbb{P}(V=j) \le \varepsilon$ and the second sum in Lemma 2.4. So the second term is at most

$$\frac{4\varepsilon m\log N}{N}\cdot\frac{2x(N-m-1)}{m(m+1)(m+2)} \le \frac{x}{m(m+1)}\,8\varepsilon\log N.$$

The first sum is $\mathbb{P}(D=m)$ without the terms $j > J$. So by Lemma B.4 it is at least $\frac{x}{m(m+1)}(1 - \frac{2x}{m+2}) - \varepsilon\sum_{j>J}\beta_j(m)$.

Step 4: the late first flips $j > J$. A family that starts after time $N/8$ rarely reaches size m. By (9), $\beta_j(m) \le \frac{j-1}{N-m}e^{-(j-1)a}$ with $a = \frac{m-1}{N-1} \le \frac14$. We sum this over $j - 1 \ge J$, using $1 - e^{-a} \ge a/2$ and $N - m \ge \frac34 N$. Since $\lfloor N/8\rfloor \ge \frac{N-1}{9}$ for $N \ge 64$, we have $aJ \ge \frac{m-1}{9}$, and so

$$\varepsilon\sum_{j>J}\beta_j(m) \le \frac{C\varepsilon}{N}\Big(\frac{J}{a} + \frac{1}{a^2}\Big)e^{-(m-1)/9} \le \frac{x}{m(m+1)}\,Cme^{-m/9}.$$

Step 5: collect. For $0 \le p \le 1$ and $q \ge 0$ we have $(1-p)(1-q) \ge 1 - p - q$. So Steps 3 and 4 give

$$f_N(m) \ge \frac{x}{m(m+1)}\Big(1 - p - \frac{2x}{m+2} - Cme^{-m/9} - 8\varepsilon\log N\Big),$$

and by the bound for p in Step 2 this is (a).

(b) Now the first flipped family has more than $N/4$ vertices and leaves only $r = N - m$ vertices outside. So we run the same argument on the scale r instead of m. We take $J = w := \lfloor r/8 \rfloor$ (so that $w \geq 2$), the event $\mathcal{E} = \{Z^{(j)}_{N-1} \leq m-1,\ Y_{m+w} \leq w\}$ and $d^\circ = m + Y_{m+w} \leq m + w$. The index d° is allowed, because $m + w + j \leq m + 2w \leq N$. As in Step 1, on $\mathcal{E}$ we have $\Gamma_1 \leq m$ and $\Gamma_{d^\circ} \geq m$, so $\Gamma_d = m$ for some $d \leq d^\circ$. Three estimates change.

The slope of β_j. For $i \leq y \leq w$ we have $N - m - i \geq r - w \geq \frac{7}{8}r$. So each fraction in the product of Step 1 is at most $\frac{8(j-2)}{7r} \leq \frac{1}{7}$, and $\beta_j(m+y) \geq \beta_j(m)(1 - \frac{8y(j-2)}{7r})$.

The chance that Y or Z is large. Since $w + 1 \geq r/8$ and $m + w \leq N$, Lemma B.2 gives $\mathbb{P}(Y_{m+w} > w) \leq 8xH_N/r$ and $\mathbb{E}Y_{m+w} \leq x \log N$, and as before $\mathbb{P}(Z^{(j)}_{N-1} \geq m) \leq xH_m/m$. So p is replaced by $p' = 8xH_N/r + xH_m/m \leq C(\frac{x \log N}{r} + \frac{x \log N}{m})$, and we may assume $p' \leq 1$. With $\mathbb{P}(V = j) \leq \varepsilon$ and the second sum in Lemma 2.4, the terms with $\mathbb{E}Y_{m+w}$ add up to at most

$$\varepsilon \cdot \frac{8x \log N}{7r} \cdot \frac{2N(r-1)}{m(m+1)(m+2)} \leq \frac{x}{m(m+1)} \cdot \frac{3x \log N}{m}.$$

The late first flips. Now $a = \frac{m-1}{N-1} \geq \frac{1}{5}$ and $N - m = r$, so (9) gives $\beta_j(m) \leq \frac{j-1}{r} e^{-(j-1)/5}$. We sum over $j - 1 \geq J$ and use $J \geq \frac{r-7}{8}$ and $J + 1 \leq r$. This gives

$$\varepsilon \sum_{j>J} \beta_j(m) \leq \frac{C\varepsilon(J+1)}{r} e^{-J/5} \leq C\varepsilon e^{-r/40} \leq \frac{x}{m(m+1)} CNe^{-r/40},$$

since $\varepsilon = \frac{x}{m(m+1)} \cdot \frac{m(m+1)}{N}$ and $m + 1 \leq N$. Collecting the terms as in Step 5, with $\frac{2x}{m+2} \leq \frac{x \log N}{m}$, gives (b). □

Proposition B.6 (cumulative upper bound)**.** *There is an absolute constant C such that for $6 \leq m \leq N/2$ with $m \geq 16x \log m$ we have $\mathbb{P}(M_N \geq m) \leq \frac{x}{m}\big(1 + C\,\frac{x \log^2 m}{m}\big)$.*

Idea of the proof. Given the first flip at time j, the count M_N is at most $D + Z_j$, the size of the first flipped family plus the count from the rest of the tree. The event $D \geq m$ gives the main term x/m. The other terms come from a shift of D by Z_j, from a large Z_j, and from a late first flip, and each is smaller by a factor of order $x \log^2 m/m$.

Proof. Step 1: split by the first flip. Let $J = \lfloor N/4 \rfloor$ and $Z_j = Z^{(j)}_{N-1}$. First let $2 \leq j \leq J$. By Lemma 3.3, given $V = j$ the count M_N has the law of $D - Y_D + Z^{(j)}_{N-D}$, where D has the law β_j and is independent of Y and $Z^{(j)}$. This is at most $D + Z_j$, because $Y \geq 0$ and $Z^{(j)}$ is nondecreasing in time. So $M_N \geq m$ needs $Z_j > m/8$, or $Z_j \leq m/8$ and $D \geq m - Z_j$. Hence

$$\mathbb{P}(M_N \geq m \mid V = j) \leq \mathbb{P}\Big(Z_j > \frac{m}{8}\Big) + \mathbb{E}\Big[\bar{\beta}_j(m - Z_j);\, Z_j \leq \frac{m}{8}\Big].$$

We compare $\bar{\beta}_j(m-z)$ with $\bar{\beta}_j(m)$ for an integer $0 \leq z \leq m/8$. By (10), each step down from s to $s-1$ with $s \leq m$ multiplies $\bar{\beta}_j$ by at most $\varrho_j := 1 + \frac{j-1}{N-m-j+2}$, and $\varrho_j \leq 1 + \frac{4(j-1)}{N}$ because $N - m - j \geq N/4$. Since $\varrho^z - 1 \leq z(\varrho - 1)\varrho^z$ for $\varrho \geq 1$, we get $\bar{\beta}_j(m-z) \leq \bar{\beta}_j(m)\varrho_j^z \leq \bar{\beta}_j(m)\big(1 + z(\varrho_j - 1)\varrho_j^{m/8}\big)$. So the mean above is at most $\bar{\beta}_j(m)\big(1 + (\varrho_j - 1)\varrho_j^{m/8}\,\mathbb{E}Z_j\big)$.

On $\{V > J\}$ there is no mark on $2, \dots, J$, so by Lemma 3.1 M_N is a copy of $M_N^{[J]}$ independent of $\{V > J\}$. Altogether

$$\mathbb{P}(M_N \geq m) \leq \sum_{j=2}^{J} \mathbb{P}(V = j)\Big[\bar{\beta}_j(m)\big(1 + (\varrho_j - 1)\varrho_j^{m/8}\,\mathbb{E}Z_j\big) + \mathbb{P}\Big(Z_j > \frac{m}{8}\Big)\Big] + \mathbb{P}(V > J)\,\mathbb{P}\big(M_N^{[J]} \geq m\big).$$

This gives four terms, which we bound in turn. The main term is $\sum_j \mathbb{P}(V=j)\bar\beta_j(m) \le \mathbb{P}(D \ge m) \le \frac{x}{m}$ by Lemma B.4.

Step 2: the shift term. This is the sum over $2 \le j \le J$ of $\mathbb{P}(V=j)\,\bar\beta_j(m)\,(\varrho_j - 1)\,\varrho_j^{m/8}\,\mathbb{E}Z_j$. We bound the five factors by

$$\mathbb{P}(V=j) \le \varepsilon, \quad \bar\beta_j(m) \le e^{-\frac{(j-1)(m-1)}{N-1}}, \quad \varrho_j - 1 \le \frac{4(j-1)}{N}, \quad \varrho_j^{m/8} \le e^{\frac{(j-1)m}{2N}}, \quad \mathbb{E}Z_j \le x \log \frac{N}{j-1}.$$

The second is (9), the fourth follows from the third and $1+y \le e^y$, and the last is Lemma B.2. The two exponentials combine. Since $m \ge 6$ we have $\frac{m-1}{N-1} - \frac{m}{2N} \ge \frac{m-2}{2N} \ge \frac{m}{3N}$. So with $a = \frac{m}{3N}$ and $i = j-1$ the shift term is at most

$$\frac{4\varepsilon x}{N} \sum_{1 \le i < N/4} i\, e^{-ai} \log \frac{N}{i}.$$

Here $i \le J - 1 < N/4$, so every term is nonnegative. In this sum only the first flips up to a time of order $1/a = 3N/m$ matter, because a later first flip rarely has a family of size m. To bound the sum we write $\log \frac{N}{i} = \log \frac{m}{3} + \log \frac{1}{ai}$. The first part is nonnegative, and $\sum_{i \ge 1} i e^{-ai} \le a^{-2}$. The second part is positive only for $i < 1/a$. There $i \log \frac{1}{ai} \le \frac{1}{ea}$, so these at most $1/a$ terms add up to at most $\frac{1}{ea^2}$. Hence the shift term is at most

$$\frac{4\varepsilon x}{N} \cdot \frac{\log m + 1}{a^2} = \frac{36x^2(\log m + 1)}{m^2} \le \frac{x}{m} \cdot \frac{Cx \log m}{m}.$$

Step 3: the big-Z term. This is $\sum_{j=2}^{J} \mathbb{P}(V=j)\mathbb{P}(Z_j > m/8)$, and again $\mathbb{P}(V=j) \le \varepsilon$. Recall that Z_j counts the disagreeing steps of a walk whose first $j-1$ steps agree. For small j we only have the general tail bound of Lemma B.2, which is of order $x \log m/m$. For large j the late start makes $Z_j \ge m/8$ exponentially unlikely (Lemma B.3). We split at $j_{\mathcal{K}} = \lfloor \mathcal{K}N/m \rfloor + 1$ with $\mathcal{K} = 32 \log(m/x)$. Note that $m \ge 16x \log 6 > 28x$, so $\mathcal{K} \ge 2$.

Let $2 \le j \le j_{\mathcal{K}}$. Lemma B.2 at the integer $\lfloor m/8 \rfloor + 1$ gives $\mathbb{P}(Z_j > m/8) \le 8xH_m/m$. There are at most $\mathcal{K}N/m$ such j, and $\mathcal{K} \le 32 \log m$. So they contribute at most

$$\varepsilon \cdot \frac{\mathcal{K}N}{m} \cdot \frac{8xH_m}{m} = \frac{x}{m} \cdot \frac{8\mathcal{K}xH_m}{m} \le \frac{x}{m} \cdot \frac{Cx \log^2 m}{m}.$$

Let $j > j_{\mathcal{K}}$, so that $j - 1 > \mathcal{K}N/m$. We use Lemma B.3 with $t = N-1$ and the level $m/8$. Its exponent contains the term $\varepsilon \log \frac{2(N-1)}{j-1}$. Here $\frac{2(N-1)}{j-1} \le \frac{2m}{\mathcal{K}} \le m$, so this term is at most $\frac{x \log m}{N} \le \frac{m}{16N}$, by the hypothesis $m \ge 16x \log m$. The other term in the exponent is $\frac{m}{8(N-1)} \ge \frac{m}{8N}$. So the lemma gives $\mathbb{P}(Z_j \ge m/8) \le e^{-(j-1)m/(32N)}$, which is a geometric sequence in j. Summing over $j - 1 \ge j_{\mathcal{K}} > \mathcal{K}N/m$, these j contribute at most

$$\varepsilon \frac{e^{-\mathcal{K}/32}}{1 - e^{-m/(32N)}} \le \varepsilon \cdot \frac{x}{m}\Big(\frac{32N}{m} + 1\Big) \le \frac{x}{m} \cdot \frac{33x}{m}.$$

Here $e^{-\mathcal{K}/32} = x/m$ by the choice of $\mathcal{K}$, and we used $\frac{1}{1-e^{-y}} \le \frac{1}{y} + 1$ for $y > 0$ and $\varepsilon \le x/m$.

Step 4: no early flip. If there is no flip up to time J, the walk agrees with X_1 for about $N/4$ steps, and Lemma B.3 makes $M_N \ge m$ exponentially unlikely in m. Namely $\mathbb{P}(V > J) = (1-\varepsilon)^{J-1} \le e^{\varepsilon - 3x/16}$, since $J \ge 3N/16$, and Lemma B.3 with $j - 1 = J \in [\frac{3N}{16}, \frac{N}{4}]$ and $t = N$ gives $\mathbb{P}(M_N^{[J]} \ge m) \le \exp\{-\frac{3m}{32} + \frac{x}{8} \log \frac{32}{3}\}$. As $x \le m/28$ and $\varepsilon \le \frac{1}{4}$, the product of the two is at most $e^{1/4 - m/13}$, and this is at most $\frac{x}{m} \cdot \frac{Cx \log^2 m}{m}$ because $x \ge 1$. □

Lemma B.7 (mode bound)**.** *There are absolute constants $\bar{c} > 0$ and A_0 such that if $\varepsilon \log N \le \bar{c}$ and $A_0 x \log(x+2) \le N/4$, then $m_h \le A_0 x \log^2(x+1)$.*

Proof. We pick a point ℓ of order $x \log x$ at which $f_N(\ell)$ is at least half of $\frac{x}{\ell(\ell+1)}$. Up to its mode f_N is nondecreasing. So if the mode were far to the right of ℓ, the bins from ℓ to the mode would carry more mass than the tail $\mathbb{P}(M_N \ge \ell)$ has.

Let $\ell = \lceil Ax \log(x+2) \rceil$. Since $\log(\ell+1) \le \log(A+2) + 2\log(x+2)$, we have $\frac{x \log(\ell+1)}{\ell} \le \frac{\log(A+2)+2}{A}$, and $\ell e^{-\ell/9}$ is small when A is large. So we can fix the absolute constants A and $\bar{c}$ such that $e_1(\ell) \le \frac{1}{2}$ in Proposition B.5(a) whenever $\varepsilon \log N \le \bar{c}$. We take $A_0 \ge A+1$, so that $2 \le \ell \le N/4$. Then $f_N(\ell) \ge \frac{x}{2\ell(\ell+1)}$. If $m_h \ge \ell$, Lemmas B.1 (with $a = \ell$, $b = m_h$) and B.2 give $(m_h - \ell + 1) f_N(\ell) \le \mathbb{P}(M_N \ge \ell) \le x H_\ell / \ell$. So $m_h \le (\ell+1)(1 + 2H_\ell)$ in every case. As $\log \ell \le C \log(x+2)$ and $\log(x+2) \le 2\log(x+1)$, this is at most $A_0 x \log^2(x+1)$ once A_0 is large. □

Proposition B.8 (local upper bound)**.** *There are absolute constants A_1 and C such that if $\varepsilon \log N \le \bar{c}$ and $A_1 x \log^2(x+1) \le m \le N/8$, then*

$$f_N(m) \le \frac{x}{m(m+1)} \big(1 + C\Delta(m)^{1/2}\big), \qquad \Delta(m) = \frac{x \log^2 m}{m} + \varepsilon \log N + \frac{m}{N}.$$

Idea of the proof. To the right of its mode f_N is nonincreasing, so $(k+1) f_N(m)$ is at most the probability of $\{m - k \le M_N \le m\}$, which is a difference of two tails. We bound the first tail from above by Proposition B.6 and the second from below by summing Proposition B.5. Then we choose k to balance the errors.

Proof. We take $A_1 \ge 2A_0$ so large that Lemma B.7 applies and every $m' \ge m/2$ satisfies $m' \ge 6$ and $m' \ge 16 x \log m'$ (as $t/\log t$ increases for $t \ge e$).

Step 1: a difference of tails. Let $1 \le k \le m/2$. Then $m - k \ge m/2 \ge m_h$, and Lemma B.1 with $a = m - k$ and $b = m$ gives $(k+1) f_N(m) \le \mathbb{P}(M_N \ge m - k) - \mathbb{P}(M_N \ge m + 1)$.

Step 2: bound each tail. Proposition B.6 at $m - k$ bounds the first term by $\frac{x}{m-k} + \frac{C x^2 \log^2 m}{m^2}$. For the second we sum Proposition B.5(a) over $m < \ell \le N/4$. Since $\sum_{m<\ell\le N/4} \frac{1}{\ell(\ell+1)} \ge \frac{1}{m+1} - \frac{4}{N}$, $\frac{\log(\ell+1)}{\ell}$ and $\ell e^{-\ell/9}$ decrease for $\ell \ge 9$, and $m e^{-m/9} \le C/m \le C\Delta(m)$, we get $\mathbb{P}(M_N \ge m+1) \ge \frac{x}{m+1}(1 - C\Delta(m))$. As $\frac{x}{m-k} - \frac{x}{m+1} = \frac{x(k+1)}{(m-k)(m+1)}$, this gives $f_N(m) \le \frac{x}{(m-k)(m+1)} + \frac{Cx\Delta(m)}{(k+1)m}$.

Step 3: choose k. If $\Delta(m) \le \frac{1}{16}$, take $k = \lceil m\Delta(m)^{1/2} \rceil \le \frac{m}{2}$. Then $\frac{m}{m-k} \le 1 + \frac{2k}{m} \le 1 + 2\Delta^{1/2} + \frac{2}{m}$ and $\frac{\Delta}{k+1} \le \frac{\Delta^{1/2}}{m}$, which gives the claim, as $\frac{1}{m} \le \Delta(m) \le \Delta(m)^{1/2}$. If $\Delta(m) > \frac{1}{16}$, take $k = \lfloor m/2 \rfloor$. Then $f_N(m) \le \frac{x}{m(m+1)}(2 + C\Delta(m))$, and the claim follows after enlarging C, since $\Delta(m)$ is bounded by an absolute constant ($\frac{x \log^2 m}{m}$ decreases for $m \ge e^2$ and is bounded at $m = A_1 x \log^2(x+1)$). □

Proof of Theorem 3.14. We check that Propositions B.5(a) and B.8 apply on the whole range of m and that their error terms tend to 0 uniformly there. Then we show that the mirror term $f_N(N-m)$ is negligible.

Step 1: the hypotheses hold. Put $\ell_x = \log(x+1) \ge \log 2$ and let $\mathcal{R}_N$ be the range $\omega_N x \ell_x^2 \le m \le \delta'_N N$. Since $x \log^2 N / N \to 0$, for large N we have $\varepsilon(1 + \log N) \le \min(\bar{c}, \frac{1}{100})$, $\omega_N \ge \max(A_1, 16)$, $\delta'_N \le \frac{1}{8}$ and $A_0 x \log(x+2) \le N/4$. The standing assumptions $N \ge 64$ and $\varepsilon \le \frac{1}{4}$ hold as well.

Step 2: the errors are small on $\mathcal{R}_N$. The function $\frac{x \log^2 m}{m}$ decreases for $m \ge e^2$. At $m = \omega_N x \ell_x^2$ we have $m \ge e^2$ and $\log m \le \log \omega_N + 3\ell_x$, as $\log x$ and $\log \ell_x$ are at most ℓ_x. So on $\mathcal{R}_N$, uniformly

in $x \ge 1$, $\frac{x\log^2 m}{m} \le \frac{(\log\omega_N + 3\ell_x)^2}{\omega_N \ell_x^2} \le \frac{1}{\omega_N}(\frac{\log\omega_N}{\log 2} + 3)^2 \to 0$. Also $m \ge \omega_N \log^2 2 \to \infty$ on $\mathcal{R}_N$. Since $\log(m+1) \le 2\log^2 m$ for $m \ge 3$, Proposition B.5(a) has $e_1(m) \le C(\frac{x\log^2 m}{m} + \varepsilon \log N + me^{-m/9}) \to 0$ uniformly on $\mathcal{R}_N$. Also $\Delta(m) \le \frac{x\log^2 m}{m} + \varepsilon \log N + \delta'_N \to 0$ there, and Proposition B.8 applies. So $f_N(m) = \frac{x}{m(m+1)}(1 + o(1))$ uniformly on $\mathcal{R}_N$.

Step 3: the mirror term. Lemma B.7 gives $m_h \le A_0 x \ell_x^2 \le h$ for large N, since $x\ell_x^2 \le x\log^2(N+1) = o(N)$. As $N - m \ge h$, Lemma 3.4 gives $0 \le f_N(N-m) \le f_N(h) = C_N$. By Theorem 2.6(b) and $\varepsilon \log N \to 0$, $C_N \le \frac{4x}{N(N+2)} N^{2\varepsilon} e^{5200\varepsilon} = \frac{4x}{N^2}(1 + o(1))$. On $\mathcal{R}_N$ this is at most $\frac{x}{m(m+1)} \cdot 8\delta'^2_N (1 + o(1))$, since $m(m+1) \le 2\delta'^2_N N^2$. Hence $P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)] = \frac{x}{2m(m+1)}(1 + o(1))$ uniformly on $\mathcal{R}_N$, and $m(m+1) = m^2(1 + o(1))$ because $m \to \infty$. □

C Proofs for the bulk

We prove Theorems 3.9 and 3.10, Lemma 3.11 and Corollary 3.12. We keep the notation of Section 3 and of Appendix B. So $\mathcal{T}$ is the tree of Lemma 3.1, $\mathcal{T}_v$ is the family of v, and μ_v is the mark of v. By Lemma 2.4 and the discussion after Lemma 3.1, $|\mathcal{T}_v|$ has the law β_v, with mean N/v, and $\sum_{v=2}^{N} \beta_v(d) = N/(d(d+1))$ for $1 \le d \le N-1$. Let

$$\Psi_N(\theta) = \sum_{d=1}^{N-1} \frac{e^{i\theta d} - 1}{d(d+1)}, \qquad \Psi_\infty(\theta) = \sum_{d \ge 1} \frac{e^{i\theta d} - 1}{d(d+1)}.$$

The law CP_N has the characteristic function $\widehat{\mathrm{CP}}_N(\theta) = e^{x\Psi_N(\theta)}$.

C.1 Proof of Theorem 3.9

Here we assume only $N \ge 2$ and $0 < \varepsilon \le \frac{1}{4}$. Let $Z_+ = \sum_{v=2}^{N} \mu_v |\mathcal{T}_v|$ be the total size of the flipped families.

Step 1: $M_N = Z_+$ with high probability. Two families are nested or disjoint. Suppose that no marked vertex has a marked strict descendant. Then the marked families are disjoint, each of their vertices has exactly one mark on its path to the root, and the other vertices have none. So $M_N = Z_+$ by Lemma 3.1. The number of pairs of a marked vertex and a marked strict descendant has mean $\varepsilon^2 \mathbb{E}\sum_v (|\mathcal{T}_v| - 1)$, since marks and tree are independent. Hence

$$\mathbb{P}(M_N \ne Z_+) \le \varepsilon^2 \sum_{v=2}^{N} \Big(\frac{N}{v} - 1\Big) \le \varepsilon^2 N(H_N - 1) \le \frac{x^2 \log N}{N}.$$

Step 2: given the tree, a product. Fix $\theta \in [-\pi, \pi]$. Given $\mathcal{T}$ the marks are independent, so $\mathbb{E}[e^{i\theta Z_+} \mid \mathcal{T}] = \prod_{v=2}^{N}(1 + a_v)$ with $a_v = \varepsilon(e^{i\theta|\mathcal{T}_v|} - 1)$. Here $|a_v| \le 2\varepsilon \le \frac{1}{2}$ and $\operatorname{Re} a_v \le 0$. Put $W = \sum_v a_v$, so that $\operatorname{Re} W \le 0$. For complex a with $|a| \le \frac{1}{2}$ the series of the logarithm gives $|\log(1+a) - a| \le |a|^2$. So $\prod_v (1 + a_v) = e^{W+\varrho}$ with $|\varrho| \le \sum_v |a_v|^2 \le 4\varepsilon^2(N-1)$. Since $|e^w - 1| \le |w| e^{|w|}$,

$$\Big|\prod_v (1 + a_v) - e^W\Big| \le e^{\operatorname{Re} W}\, 4\varepsilon^2 (N-1)\, e^{4\varepsilon^2(N-1)} \le \frac{4x^2}{N} e^{4x^2/N}.$$

Step 3: replace W by its mean. Write $W = \varepsilon F$ with $F = \sum_{v=2}^{N}(e^{i\theta|\mathcal{T}_v|} - 1)$. Then $\mathbb{E}W = \varepsilon \sum_d (e^{i\theta d} - 1) \sum_v \beta_v(d) = x\Psi_N(\theta)$, which is $\log \widehat{\mathrm{CP}}_N(\theta)$. Put $\delta = W - \mathbb{E}W$. Taylor's formula for

$s \mapsto e^{\mathbb{E}W + s\delta}$ on $[0,1]$ gives

$$e^W - e^{\mathbb{E}W} - e^{\mathbb{E}W}\delta = \delta^2 \int_0^1 (1-s)\, e^{\mathbb{E}W + s\delta}\, ds.$$

The exponent $\mathbb{E}W + s\delta = (1-s)\mathbb{E}W + sW$ has real part at most 0, so the right side has modulus at most $\frac{1}{2}|\delta|^2$. As $\mathbb{E}\delta = 0$, we get $|\mathbb{E}e^W - e^{\mathbb{E}W}| \le \frac{1}{2}\varepsilon^2 \operatorname{Var} F$, where $\operatorname{Var} F = \mathbb{E}|F - \mathbb{E}F|^2$.

Step 4: the variance of F. The variable F is a function of the independent parents $U_1, \dots, U_{N-1}$. Let $F^{(k)}$ be F with U_k replaced by an independent copy U_k'. The Efron–Stein inequality [15] says that $\operatorname{Var} F \le \frac{1}{2}\sum_k \mathbb{E}|F - F^{(k)}|^2$. We use it for a complex function of independent variables that are not identically distributed, so we recall the proof. Let $\mathbb{E}_k$ be the average over U_k alone, and let $\zeta_k = \mathbb{E}[F - \mathbb{E}_k F \mid U_1, \dots, U_k]$. Then $\zeta_k = \mathbb{E}[F \mid U_1, \dots, U_k] - \mathbb{E}[F \mid U_1, \dots, U_{k-1}]$, so $F - \mathbb{E}F = \sum_k \zeta_k$ and the ζ_k are orthogonal. Hence $\operatorname{Var} F = \sum_k \mathbb{E}|\zeta_k|^2 \le \sum_k \mathbb{E}|F - \mathbb{E}_k F|^2$, and $\mathbb{E}|F - \mathbb{E}_k F|^2 = \frac{1}{2}\mathbb{E}|F - F^{(k)}|^2$, because given the other parents F and $F^{(k)}$ are independent with the same law.

Replacing U_k by U_k' moves the family of $k+1$ from U_k to U_k'. So $|\mathcal{T}_v|$ changes only for vertices $v \ge 2$ on the path from U_k to the root or on the path from U_k' to the root. These paths do not depend on the choice at step k, since $U_k, U_k' \le k$. Let $\operatorname{dp}(u)$ be the depth of u, namely the number of edges between u and the root. There are at most $\operatorname{dp}(U_k) + \operatorname{dp}(U_k')$ such vertices, and each term of F moves by at most 2. Hence $\mathbb{E}|F - F^{(k)}|^2 \le 8\,\mathbb{E}[\operatorname{dp}(U_k)^2 + \operatorname{dp}(U_k')^2] = 16\,\mathbb{E}\operatorname{dp}(U_k)^2$.

Since $\operatorname{dp}(u+1) = 1 + \operatorname{dp}(U_u)$, the generating function $g_u(s) = \mathbb{E}s^{\operatorname{dp}(u)}$ satisfies $g_1 = 1$ and $u\, g_{u+1}(s) = s\sum_{j \le u} g_j(s)$. Subtracting the same identity for $u-1$ gives $u\, g_{u+1}(s) = (u - 1 + s)\, g_u(s)$ for $u \ge 2$, and $g_2(s) = s$. So $g_u(s) = \prod_{i=1}^{u-1} \frac{i-1+s}{i}$. Thus $\operatorname{dp}(u)$ is a sum of independent Bernoulli$(1/i)$ variables with $1 \le i < u$, and $\mathbb{E}\operatorname{dp}(u)^2 \le H_{u-1}^2 + H_{u-1}$. The parent U_k is independent of the depths of $1, \dots, k$, so $\mathbb{E}\operatorname{dp}(U_k)^2 \le H_N^2 + H_N$. We get $\operatorname{Var} F \le \frac{1}{2}(N-1)\cdot 16(H_N^2 + H_N) \le 8N(H_N^2 + H_N)$, and the bound of Step 3 is at most $4x^2(H_N^2 + H_N)/N$.

Step 5: Fourier inversion. For every integer m,

$$\mathbb{P}(Z_+ = m) - \mathrm{CP}_N(m) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-im\theta}\big(\mathbb{E}e^{i\theta Z_+} - \widehat{\mathrm{CP}}_N(\theta)\big)\, d\theta,$$

so the left side is at most $\sup_\theta |\mathbb{E}e^{i\theta Z_+} - \widehat{\mathrm{CP}}_N(\theta)|$ in absolute value. By Steps 2 to 4 this is at most $\frac{4x^2}{N}e^{4x^2/N} + \frac{4x^2(H_N^2 + H_N)}{N}$. Finally $|f_N(m) - \mathbb{P}(Z_+ = m)| \le \mathbb{P}(M_N \ne Z_+)$, and Step 1 bounds it. This proves Theorem 3.9. □

C.2 Proof of Theorem 3.10

We write $B(t) = -it\log|t| - \frac{\pi}{2}|t|$ for real $t \ne 0$, and $B(0) = 0$. For $t > 0$ this is the value of $u\log u$ at $u = -it$, with the principal logarithm, and $B(-t) = \overline{B(t)}$. We need three facts about Ψ_N and Ψ_∞.

Fact 1. For real θ, $|\Psi_N(\theta) - \Psi_\infty(\theta)| \le \sum_{d \ge N} \frac{2}{d(d+1)} = \frac{2}{N}$.

Fact 2. For $0 < |\theta| \le \pi$ we have $\Psi_\infty(\theta) = -i\theta\log(-i\theta) + R(\theta)$ with $|R(\theta)| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big)$. The reason is that Ψ_∞ has a closed form in which θ enters only through $1 - e^{i\theta}$, and $1 - e^{i\theta}$ is close to $-i\theta$ for small θ. The term R is what this replacement changes. We prove it in the next four paragraphs.

We start with the closed form. For $0 < |s| \le 1$, $s \ne 1$,

$$\sum_{d \ge 1} \frac{s^d - 1}{d(d+1)} = \chi(s) := \frac{(1-s)\log(1-s)}{s},$$

because $\frac{1}{d(d+1)} = \frac{1}{d} - \frac{1}{d+1}$, $\sum_d s^d/d = -\log(1-s)$, $\sum_d s^d/(d+1) = (-\log(1-s) - s)/s$ and $\sum_d \frac{1}{d(d+1)} = 1$. So $\Psi_\infty(\theta) = \chi(e^{i\theta})$.

Next we compare $1 - e^{i\theta}$ with $-i\theta$. Write $1 - e^{i\theta} = (-i\theta)\,\mathsf{u}$ with $\mathsf{u} = \operatorname{sinc}(\theta/2)\, e^{i\theta/2}$, where $\operatorname{sinc} y = \sin y/y$ is positive for $|y| \le \frac{\pi}{2}$. So u is close to 1 when θ is small. The argument of $-i\theta$ is $\mp\frac{\pi}{2}$ for $\pm\theta > 0$ and that of u is $\theta/2$. They add up to a number in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\log(1 - e^{i\theta}) = \log(-i\theta) + \log \mathsf{u}$ for principal logarithms.

Now we get a formula for R. By the two steps above, $\chi(e^{i\theta}) = (-i\theta)\,\mathsf{u}e^{-i\theta}\big(\log(-i\theta) + \log \mathsf{u}\big)$. Subtracting $-i\theta \log(-i\theta)$ gives

$$R(\theta) = -i\theta \log(-i\theta)\big(\mathsf{u}e^{-i\theta} - 1\big) - i\theta\, \mathsf{u}e^{-i\theta} \log \mathsf{u}.$$

It remains to bound the three quantities in this formula. We show that

$$|\mathsf{u}e^{-i\theta} - 1| \le 0.631\,|\theta|, \qquad |\log \mathsf{u}| \le 0.53\,|\theta|, \qquad |\log(-i\theta)| \le |\log|\theta|| + \frac{\pi}{2}.$$

For the first, put $y = \theta/2$. Then $|\mathsf{u}e^{-i\theta} - 1| = |\operatorname{sinc} y - e^{iy}| \le (1 - \operatorname{sinc} y) + |1 - e^{iy}| \le \frac{y^2}{6} + |y| \le (\frac{1}{2} + \frac{\pi}{24})|\theta|$, where the last step uses $|\theta| \le \pi$, and $\frac{1}{2} + \frac{\pi}{24} \le 0.631$. For the second, $|\log \mathsf{u}|^2 = \log^2 \operatorname{sinc}(\theta/2) + \theta^2/4$, and $-\log \operatorname{sinc} y = \sum_{k\ge1} \frac{\zeta(2k)}{k\pi^{2k}} y^{2k}$ has positive coefficients. So $|\log \operatorname{sinc}(\theta/2)|/|\theta|$ increases in $|\theta|$, and its value at $|\theta| = \pi$ gives $|\log \mathsf{u}| \le \big(\frac{1}{4} + \frac{\log^2(\pi/2)}{\pi^2}\big)^{1/2}|\theta| \le 0.53\,|\theta|$. The third holds because $-i\theta$ has modulus $|\theta|$ and argument $\pm\frac{\pi}{2}$. Since $|\mathsf{u}e^{-i\theta}| \le 1$, the formula for R now gives

$$|R(\theta)| \le |\theta|\Big(|\log|\theta|| + \frac{\pi}{2}\Big) \cdot 0.631\,|\theta| + |\theta| \cdot 0.53\,|\theta| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big),$$

as $0.631 \cdot \frac{\pi}{2} + 0.53 \le 1.53$.

Fact 3. For $|\theta| \le \pi$, $-\operatorname{Re}\Psi_\infty(\theta) = \sum_d \frac{1-\cos\theta d}{d(d+1)} \ge \frac{|\theta|}{2\pi}$. We may take $\theta \ne 0$. Keep only the terms with $d \le D := \lfloor \pi/|\theta| \rfloor$ and use $1 - \cos v \ge 2v^2/\pi^2$ for $|v| \le \pi$. This gives at least $\frac{2\theta^2}{\pi^2} \sum_{d\le D} \frac{d}{d+1} \ge \frac{\theta^2 D}{\pi^2} \ge \frac{|\theta|}{2\pi}$, since $D \ge \frac{\pi}{2|\theta|}$.

The function e^B is a characteristic function of class L. Let S_x have the law CP_∞ with parameter x, namely the law of $\sum_{d\ge1} d\,\mathcal{N}_d$. Its characteristic function is $e^{x\Psi_\infty(\theta)}$. By Fact 2, for $t \ne 0$ and $x \ge \max(1, |t|/\pi)$,

$$x\Psi_\infty(t/x) - it\log x = B(t) + xR(t/x), \qquad |xR(t/x)| \le \frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big). \quad (11)$$

So the characteristic function of $(S_x - x\log x)/x$ tends to $e^{B(t)}$ as $x \to \infty$, for every real t. The limit is continuous at 0, so by Lévy's continuity theorem it is the characteristic function of a probability law. It is integrable, so this law has the bounded continuous density f_L of Theorem 3.10. For $0 < c < 1$ and real t we have $B(ct) = cB(t) - ict\log c$, and hence

$$e^{B(t)} = e^{B(ct)} \cdot e^{B((1-c)t)}\, e^{it[(1-c)\log(1-c) + c\log c]}.$$

The last two factors form the characteristic function of a scaled and shifted copy of the law. So the law is of class L (self-decomposable), and f_L is unimodal by [14].

Proof of (a). By Fourier inversion on $\mathbb{Z}$ and the substitution $\theta = t/x$,

$$x\,\mathrm{CP}_N(m) = \frac{1}{2\pi}\int_{-\pi x}^{\pi x} e^{-i\lambda_m t} e^{A(t)}\,dt, \qquad A(t) = x\Psi_N(t/x) - it\log x.$$

Let $|t| \le \pi x$. By (11) and Fact 1,

$$|A(t) - B(t)| \le \frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big) + \frac{2x}{N},$$

and by Facts 1 and 3, $\max(\operatorname{Re} A(t), \operatorname{Re} B(t)) \le -\frac{|t|}{2\pi} + \frac{2x}{N}$. Now $|e^A - e^B| \le |A - B|\, e^{\max(\operatorname{Re} A, \operatorname{Re} B)}$, and $\frac{2x}{N} \le \frac{1}{2}$. The integrals of $t^2 e^{-|t|/2\pi}$, of $t^2|\log|t||e^{-|t|/2\pi}$ and of $e^{-|t|/2\pi}$ over the real line are finite. So

$$\frac{1}{2\pi}\int_{-\pi x}^{\pi x} |e^{A(t)} - e^{B(t)}|\, dt \le C\Big(\frac{1 + \log x}{x} + \frac{x}{N}\Big).$$

Beyond πx we use $|e^{B(t)}| = e^{-\pi|t|/2}$, which gives $\frac{1}{2\pi}\int_{|t|>\pi x} |e^{B(t)}|\, dt = \frac{2}{\pi^2} e^{-\pi^2 x/2} \le \frac{C}{x}$. Since $f_L(\lambda_m) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-i\lambda_m t} e^{B(t)}\, dt$, this proves (a).

Proof of (b). By Theorem 3.9, $|x f_N(m) - x\,\mathrm{CP}_N(m)| \le x\delta_B$. Add (a).

Proof of (c). Let $x \to \infty$ and $x^3 \log^2 N/N \to 0$. Then $x\delta_B \to 0$ and $x/N \to 0$, so the bound of (b) tends to 0. Let U be uniform on $[0, 1)$ and independent of M_N. Then $(M_N + U - x\log x)/x$ has the density g_N with $g_N(\lambda) = x f_N(m)$ for $\lambda_m \le \lambda < \lambda_m + \frac{1}{x}$. On a compact set of λ the relevant m are nonnegative once x is large, and $|\lambda - \lambda_m| < \frac{1}{x}$. As f_L is uniformly continuous, $g_N \to f_L$ uniformly on compact sets. By Scheffé's lemma $g_N \to f_L$ in L^1, and since $U/x \to 0$ this gives the convergence in law. At the crossing $x_N \sim 2\log N$ by Theorem 2.7, so both conditions hold. □

C.3 Proof of Lemma 3.11

Step 1: a real integral. We claim that for every real λ and $k \ge 0$,

$$f_L^{(k)}(\lambda) = \frac{1}{\pi}\int_0^\infty (-s)^k\, e^{-s\log s - \lambda s} \sin(\pi s)\, ds. \tag{12}$$

Since $B(-t) = \overline{B(t)}$, the definition of f_L gives $f_L(\lambda) = \frac{1}{\pi}\operatorname{Re}\int_0^\infty \Phi(-it)\, dt$ with $\Phi(u) = \exp(u\log u + \lambda u)$. The function Φ is holomorphic off the negative real axis. Put $u = \rho e^{i\varphi}$ with $-\pi < \varphi \le -\frac{\pi}{2}$. Then $\operatorname{Re}(u\log u) = \rho(\log\rho\cos\varphi - \varphi\sin\varphi)$, where $\cos\varphi \le 0$ and $\varphi\sin\varphi \ge 0$. For $\log\rho \ge |\lambda| + 1$ this gives $\operatorname{Re}(u\log u + \lambda u) \le -\rho(|\cos\varphi| + \varphi\sin\varphi) \le -\rho$, because $|\cos\varphi| + \varphi\sin\varphi \ge 1$ on $[-\pi, -\frac{\pi}{2}]$ (for $\psi = \varphi + \pi \in [0, \frac{\pi}{2}]$ it equals $\cos\psi + (\pi - \psi)\sin\psi \ge \cos\psi + \sin\psi \ge 1$). Also Φ is bounded near 0. So by Cauchy's theorem we may turn the ray $u = -it$ to the lower side of the negative real axis, where $u = -s$ and $\log u = \log s - i\pi$. There $u\log u = -s\log s + i\pi s$, and $dt = i\, du = -i\, ds$. Hence

$$f_L(\lambda) = \frac{1}{\pi}\operatorname{Re}\Big[-i\int_0^\infty e^{-s\log s - \lambda s}\, e^{i\pi s}\, ds\Big] = \frac{1}{\pi}\int_0^\infty e^{-s\log s - \lambda s}\sin(\pi s)\, ds.$$

The integrand decays faster than every exponential, so we may differentiate under the integral sign. This proves (12). Also f_L'' is bounded on the real line, because $t^2 e^{B(t)}$ is integrable and we may differentiate the Fourier formula for f_L twice.

Step 2: certified enclosures. Let $-0.223 \le \lambda \le 0$ and $k \in \{1, 2\}$, and split the integral in (12) at $\delta = 2^{-20}$ and at $T = 30$. On $[0, \delta]$ we have $-s\log s \le 1/e$, $-\lambda s \le 0.223\, s$ and $|\sin\pi s| \le \pi s$, so this part is at most $\pi e^{1/e + 0.223\delta}\delta^{k+2}/(k+2)$ in absolute value. On $[T, \infty)$ we have $s\log s + \lambda s \ge cs$ with $c = \log 30 - 0.223 > 3$, so this part is at most $\int_T^\infty s^2 e^{-cs}\, ds \le T^2 e^{-cT}$. Both are below 10^{-17}. On $[\delta, T]$ the integrand is holomorphic in a neighborhood of the path, and we enclose the integral with the rigorous integration algorithm of [18], in ball arithmetic with 128 bits. The parameter λ enters as a ball, so one run encloses the integral for every λ in an interval. The script `certify_elephant.py` does this and reports

$$f_L'(-0.2229) \in [9.232 \cdot 10^{-6}, 9.234 \cdot 10^{-6}], \qquad f_L'(-0.2227) \in [-6.548 \cdot 10^{-6}, -6.546 \cdot 10^{-6}],$$

and $f_L''(\lambda) \in [-0.0792, -0.0786]$ for every $\lambda \in [-0.2229, -0.2227]$. In the same way it finds that f_L' is positive at -0.22278299 and negative at -0.22278297. The two enclosures are $6.8984026 \cdot 10^{-10}$ and $-8.8809315 \cdot 10^{-10}$ with radii below 10^{-17}, and these are the smallest margins in the proof.

Step 3: the mode. Write $[a, b] = [-0.2229, -0.2227]$. By Step 2, f_L' is strictly decreasing on $[a, b]$ with $f_L'(a) > 0 > f_L'(b)$. So f_L' has exactly one zero λ_0 in $[a, b]$, and f_L is strictly increasing on $[a, \lambda_0]$ and strictly decreasing on $[\lambda_0, b]$. Since f_L is unimodal, there is a point λ_* such that f_L is nondecreasing on $(-\infty, \lambda_*]$ and nonincreasing on $[\lambda_*, \infty)$. If $\lambda_* > \lambda_0$, then f_L would be nondecreasing just to the right of λ_0, which is false. In the same way $\lambda_* < \lambda_0$ is impossible. So $\lambda_* = \lambda_0$. By Step 2 again, $\lambda_0 \in (-0.22278299, -0.22278297)$. This proves (a) and (b).

Step 4: the constant κ_L. Let $\delta_0 = \min(\lambda_0 - a, b - \lambda_0) > 0$. For $|\lambda - \lambda_0| \le \delta_0$, Taylor's formula with $f_L'(\lambda_0) = 0$ and $f_L'' \le -0.078$ on $[a, b]$ gives $f_L(\lambda_0) - f_L(\lambda) \ge 0.039(\lambda - \lambda_0)^2$. For $|\lambda - \lambda_0| > \delta_0$ the monotonicity of Step 3 gives $f_L(\lambda) \le f_L(\lambda_0 \pm \delta_0) \le f_L(\lambda_0) - 0.039\,\delta_0^2$. Since $\delta_0 < 1$, the bound in (c) holds with $0.039\,\delta_0^2$ in place of κ_L. By (a), $\lambda_0 - a > 0.2229 - 0.22278299 > 1.17 \cdot 10^{-4}$ and $b - \lambda_0 > 0.22278297 - 0.2227 = 8.297 \cdot 10^{-5}$, so $0.039\,\delta_0^2 > 0.039 \cdot (8.297 \cdot 10^{-5})^2 > 2.68 \cdot 10^{-10}$. So (c) holds with $\kappa_L = 2.6 \cdot 10^{-10}$. □

C.4 Proof of Corollary 3.12

The method is to compare f_N at a mode with f_N at the integer m' nearest to the predicted place $x \log x + \lambda_0 x$. By Theorem 3.10(b) both values are close to values of f_L, and f_L falls off quadratically away from λ_0. So a mode cannot be far from m'.

We first prove a statement at finite N. Assume the standing assumptions of Section 3, let κ_L be as in Lemma 3.11(c), and let E_1 be the bound of Theorem 3.10(b). Put

$$E' = E_1 + \frac{16x^2 H_N}{N^2} + \frac{\|f_L''\|_\infty}{8x^2}.$$

We claim that if $E' < \kappa_L/2$ and $m_h \le N/4$, then every mode m of f_N satisfies $|m - x\log x - \lambda_0 x| \le x\sqrt{2E'/\kappa_L}$, and so does m^* for every mode $a^* \ge N/2$ of P_N.

Let m' be the integer nearest to $x \log x + \lambda_0 x$. This number increases in $x \ge 1$ and is at least $\lambda_0 > -\frac{1}{2}$, so $m' \ge 0$. Also $|\lambda_{m'} - \lambda_0| \le \frac{1}{2x}$, and since $f_L'(\lambda_0) = 0$, $f_L(\lambda_{m'}) \ge f_L(\lambda_0) - \|f_L''\|_\infty/(8x^2)$. Let $\tilde{m}$ be a mode of f_N or $\tilde{m} = m^*$. We claim that $f_N(\tilde{m}) \ge f_N(m') - 16xH_N/N^2$. For a mode of f_N this is clear. For m^* we have $f_N(m^*) + f_N(N - m^*) = 2P_N(N - m^*) \ge 2P_N(N - m') \ge f_N(m')$. Here $N - m^* \ge \lceil N/2 \rceil \ge \lceil N/4 \rceil \ge m_h$. So f_N is nonincreasing on $\{\lceil N/4 \rceil, \dots, N\}$ by Lemma 3.4. There are at least $N/4$ integers from $\lceil N/4 \rceil$ to $\lceil N/2 \rceil$. So Lemma B.1 (with $a = \lceil N/4 \rceil \ge m_h$ and $b = \lceil N/2 \rceil$) gives the second inequality below, and Lemma B.2 gives the third,

$$f_N(N - m^*) \le f_N(\lceil N/2 \rceil) \le \frac{\mathbb{P}(M_N \ge \lceil N/4 \rceil)}{N/4} \le \frac{16xH_N}{N^2}.$$

Now Theorem 3.10(b), used at $\tilde{m}$ and at m', gives

$$f_L(\lambda_{\tilde{m}}) \ge x f_N(\tilde{m}) - E_1 \ge x f_N(m') - \frac{16x^2 H_N}{N^2} - E_1 \ge f_L(\lambda_{m'}) - 2E_1 - \frac{16x^2 H_N}{N^2} \ge f_L(\lambda_0) - 2E'.$$

Since $2E' < \kappa_L$, Lemma 3.11(c) gives $(\lambda_{\tilde{m}} - \lambda_0)^2 \le 2E'/\kappa_L$, which is the claim.

Now take $x = x_N$ at the crossing. By Theorem 2.7, $x \sim 2\log N$. So $x\delta_B$, x/N, $16x^2 H_N/N^2$ and $\|f_L''\|_\infty/(8x^2)$ are all $O(\log x/x)$, and $E' = O(\log x/x)$. For large N the standing assumptions hold, $E' < \kappa_L/2$, and Lemma B.7 gives $m_h \le A_0 x \log^2(x+1) \le N/4$. Hence $|m^* - x\log x - \lambda_0 x| \le x\sqrt{2E'/\kappa_L} = O(\sqrt{x \log x})$, and the same holds for m_h. □

Statements and declarations

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Data availability. All code, data and Lean files supporting this paper are in the folder `kagey-problems/problem131` of `https://github.com/apemm/Kagey-Problems`, at the release `paperD-arxiv-v1`. To rerun the checks, run `python -B verify_all.py --paper d` in that folder, which takes several minutes. The scripts are in the subfolder `paperD` and need `numpy`, `scipy`, `mpmath` and `python-flint`. The file `SUPPLEMENT.md` says what each script checks and where to find the predictions that we wrote down before each computation. Section 4 describes the checks.

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